

Risk as Alpha

ML for Cross-Asset Signal Discovery

A Learning Guide for the
SecDB Signal Discovery Project

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This document teaches the theory, methods, and methodology needed to execute and defend the Risk-as-Alpha internship project.

Part I covers foundations: asset pricing, risk systems, ML methods.

Part II covers applied methodology: features, validation, backtesting.

Every chapter connects to a specific project phase.

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Part I

Foundations

Chapter 1

Asset Pricing Foundations

Project Connection

This chapter provides the theoretical foundation for why risk signals should predict returns. Understanding the SDF framework is essential for framing your project thesis (Phase 0 pitch) and interpreting results (Phase 5 presentation). Every feature you engineer in Phase 2 is, at its core, a proxy for a component of the pricing kernel. When your sponsor asks “why should VaR dynamics predict returns?” the answer lives here.

1.1 The Stochastic Discount Factor

Conditional Expectation

If X and Y are random variables, the conditional expectation $\mathbb{E}[X | Y]$ is itself a random variable—it gives the “best prediction” of X given knowledge of Y . The law of iterated expectations says $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$. You will use this constantly in asset pricing: today’s price reflects the conditional expectation of tomorrow’s discounted payoff.

Law of One Price

Two portfolios with identical future payoffs in every state of the world must have the same price today. If they did not, you could buy the cheap one, sell the expensive one, and pocket the difference risklessly. This principle is weaker than no-arbitrage (it does not rule out “free lottery tickets”), but it is enough to guarantee the existence of a linear pricing rule.

Gross vs. Net Returns

Finance uses two return conventions. A **net return** r_i is the percentage gain: if you invest \$100 and end with \$108, $r_i = 0.08$ (8%). A **gross return** $R_i = 1 + r_i = 1.08$ is the wealth multiplier: \$100 becomes $100 \times 1.08 = \$108$. The SDF equation uses gross returns because the algebra is cleaner. When you see $\mathbb{E}[\mathcal{M} \cdot R_i] = 1$, remember R_i includes the principal repayment.

The central object in modern asset pricing is the **stochastic discount factor** (SDF), also called the pricing kernel or state-price density. It is a single random variable that prices every asset in the economy.

Stochastic Discount Factor

A random variable $\mathcal{M}_{t+1}$ is a stochastic discount factor if, for every asset i with gross return $R_{i,t+1}$,

$$\mathbb{E}_t[\mathcal{M}_{t+1} R_{i,t+1}] = 1 \quad (1.1)$$

where $\mathbb{E}_t[\cdot]$ denotes the expectation conditional on information available at time t .

- $\mathcal{M}_{t+1}$ — the SDF, a random variable realized at $t+1$. High in “bad” states, low in “good” states.
- $R_{i,t+1}$ — gross return on asset i (i.e., $1 + r_i$ where r_i is the net return).
- $\mathbb{E}_t[\cdot]$ — expectation conditional on all information known at time t .

Equivalently, if the asset has a random payoff x_{t+1} and a price p_t today:

$$p_t = \mathbb{E}_t[\mathcal{M}_{t+1} x_{t+1}] \quad (1.2)$$

Price equals expected discounted payoff. Every model in finance—CAPM, Fama-French, option pricing, term structure models—is a different claim about what $\mathcal{M}$ looks like (Cochrane, 2005, Ch. 1).

The SDF as “State Dislike”

The SDF is a single number that tells you how much the market “dislikes” each state of the world. States where you are already losing money—recessions, financial crises, liquidity dry-ups—get high SDF values. Assets that pay off precisely in those painful states are valuable (they provide insurance), so their prices get bid up and their expected returns are low. Assets that only pay off in good times are cheap and must offer high expected returns to attract buyers. The SDF is the exchange rate between dollars today and dollars tomorrow, where the rate depends on which “tomorrow” you are in.

1.1.1 The Fundamental Theorem of Asset Pricing

The deep connection between no-arbitrage and the SDF is:

Fundamental Theorem of Asset Pricing

The following are equivalent (Harrison and Kreps, 1979; Harrison and Pliska, 1981):

1. There are no arbitrage opportunities (no way to get something for nothing).
2. There exists a strictly positive SDF: $\mathcal{M} > 0$ such that $\mathbb{E}[\mathcal{M} R_i] = 1$ for all traded assets i .

Strictly positive means $\mathcal{M} > 0$ in every state—it never assigns zero value to any possible outcome.

This theorem is the bedrock of quantitative finance. It says: if markets make any sense at all (no free money), then there *must* exist a positive random variable that prices everything. The entire research program of asset pricing is figuring out what drives $\mathcal{M}$.

No-Arbitrage

An **arbitrage** is a trading strategy that (i) costs nothing to enter, (ii) has no possibility of losing money, and (iii) has a positive probability of making money. Example: gold trades at \$2,000 in New York and \$2,010 in London. Buy in New York, sell in London, pocket \$10 risk-free. No-arbitrage means these opportunities do not persist—they get traded away instantly. This is the weakest assumption in finance: even if investors are irrational, as long as *someone* is willing to pick up free money, arbitrage opportunities vanish.

Note that the fundamental theorem does *not* say $\mathcal{M}$ is unique. In incomplete markets (where not every payoff can be replicated by traded assets), many different SDFs are consistent with no-arbitrage. Each corresponds to a different way of pricing payoffs that are not currently traded. The CAPM, Fama-French, and intermediary models (Chapter 2) each propose a specific $\mathcal{M}$; they disagree on which one is “correct.”

1.1.2 The Risk-Free Rate and the SDF

If a risk-free asset exists with certain gross return R_f , then since R_f is non-random:

$$\mathbb{E}_t[\mathcal{M}_{t+1}] \cdot R_f = 1 \implies R_f = \frac{1}{\mathbb{E}_t[\mathcal{M}_{t+1}]} \quad (1.3)$$

The risk-free rate is the inverse of the expected SDF. When the expected SDF is high (the market expects bad times ahead and “discounts the future heavily”), the risk-free rate is low.

1.2 Risk Premia

Covariance Decomposition

For any two random variables X and Y : $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \text{Cov}(X, Y)$. This identity is the algebraic engine behind most of asset pricing. When you see $\mathbb{E}[\mathcal{M} \cdot R] = 1$, splitting it into mean and covariance components gives you the risk premium formula.

Why do different assets earn different expected returns? Apply the covariance decomposition to equation (1.1):

$$\begin{aligned} 1 &= \mathbb{E}_t[\mathcal{M}_{t+1} R_{i,t+1}] \\ &= \mathbb{E}_t[\mathcal{M}_{t+1}] \mathbb{E}_t[R_{i,t+1}] + \text{Cov}_t(\mathcal{M}_{t+1}, R_{i,t+1}) \end{aligned} \quad (1.4)$$

Substituting $\mathbb{E}_t[\mathcal{M}_{t+1}] = 1/R_f$ from equation (1.3) and rearranging:

Risk Premium Formula

$$\mathbb{E}_t[R_{i,t+1}] - R_f = -R_f \text{Cov}_t(\mathcal{M}_{t+1}, R_{i,t+1}) \quad (1.5)$$

- $\mathbb{E}_t[R_{i,t+1}] - R_f$ — the **risk premium** (expected excess return) on asset i .
- R_f — the gross risk-free rate (a positive constant).
- $\text{Cov}_t(\mathcal{M}_{t+1}, R_{i,t+1})$ — covariance between the SDF and the asset’s return.
- The minus sign: assets with *negative* covariance with $\mathcal{M}$ (they pay off when the SDF is low, i.e., in good times) earn *positive* risk premia.

This single equation explains all of asset pricing. An asset earns a risk premium if and only if its return covaries negatively with the SDF. No covariance, no premium—regardless of how volatile the asset is.

Risk vs. Volatility

Volatility alone does not earn you a risk premium. Only *systematic* risk—covariance with the pricing kernel—is compensated. A coin flip that doubles or halves your money is extremely volatile but has zero SDF covariance (coin flips are unrelated to aggregate economic conditions), so it earns zero premium. This is why idiosyncratic risk is not priced in equilibrium.

Insurance Analogy

Think of risk premia like insurance premiums. Homeowner's insurance costs money (negative expected return for the buyer) because it pays off precisely when you need it most—when your house burns down. The insurance company earns a positive expected return by bearing that risk. In asset pricing, Treasury bonds are like insurance: they pay off in recessions (when the SDF is high), so their expected returns are low. Equities are like being the insurance company: they lose value in recessions, so investors demand a premium to hold them. The market risk premium ($\sim 6\text{--}8\%$ /year historically) is the “premium” investors earn for bearing aggregate economic risk.

Worked Example:

Two Assets in Two States Consider a simple economy with two equally likely states: **Boom** and **Recession**.

	Probability	SDF ($\mathcal{M}$)
Boom	0.5	0.8
Recession	0.5	1.2

The SDF is higher in recessions (the market dislikes recessions). The risk-free rate is:

$$R_f = \frac{1}{\mathbb{E}[\mathcal{M}]} = \frac{1}{0.5 \times 0.8 + 0.5 \times 1.2} = \frac{1}{1.0} = 1.0 \quad (\text{i.e., } r_f = 0\%)$$

Asset A (“procyclical”): pays $R_A = 1.3$ in Boom, $R_A = 0.7$ in Recession.

$$\mathbb{E}[R_A] = 0.5(1.3) + 0.5(0.7) = 1.0$$

$$\text{Cov}(\mathcal{M}, R_A) = \mathbb{E}[\mathcal{M} \cdot R_A] - \mathbb{E}[\mathcal{M}]\mathbb{E}[R_A] = [0.5(0.8)(1.3) + 0.5(1.2)(0.7)] - (1.0)(1.0) = 0.94 - 1.0 = -0.06$$

Risk premium = $-R_f \cdot \text{Cov}(\mathcal{M}, R_A) = -(1.0)(-0.06) = +0.06$, i.e., +6%.

But wait— $\mathbb{E}[R_A] = 1.0$ and $R_f = 1.0$, so the premium is 0%, not 6%. That is because Asset A is mispriced at our assumed returns. To be consistent with the SDF, Asset A's price must adjust so that $p_A = \mathbb{E}[\mathcal{M} \cdot x_A] = 0.94$, giving an expected return of $\mathbb{E}[x_A]/p_A = 1.0/0.94 = 1.064$, or a 6.4% expected return.

Asset B (“countercyclical”): pays $R_B = 0.7$ in Boom, $R_B = 1.3$ in Recession.

$$\text{Cov}(\mathcal{M}, R_B) = [0.5(0.8)(0.7) + 0.5(1.2)(1.3)] - (1.0)(1.0) = 1.06 - 1.0 = +0.06$$

Risk premium = $-(1.0)(+0.06) = -0.06$, i.e., -6%.

Asset B pays off in recessions when the SDF is high—it provides insurance. Investors bid up its price, driving its expected return *below* the risk-free rate. Asset A does the opposite: it fails you in recessions, so you demand compensation.

Takeaway: Same volatility (30% spread between states), opposite risk premia. Covariance with $\mathcal{M}$, not variance, determines expected returns.

1.3 Factor Models

Linear Regression and OLS

A factor model is just a multivariate linear regression. If you regress asset returns R_i on factors $f_1, \dots, f_K$:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

The β 's are OLS coefficients measuring the asset's sensitivity ("loading") on each factor. The R^2 tells you what fraction of return variation the factors explain. The intercept α_i is the return left unexplained—the "alpha."

The SDF framework is powerful but abstract. In practice, we need to specify *what drives* $\mathcal{M}$. Factor models do this by saying: the SDF is a linear function of a small number of observable factors.

Linear Factor Model

A linear factor model assumes the SDF is:

$$\mathcal{M}_{t+1} = a - \mathbf{b}^\top \mathbf{f}_{t+1} \quad (1.6)$$

where $\mathbf{f}_{t+1}$ is a $K \times 1$ vector of factor returns and $a, \mathbf{b}$ are constants.

- a — a constant ensuring $\mathbb{E}[\mathcal{M}] = 1/R_f$.
- $\mathbf{b}$ — vector of factor "prices of risk." Larger b_k means factor k matters more for pricing.
- $\mathbf{f}_{t+1}$ — realized factor returns (e.g., market return, size factor, value factor).

Substituting equation (1.6) into the risk premium formula (1.5) and using the covariance decomposition:

$$\mathbb{E}[R_i] - R_f = \sum_{k=1}^K \beta_{i,k} \lambda_k \quad (1.7)$$

where $\beta_{i,k} = \text{Cov}(R_i, f_k) / \text{Var}(f_k)$ is asset i 's loading on factor k , and λ_k is the risk premium for factor k (the expected excess return on a portfolio with unit beta on factor k and zero on all others). This is the **beta pricing equation**—the operational heart of factor models. Every specific model is a claim about which factors belong in $\mathbf{f}$.

Factors as "Bad Times" Indicators

Why should loading on a factor earn you a premium? Because factors represent states of the world that investors dislike. The market factor is obvious: it drops in recessions. The size factor captures the idea that small firms are particularly vulnerable in downturns (they have less access to credit, thinner margins). The value factor captures the idea that cheap, distressed firms are risky precisely because they are already in trouble and could get worse. Each factor is, in SDF language, a component of $\mathcal{M}$ —a dimension of "bad times."

1.3.1 CAPM: The One-Factor Benchmark

The **Capital Asset Pricing Model** (Sharpe, 1964; Lintner, 1965) is the simplest factor model. It says the market portfolio is the only factor:

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f) \quad (1.8)$$

- $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$ — asset i 's sensitivity to the market.
- $\mathbb{E}[R_m] - R_f$ — the **market risk premium** (historically $\approx 6\text{--}8\%$ per year for U.S. equities).
- The equation says: your expected excess return is proportional to your market beta.

In SDF language, the CAPM says $\mathcal{M}_{t+1} = a - b R_{m,t+1}$: the pricing kernel depends *only* on the market return. This is equivalent to assuming all investors hold mean-variance efficient portfolios and have homogeneous expectations. Under these assumptions, the market portfolio is mean-variance efficient, and market beta is the only source of priced risk.

The CAPM is elegant but empirically incomplete. Early tests by Black, Jensen, and Scholes (1972) found a positive relation between beta and average return, but the relationship is “too flat”: low-beta stocks earn more than the CAPM predicts, and high-beta stocks earn less. Fama and French (1992) went further, showing that beta has essentially no explanatory power for the cross-section of returns once you control for firm size. Fama and French (2004) summarized decades of evidence with the blunt assessment that “the failure of the CAPM in empirical tests implies that most applications of the model are invalid.”

Despite these failures, the CAPM remains your default benchmark. When your sponsor asks “does this signal have alpha?” they mean alpha relative to the market at minimum. You will always include market beta as a control.

1.3.2 Fama-French Three-Factor Model

Fama and French (1993) proposed three factors:

$$\mathbb{E}[R_i] - R_f = \beta_i^{\text{MKT}} (\mathbb{E}[R_m] - R_f) + \beta_i^{\text{SMB}} \mathbb{E}[\text{SMB}] + \beta_i^{\text{HML}} \mathbb{E}[\text{HML}] \quad (1.9)$$

- **MKT** — excess return on the market portfolio (same as CAPM).
- **SMB** (Small Minus Big) — return on small-cap stocks minus large-cap stocks. Captures the **size premium**: small firms earn higher average returns.
- **HML** (High Minus Low) — return on high book-to-market stocks minus low book-to-market stocks. Captures the **value premium**: cheap (“value”) firms earn higher average returns than expensive (“growth”) firms.

Fama and French (1993)

The three-factor model explains over 90% of diversified portfolio return variation, compared with roughly 70% for the CAPM alone. In their 1963–1991 U.S. sample, the average monthly SMB premium was approximately 0.27% ($\approx 3.2\%$ annualized) and the average monthly HML premium was approximately 0.40% ($\approx 4.8\%$ annualized). After controlling for size and book-to-market, market beta has no residual explanatory power for the cross-section of expected returns.

Why Does This Model “Work”?

Fama and French (1993) is a statistical fact, not an explanation. It tells you that size and value loadings predict returns, but it does not tell you *why*. Fama and French argue these are risk factors (small/value firms are riskier in recessions). Others argue they reflect mispricing (investors systematically undervalue cheap, boring firms). For your project, this debate matters less than the practical point: any “alpha” you claim must survive as alpha *after* controlling for known factors. If your signal just loads on HML, you have not

found anything new.

1.3.3 Arbitrage Pricing Theory (APT)

Ross (1976) took a different approach. Instead of deriving factors from an equilibrium model, the **Arbitrage Pricing Theory** starts from two assumptions:

1. Returns are generated by a linear factor model:

$$R_{i,t} = a_i + \beta_{i,1}f_{1,t} + \beta_{i,2}f_{2,t} + \cdots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t} \quad (1.10)$$

2. No-arbitrage: you cannot construct a zero-cost, zero-risk portfolio with a positive expected return.

From these, Ross proved:

$$\mathbb{E}[R_i] - R_f \approx \beta_{i,1}\lambda_1 + \beta_{i,2}\lambda_2 + \cdots + \beta_{i,K}\lambda_K \quad (1.11)$$

where λ_k is the **risk premium** for factor k —the expected return per unit of exposure to that factor.

- $\beta_{i,k}$ — asset i 's loading (sensitivity) on factor k . Estimated via time-series regression.
- λ_k — the price of risk for factor k . Estimated cross-sectionally.
- $\varepsilon_{i,t}$ — idiosyncratic return, diversifiable and unpriced.
- The $\approx$ sign: the APT pricing relation holds exactly only as the number of assets grows to infinity. For finite portfolios, there is an approximation error.

APT vs. CAPM: What Changed

The CAPM tells you *which* factor matters (the market) but requires strong assumptions (mean-variance investors, a single period, etc.). The APT tells you that *some* factors are priced but does not tell you which ones—that is an empirical question. The APT is more general but less specific. In practice, applied researchers (and your project) take the APT perspective: propose factors motivated by theory, test whether their λ 's are significantly different from zero.

1.3.4 Comparison of Factor Models

	CAPM	Fama-French 3	APT
Factors	Market only	Market, SMB, HML	Unspecified (empirical)
Derivation	Equilibrium (mean-variance optimization)	Empirical observation	No-arbitrage
Assumptions	Homogeneous expectations, single period, all assets tradable	Factors capture risk (debated)	Factor structure in returns, no-arbitrage
Testable?	Yes (but market portfolio unobservable—Roll’s critique)	Yes (time-series and cross-sectional regressions)	Weakly (does not specify factors)
R^2 for portfolios	~70%	~90%+	Depends on factor choice
Practical use	Benchmark; cost of equity	Performance attribution; alpha testing	Theoretical justification for multi-factor models

1.4 Alpha, Beta, and Information Ratio

These three concepts are the vocabulary of quantitative investing. You will use them every day on the desk.

Beta

The **beta** of asset i with respect to factor k is:

$$\beta_{i,k} = \frac{\text{Cov}(R_i, f_k)}{\text{Var}(f_k)} \quad (1.12)$$

It is the OLS slope coefficient from regressing R_i on f_k . Beta measures sensitivity: if $\beta_{i,\text{MKT}} = 1.2$, then when the market rises 1%, asset i rises 1.2% on average (and vice versa for declines).

Alpha

The **alpha** of asset i (or a portfolio/strategy) relative to a set of factors $\mathbf{f}$ is the intercept in the regression:

$$R_{i,t} - R_f = \alpha_i + \beta_i^\top \mathbf{f}_t + \varepsilon_{i,t} \quad (1.13)$$

Alpha is the average return that is *not* explained by the asset’s factor exposures. Positive alpha means the asset earns more than its risk (as measured by the factor model) justifies.

Alpha Is Always Relative

Alpha is always relative to a factor model. A strategy might show $\alpha = 3\%$ per year relative to CAPM (one factor), but $\alpha = 0\%$ relative to Fama-French (three factors)—because the strategy was simply loading on SMB and HML. When you present results in Phase 5, always specify the benchmark. Your sponsor will ask: “Is this alpha relative to what?” If

you cannot answer precisely, your result is uninterpretable. In Chapter 12, you will build this into your reporting pipeline.

Information Ratio

The **information ratio** (IR) of a strategy is:

$$\text{IR} = \frac{\alpha}{\sigma(\varepsilon)} \quad (1.14)$$

- α — the strategy’s alpha (annualized excess return over the benchmark).
- $\sigma(\varepsilon)$ — the **tracking error**, the standard deviation of the strategy’s residual returns (i.e., the volatility of the component not explained by factors).

The IR measures how much excess return you earn per unit of active risk. It is the active manager’s analogue of the Sharpe ratio.

How large is a “good” IR? Top-quartile active managers achieve annualized information ratios of roughly 0.5 (Grinold and Kahn, 2000). An IR of 1.0 is exceptional and rare. Your project signals, which are research-stage and pre-transaction-cost, might show higher in-sample IRs, but always apply the Deflated Sharpe Ratio (Chapter 11) to check whether that is real.

The Fundamental Law of Active Management

Grinold (1989) showed:

$$\text{IR} \approx \text{IC} \times \sqrt{BR} \quad (1.15)$$

- IC — **information coefficient**, the correlation between your signal’s forecast and the realized return. Even small ICs (0.02–0.05) are valuable.
- BR — **breadth**, the number of independent bets per year.

A mediocre signal ($\text{IC} = 0.03$) applied to 500 independent bets per year gives $\text{IR} \approx 0.03 \times \sqrt{500} \approx 0.67$ —a respectable result. This is why cross-asset strategies (more assets = more bets) are attractive, and why your project targets multiple asset classes.

Worked Example:

Computing Alpha and IR Suppose you run a strategy that earns 8% per year with a volatility of 12%. The risk-free rate is 4%. You regress your strategy returns on the Fama-French three factors and estimate:

$$R_{\text{strat},t} - R_f = \underbrace{0.02}_{\alpha} + 0.6 \cdot \text{MKT}_t + 0.3 \cdot \text{SMB}_t - 0.1 \cdot \text{HML}_t + \varepsilon_t$$

- Your alpha is $\alpha = 2\%$ per year. The remaining 2% of your 4% excess return ($8\% - 4\%$) is explained by your market, size, and value exposures.
- If $\sigma(\varepsilon) = 5\%$ per year, then $\text{IR} = 2\%/5\% = 0.40$.
- Interpretation: your strategy generates 0.40 units of excess return per unit of active risk—roughly median for active managers. If your sponsor expected top-quartile performance, you would need $\text{IR} \geq 0.5$.

1.4.1 Sharpe Ratio

Annualizing Returns and Volatility

If you observe monthly returns, annualize the mean by multiplying by 12: $\mu_{\text{ann}} = 12\mu_{\text{monthly}}$. Annualize volatility by multiplying by $\sqrt{12}$: $\sigma_{\text{ann}} = \sqrt{12}\sigma_{\text{monthly}}$. The square-root rule assumes returns are independently distributed across months. If they are autocorrelated, the annualized volatility will be higher (positive autocorrelation) or lower (mean reversion) than the square-root rule suggests.

A closely related measure is the **Sharpe ratio**:

$$\text{SR} = \frac{\mathbb{E}[R_p] - R_f}{\sigma(R_p)} \quad (1.16)$$

The Sharpe ratio uses total excess return and total volatility, while the IR uses alpha (factor-adjusted excess return) and tracking error. If you have no factor exposures ($\beta = 0$), the two are identical. In practice, you will report both. The Sharpe ratio tells you how good the strategy is in isolation; the IR tells you how much value it adds beyond existing exposures.

1.5 Cross-Sectional vs. Time-Series Tests

Panel Data

A **panel** (or longitudinal dataset) has observations indexed by both entity i and time t . In finance, you might have monthly returns for 500 stocks over 20 years: $500 \text{ entities} \times 240 \text{ months} = 120,000 \text{ observations}$. Cross-sectional analysis looks across entities at a point in time. Time-series analysis looks at one entity across time. Panel methods combine both dimensions; see Chapter 8.

Asset pricing tests come in two flavors, and understanding the distinction matters for your project.

1.5.1 Time-Series Tests

A time-series test asks: does a factor explain the *time variation* in an asset's returns? You run:

$$R_{i,t} - R_f = \alpha_i + \beta_i f_t + \varepsilon_{i,t} \quad (1.17)$$

for each asset i separately. The test is whether $\alpha_i = 0$: if the factor model is correct, the intercept should be zero (all expected return is explained by factor exposure). A joint test across all assets (e.g., the GRS test of Gibbons, Ross, and Shanken, 1989) checks whether all alphas are simultaneously zero.

Worked Example:

Time-Series Alpha Test You regress monthly excess returns of a high-yield bond portfolio on the three Fama-French factors over 120 months and estimate $\hat{\alpha} = 0.15\%$ per month with a standard error of 0.08% . The t -statistic is $0.15/0.08 = 1.875$. At the 5% significance level ($t_{\text{crit}} \approx 1.98$ for 116 degrees of freedom), you cannot reject $\alpha = 0$. Conclusion: the portfolio's average return is fully explained by its market, size, and value exposures. There is no evidence of "skill" or mispricing beyond these factors.

1.5.2 Cross-Sectional Tests

A cross-sectional test asks: do assets with higher betas earn higher average returns? The standard procedure is the **Fama-MacBeth regression** (Fama and MacBeth, 1973):

Fama-MacBeth Two-Pass Regression

Pass 1 (time-series): For each asset i , regress its full return history on factors to estimate $\hat{\beta}_i$:

$$R_{i,t} - R_f = \alpha_i + \hat{\beta}_i f_t + \varepsilon_{i,t}, \quad t = 1, \dots, T$$

Pass 2 (cross-sectional): For each month t , regress all assets' returns on their estimated betas:

$$R_{i,t} - R_f = \gamma_{0,t} + \gamma_{1,t} \hat{\beta}_i + \eta_{i,t}, \quad i = 1, \dots, N$$

Inference: Average the monthly $\gamma_{1,t}$ estimates across time:

$$\hat{\lambda} = \frac{1}{T} \sum_{t=1}^T \gamma_{1,t}$$

and compute a t -statistic using the time-series standard error of $\{\gamma_{1,t}\}$. If $\hat{\lambda}$ is significantly positive, the factor carries a positive risk premium.

Why average across months? Because each monthly cross-sectional regression gives a noisy estimate of λ . By averaging across T months, you reduce noise by $1/\sqrt{T}$. The time-series variation in $\gamma_{1,t}$ naturally accounts for the fact that some months the factor works well and other months it does not. This is a key advantage of Fama-MacBeth over a single pooled regression.

Limitation: Fama-MacBeth standard errors correct for cross-sectional correlation (returns of different stocks in the same month are correlated) but do *not* correct for time-series autocorrelation. If the risk premium is persistent, standard errors may be understated. Newey-West or Shanken (1992) corrections are standard fixes.

Errors-in-Variables in Two-Pass Regressions

The second-pass regression uses *estimated* betas from the first pass, not true betas. Since $\hat{\beta}_i$ contains estimation error, the second-pass coefficients are biased (attenuation bias—the λ estimates are pushed toward zero). This is why Fama and MacBeth (1973) and most subsequent work form *portfolios* sorted on characteristics (e.g., 25 size-value portfolios) rather than using individual stocks: portfolio betas are estimated more precisely because idiosyncratic noise averages out.

1.5.3 Which Test Does Your Project Use?

Your project is primarily a **time-series prediction** problem: you observe risk-system features today and predict returns/volatility tomorrow. This is closer to time-series tests. However, because you work across asset classes (equities, rates, FX, credit), you also have a cross-sectional dimension: do assets with higher VaR utilization today earn different returns tomorrow? Chapter 8 covers how to combine both dimensions using panel regression and the fixed-effects estimators you will need.

Time-Series Prediction and the SDF

Your Phase 2 features (VaR dynamics, factor concentration, scenario P&L) are, in SDF language, proxies for the state variables that drive $\mathcal{M}$. Intermediary asset pricing theory (Chapter 2) argues that dealer balance-sheet constraints *are* the relevant state variables. When you test whether VaR utilization predicts returns, you are testing whether dealer risk capacity is a priced state variable—i.e., whether it belongs in equation (1.6). Frame it this way in your pitch and your sponsor will immediately see the intellectual foundation.

1.6 Connecting SDF Theory to Your Features

Let us make the link between abstract theory and your concrete project explicit. The SDF framework says expected returns are determined by covariance with the pricing kernel. Your project hypothesis is:

SecDB risk-system outputs (VaR, Greeks, scenario P&L) are observable proxies for the state variables driving $\mathcal{M}$, and therefore predict returns cross-sectionally and over time.

How does each feature family map to this framework?

Feature Family	SDF Interpretation	Theory Source
VaR utilization	Proxy for dealer risk capacity; when VaR usage is high, dealer SDF is high (risk-averse state)	He and Krishnamurthy (2013), Adrian and Shin (2010)
Factor concentration	Measures crowding; high Herfindahl $\Rightarrow$ concentrated risk $\Rightarrow$ fragile SDF	Coval and Stafford (2007)
Scenario P&L	Directly measures tail exposures—the states where SDF is highest	Cochrane (2005), Ch. 1
Cross-asset flow	Captures risk reallocation; component VaR shifts signal changing SDF loadings	He, Kelly, and Manela (2017)

This mapping is what transforms your project from “I ran some ML on risk data” into “I tested whether dealer balance-sheet state variables are priced,” which is a much stronger pitch. Chapter 2 develops the intermediary asset pricing theory in detail.

From Theory to Features: The Pipeline

The logic chain connecting this chapter to your code is:

1. SDF theory says expected returns are driven by covariance with a pricing kernel.
2. Intermediary theory (Chapter 2) says dealer constraints are a major component of that kernel.
3. SecDB risk outputs are daily, high-frequency measurements of those constraints.
4. You engineer features from SecDB data (Chapter 9) that proxy for the state variables in the intermediary SDF.
5. You test whether those features predict returns/volatility using proper validation (Chapter 11).
6. If they do, you have evidence that dealer risk capacity is a priced state variable—a

publishable, defensible result.
Every step after this chapter builds on this foundation.

1.7 Summary

1. The **stochastic discount factor** $\mathcal{M}$ is a single random variable that prices all assets: $\mathbb{E}[\mathcal{M} \cdot R_i] = 1$.
2. No-arbitrage is equivalent to the existence of a strictly positive SDF (the Fundamental Theorem).
3. The **risk premium** on any asset equals $-R_f \text{Cov}(\mathcal{M}, R_i)$: only covariance with the pricing kernel is compensated.
4. Volatility alone does not earn a risk premium. Idiosyncratic risk is not priced.
5. **Factor models** parametrize $\mathcal{M}$ as linear in observable factors. The CAPM uses one factor (market); Fama-French uses three (market, size, value); the APT leaves factors unspecified.
6. Fama-French (1993) three factors explain $> 90\%$ of diversified portfolio return variation, vs. $\sim 70\%$ for CAPM.
7. **Alpha** is the return unexplained by a factor model—always specify the benchmark.
8. **Beta** measures sensitivity to a factor; it is an OLS regression coefficient.
9. The **information ratio** $= \alpha/\sigma(\varepsilon)$ measures risk-adjusted active return. Top-quartile managers achieve $\text{IR} \approx 0.5$.
10. The **Fundamental Law**: $\text{IR} \approx \text{IC} \times \sqrt{BR}$. Small edge $\times$ many bets = good performance.
11. **Time-series tests** check if alpha is zero for a given factor model. **Cross-sectional tests** (Fama-MacBeth) check if betas explain return differences across assets.
12. Your project is primarily time-series prediction but with cross-asset panel structure.
13. The **Fama-MacBeth regression** is the standard tool for cross-sectional tests: estimate betas in time series, then regress returns on betas cross-sectionally each period.
14. Every feature you build is, in SDF language, a candidate state variable for the pricing kernel. Frame your pitch accordingly.

Key Results Reference

Paper	Key Claim	Headline Result
Sharpe (1964)	Expected return is linear in market beta	CAPM: $\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f)$
Black, Jensen, Scholes (1972)	CAPM relationship is “too flat”	Low-beta stocks earn more than predicted; high-beta stocks earn less
Ross (1976)	No-arbitrage implies linear factor pricing	APT: $\mathbb{E}[R_i] - R_f \approx \sum_k \beta_{i,k} \lambda_k$
Fama & French (1993)	Size and value factors explain cross-section	3-factor $R^2 > 90\%$ vs. CAPM $R^2 \approx 70\%$; $\text{SMB} \approx 3.2\%/yr$, $\text{HML} \approx 4.8\%/yr$
Harrison & Kreps (1979)	No-arbitrage $\Leftrightarrow$ positive SDF exists	Fundamental Theorem of Asset Pricing
Grinold (1989)	IR decomposes into skill $\times$ breadth	$\text{IR} \approx \text{IC} \times \sqrt{BR}$

Chapter 2

Intermediary Asset Pricing

This Is the Core Thesis of Your Project

Every feature you engineer in Phase 2 is a proxy for the intermediary constraints described in this chapter. He and Krishnamurthy (2013) predict that dealer capital constraints drive risk premia. Adrian, Etula, and Muir (2014) show that a single intermediary-leverage factor prices the cross-section of equities, momentum, and bonds with $R^2 = 77\%$. He, Kelly, and Manela (2017) extend this result across seven asset classes—equities, government and corporate bonds, sovereign bonds, options, CDS, commodities, and FX—using a single intermediary capital ratio. Your project tests this theory with proprietary daily data from SecDB: the data the academics wished they had.

2.1 Classical vs. Intermediary Asset Pricing

Representative Agent

Classical macro-finance models assume a single “representative household” whose preferences determine all asset prices. This is a modeling device: it does not mean everyone is identical, but rather that aggregate prices behave *as if* a single agent with average preferences were trading. The SDF from Chapter 1 is this agent’s marginal rate of substitution between consumption today and tomorrow.

Chapter 1 developed the SDF framework: $\mathbb{E}_t[\mathcal{M}_{t+1}R_{i,t+1}] = 1$, where the pricing kernel $\mathcal{M}$ is driven by the marginal investor’s state-dependent preferences. The classical approach assumes the marginal investor is a representative household, giving:

$$\mathcal{M}_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \quad (2.1)$$

- β — subjective discount factor (how impatient the household is; typically ≈ 0.99 quarterly).
- $u'(C_t)$ — marginal utility of consumption at time t . High when consumption is low (bad times).
- The ratio $u'(C_{t+1})/u'(C_t)$ — marginal rate of substitution. High when consumption falls (recession).

This is elegant, but it fails empirically. The problem is known as the **equity premium puzzle** (Mehra and Prescott, 1985).

Risk Aversion and Utility

A **utility function** $u(C)$ assigns a “happiness value” to consumption level C . Risk aversion means u is concave: the marginal utility of an extra dollar is higher when you are poor than when you are rich. The coefficient of **relative risk aversion** (RRA) $\gamma = -Cu''(C)/u'(C)$ measures curvature. Empirical estimates from labor supply and insurance data suggest $\gamma \approx 1\text{--}5$ for typical households. Higher γ means the agent cares more about consumption fluctuations.

The Equity Premium Puzzle

U.S. equities earned roughly 6–8% per year more than Treasury bills over 1926–2023 (Dimson, Marsh, and Staunton, 2023). But aggregate consumption growth is extremely smooth: its annual standard deviation is only about 1–2%. To generate a 6% equity premium from equation (2.1), you need the household’s risk aversion γ to exceed 50—an order of magnitude above plausible estimates. Alternatively, you need the correlation between consumption and stock returns to be much higher than it is in the data. Neither works. The household SDF simply does not move enough to explain observed risk premia.

The intermediary approach offers a resolution. Instead of asking “what does the representative household fear?” it asks: *who is actually the marginal investor in most markets?*

Marginal Investor

The **marginal investor** is the agent whose trading activity sets the price at the margin. In equity markets, this might plausibly be households (via mutual funds). But in derivatives, credit, FX, and fixed income, the marginal investor is almost always a **financial intermediary**—a broker-dealer, market maker, or bank. These institutions hold the inventory, provide liquidity, and absorb order-flow imbalances. If their constraints tighten, they pull back, and risk premia rise.

The Intermediary Asset Pricing Hypothesis

If financial intermediaries—not households—are the marginal investors in most asset markets, then the relevant SDF is the intermediary’s marginal value of wealth, not the household’s marginal utility of consumption. The intermediary SDF depends on the intermediary’s *capital constraints*: its equity, leverage, VaR limits, and regulatory requirements. When capital is scarce, the intermediary’s marginal value of a dollar is high, the SDF is high, and risk premia rise. This is the theoretical foundation for every feature you will build.

Dealers as the Fire Department

Think of dealers as a city’s fire department. In normal times, the department has ample capacity—trucks sit idle, firefighters train, response times are fast. When a major fire breaks out, the department is stretched. If a *second* fire starts while the first is raging, the city has a serious problem: the marginal cost of deploying another truck is enormous. In financial markets, “fires” are large sell-offs or liquidity demands. In normal times, dealers absorb these easily. But when their capital is depleted (the first fire), even a modest additional shock (the second fire) causes outsized price moves. VaR utilization near 100% is the financial equivalent of all trucks deployed. Your project measures exactly this.

The key difference between classical and intermediary pricing can be stated precisely. Classi-

cal models write $\mathcal{M}_{t+1} = g(C_{t+1}/C_t)$ for some function g of aggregate consumption growth. Intermediary models write:

$$\mathcal{M}_{t+1} = h(w_{t+1}) \quad (2.2)$$

where w_{t+1} is the intermediary's wealth or capital ratio, and h is a function that increases sharply as w falls toward the constraint boundary. Both satisfy $\mathbb{E}_t[\mathcal{M}_{t+1}R_{i,t+1}] = 1$ from Chapter 1. They differ in *whose* state variable drives $\mathcal{M}$.

2.2 He and Krishnamurthy (2013): The Theoretical Foundation

He and Krishnamurthy (2013, AER) provide the theoretical model that your project is built on. The model has two agents: households and intermediaries.

Balance Sheet Mechanics

A firm's balance sheet satisfies: Assets = Equity + Debt. **Equity** (or “capital”) is the firm's own money; **debt** is borrowed money. **Leverage** is the ratio Assets/Equity. A firm with \$100 in assets, \$10 in equity, and \$90 in debt has leverage of 10×. If asset values drop by 5% (to \$95), equity drops by 50% (to \$5)—leverage amplifies losses. This is why bank capital matters: small asset losses can wipe out equity when leverage is high.

2.2.1 Model Setup

Households receive labor income and can invest in a risk-free asset or in intermediary equity. They cannot directly hold risky assets—they must invest through intermediaries.

Intermediaries hold risky assets (equities, bonds, derivatives) funded by household deposits (debt) and their own equity capital. They face an **equity capital constraint**: their equity must remain positive, and a tighter version requires equity to exceed a fraction of assets.

The capital ratio—the intermediary's equity divided by total assets—is the state variable:

$$w_t = \frac{E_t}{A_t} \quad (2.3)$$

- w_t — the intermediary's capital ratio at time t .
- E_t — equity capital (the intermediary's “skin in the game”).
- A_t — total assets under management.

2.2.2 Two Regimes

The model generates two regimes, determined by how close w_t is to the constraint boundary.

When capital is ample (w_t far from the constraint): the intermediary is unconstrained. It can absorb shocks freely, and its SDF approximately equals the household SDF. Risk premia are moderate and stable.

When capital is scarce (w_t near the constraint boundary): the intermediary's marginal value of wealth spikes. It must either sell risky assets (pushing prices down) or raise costly equity. Risk premia rise sharply. The SDF becomes:

$$\mathcal{M}_{t+1}^{\text{int}} \approx \mathcal{M}_{t+1}^{\text{HH}} + \mu_t \cdot \mathbf{1}\{\text{constraint near binding}\} \quad (2.4)$$

where $\mu_t > 0$ is the shadow cost of the capital constraint (a Lagrange multiplier). The closer the intermediary is to the constraint, the larger μ_t , and the more the intermediary SDF exceeds the household SDF.

Lagrange Multipliers

In constrained optimization, a **Lagrange multiplier** μ measures the “shadow price” of a constraint: how much the objective improves if you relax the constraint by one unit. If $\mu = 0$, the constraint is slack (non-binding) and irrelevant. If $\mu > 0$, the constraint is active and costly. In He-Krishnamurthy, μ_t is the shadow cost of the capital constraint. When $\mu_t = 0$, the intermediary is unconstrained and prices assets like a household. When $\mu_t > 0$, the constraint binds and the intermediary demands higher risk premia to compensate for its constrained state.

The nonlinearity is critical. Risk premia are *not* a smooth, linear function of the capital ratio. Instead, they are nearly flat when capital is abundant and then rise steeply—almost like a hockey stick—as the constraint boundary approaches. This matches the empirical observation that risk premia are roughly stable during normal periods but spike violently during crises.

Worked Example:

The Capital Constraint in Action Consider a simplified version of the He-Krishnamurthy setup. A dealer has \$10 billion in equity and \$90 billion in debt, for \$100 billion in total assets and a capital ratio of $w = 10\%$. Suppose the VaR limit is set so that the dealer must maintain $w \geq 5\%$.

Normal times: Assets drop 2% (\$2B). Equity falls from \$10B to \$8B. Capital ratio: $8/98 = 8.2\%$ —still well above the 5% floor. The dealer can absorb the loss and continue trading. Risk premia change little.

After a prior crisis: Equity has already fallen to \$6B (capital ratio = $6/96 = 6.25\%$, near the floor). The same 2% asset decline now pushes equity to \$4.08B and the capital ratio to $4.08/94.08 = 4.3\%$ —below the constraint. The dealer *must* sell \$20B+ in risky assets to restore the ratio, flooding the market with supply. These forced sales push prices down further, creating a feedback loop. Risk premia spike.

Key insight: The same shock (2% asset decline) produces a radically different risk-premium response depending on the starting capital ratio. This is the nonlinearity that He-Krishnamurthy’s model captures, and it is what your VaR utilization feature is designed to measure.

He and Krishnamurthy (2013)

The model produces three predictions testable with SecDB data:

1. Risk premia are a **nonlinear** function of the intermediary capital ratio: low and stable when capital is ample, sharply elevated when capital is scarce.
2. The speed of **reversion** from crisis to normal risk premia depends on the rate at which intermediaries rebuild capital (retained earnings, new equity issuance).
3. **Policy implications:** injecting equity capital into intermediaries is more effective than reducing borrowing costs or purchasing distressed assets, because it directly relaxes the binding constraint.

The calibrated model matches both the nonlinearity of risk premia during the 2007–2009 crisis and the speed of reversion back to normal levels.

VaR Utilization as the Capital Constraint Proxy

He and Krishnamurthy’s capital ratio w_t is exactly what your VaR utilization feature measures. VaR utilization = current VaR / VaR limit. When utilization approaches 100%, the desk is at its constraint boundary—the model predicts risk premia should spike. You

observe this daily from SecDB; the academics could only calibrate to aggregate quarterly data.

2.3 Adrian, Etula, and Muir (2014): Empirical Evidence

He and Krishnamurthy (2013) is a theoretical model. Adrian, Etula, and Muir (2014, JF) take it to the data with a remarkable result.

Cross-Sectional Pricing Test

A cross-sectional pricing test asks: can a proposed factor explain *why* different assets earn different average returns? You sort assets into portfolios, compute each portfolio's exposure (beta) to the factor, and check whether higher-beta portfolios earn higher average returns. The cross-sectional R^2 measures how much of the variation in average returns across portfolios is explained by the factor. Chapter 1 covered the Fama-MacBeth procedure for this.

2.3.1 The Leverage Factor

Adrian, Etula, and Muir construct a single factor: the **growth rate of broker-dealer leverage**. Their leverage measure comes from the Federal Reserve's Flow of Funds (Z.1) data:

$$\text{Leverage}_t = \frac{\text{Total financial assets}_t}{\text{Total financial assets}_t - \text{Total liabilities}_t} \quad (2.5)$$

The numerator is total assets; the denominator is equity (assets minus liabilities). The factor is the innovation (unexpected change) in leverage growth, estimated quarterly from 1968Q1 to 2009Q4.

The logic: when funding conditions deteriorate, broker-dealers deleverage (sell risky assets, shrink balance sheets). This deleveraging *is* the realization of the intermediary SDF being high. Assets that lose value precisely when dealers are deleveraging—i.e., assets with positive beta to leverage growth—are risky and should earn a premium.

2.3.2 Headline Result

Adrian, Etula, and Muir (2014)

A single factor—broker-dealer leverage growth—prices 25 size and book-to-market equity portfolios, 10 momentum portfolios, and 6 Treasury bond portfolios (41 test assets total) with a cross-sectional R^2 of 77% and an average annual pricing error of approximately 1%. For comparison on the same test assets: the CAPM achieves $R^2 \approx 24\%$, and the Fama-French three-factor model achieves $R^2 \approx 73\%$. In the pre-crisis subsample (1968–2005), the intermediary factor still achieves $R^2 = 67\%$ with pricing errors of 37 basis points per year.

Model	Factors	Cross-sectional R^2	Avg. pricing error
CAPM	1 (market)	24%	—
Fama-French 3	3 (MKT, SMB, HML)	73%	—
Intermediary (AEM)	1 (leverage growth)	77%	~1%/yr

Sample: 1968Q1–2009Q4, quarterly. Data source: Fed Flow of Funds (Z.1), security broker-dealer sector.

This is a striking result. One factor derived from broker-dealer balance sheets outperforms a three-factor model that was specifically *designed* to price these portfolios. It strongly suggests that intermediary constraints, not household consumption, are what drives the cross-section of risk premia.

Data Limitations of the Leverage Factor

The Fed Z.1 data is quarterly with a roughly two-month publication lag. This means the leverage factor is measured at a very low frequency and with stale information. For a trading signal, quarterly data with a two-month lag is nearly useless—by the time you observe the leverage change, markets have already moved. This is precisely the bottleneck your project solves: SecDB gives you the *same economic information* (dealer balance-sheet constraints) at daily frequency with no publication lag.

2.4 He, Kelly, and Manela (2017): Cross-Asset Evidence

Adrian, Etula, and Muir (2014) tested their factor on equities and Treasuries. He, Kelly, and Manela (2017, JFE) ask the harder question: does a single intermediary factor price *all* asset classes?

Primary Dealers

Primary dealers are the roughly 20 financial institutions authorized to trade directly with the Federal Reserve Bank of New York in open-market operations. They include firms like Goldman Sachs, J.P. Morgan, and Barclays. They are the backbone of U.S. fixed-income markets, intermediating trillions of dollars in Treasury, agency, and corporate bonds. Because they span multiple asset classes, their aggregate capital position reflects economy-wide intermediation capacity.

2.4.1 The Capital Ratio Factor

He, Kelly, and Manela construct an alternative to Adrian, Etula, and Muir’s leverage measure. Instead of aggregate Flow of Funds leverage, they use the **equity capital ratio** of primary dealer holding companies:

$$\text{Capital ratio}_t = \frac{\text{Market equity}_t}{\text{Market equity}_t + \text{Book debt}_t} \quad (2.6)$$

- Market equity is the stock-market capitalization of the primary dealer holding companies (from CRSP/Compustat).
- Book debt is total liabilities from financial statements.
- The factor is the *innovation* (shock) to this ratio, estimated as the residual from an AR(1) model.
- Sample: 1970Q1–2012Q4, quarterly.

AR(1) Models and Innovations

An **AR(1)** (first-order autoregressive) model predicts a variable from its own lagged value: $x_t = \phi x_{t-1} + \varepsilon_t$, where ϕ is the persistence parameter and ε_t is the residual (“innovation” or “shock”). The innovation $\varepsilon_t = x_t - \phi x_{t-1}$ is the *unexpected* component of x_t —the part not predicted by the previous value. In asset pricing, innovations matter more than levels because markets anticipate persistent trends. A capital ratio of 12% is not informative on its own; a *surprise drop* from the expected 12% to 10% is what moves risk premia.

Procyclical vs. Countercyclical

A variable is **procyclical** if it rises during economic expansions and falls during contractions. GDP, employment, and stock prices are procyclical. A variable is **countercyclical** if it moves opposite to the cycle: unemployment, credit spreads, and the VIX are countercyclical. In intermediary asset pricing, the capital ratio is procyclical (rises in booms), while leverage = 1/capital ratio is countercyclical (rises in crises). The He-Krishnamurthy model predicts this: dealers expand their balance sheets in good times and are forced to contract in bad times.

The capital ratio is procyclical: it rises in booms (equity values increase) and falls in crises (equity values collapse). This implies that intermediary *leverage* is countercyclical—exactly as the He-Krishnamurthy model predicts.

2.4.2 Seven Asset Classes

He, Kelly, and Manela (2017)

The capital ratio of primary dealers prices seven asset classes simultaneously:

1. U.S. equities (size, value, momentum portfolios)
2. U.S. government bonds (maturity-sorted)
3. U.S. corporate bonds (rating-sorted)
4. Foreign sovereign bonds
5. Options (volatility-sorted)
6. Credit default swaps (CDS)
7. Commodities and foreign exchange (FX)

The capital risk premium—the compensation for bearing intermediary capital risk—ranges from 7% to 11% per year across five of the seven asset classes, with a pooled estimate of approximately 9% per year. The price of risk is consistently positive and of similar magnitude when estimated separately for each asset class. Results are qualitatively stable in the pre-crisis subsample (1970Q1–2006Q4) and at monthly frequency.

The fact that a *single* intermediary factor works across equities, bonds, credit, FX, commodities, options, and sovereign debt is powerful evidence that intermediaries are the marginal investors in all these markets. This directly motivates your project’s cross-asset panel design (Chapter 8).

Cross-Asset Pricing Implies Cross-Asset Prediction

If one intermediary factor prices seven asset classes simultaneously, then a shock to intermediary capital should predict returns across *all* those asset classes—not just the one where the shock originated. Concretely: if the rates desk hits its VaR limit and is forced to sell Treasuries, the resulting capital depletion should also affect expected returns in credit, FX, and equities—because the same firm’s capital backs all of its trading desks. This cross-asset spillover is what your “cross-asset component VaR flow” feature (Chapter 9) is designed to capture: you measure how VaR contributions shift across asset classes within the firm, and test whether those shifts predict returns in the receiving asset class.

2.4.3 AEM vs. HKM: Two Intermediary Measures

Adrian, Etula, and Muir (AEM) and He, Kelly, and Manela (HKM) measure intermediary health differently:

	AEM (2014)	HKM (2017)
Measure	Book leverage of broker-dealer sector	Market equity capital ratio of primary dealers
Data source	Fed Z.1 Flow of Funds	CRSP/Compustat + Datas-tream
Level/growth	Leverage <i>growth</i> innovations	Capital ratio <i>level</i> innovations
Cyclical	Procyclical leverage (rises in booms)	Procyclical capital ratio (rises in booms)
Asset classes	Equities + Treasuries (41 portfolios)	7 asset classes
Frequency	Quarterly	Quarterly (monthly robust-ness)

The two measures are conceptually related—both capture intermediary balance-sheet health—but they move differently in the data and can even give contradictory signals in some periods. He, Kelly, and Manela (2017) provide evidence that their capital ratio dominates AEM’s leverage growth in joint cross-sectional tests across the broader set of asset classes.

The AEM vs. HKM Debate Is Unresolved

The academic literature has not settled whether leverage growth (AEM) or the capital ratio level (HKM) is the “correct” intermediary state variable. The two can give different signs: AEM leverage is procyclical (leverage rises in booms), while HKM’s capital ratio is also procyclical (it rises in booms)—but the innovations (shocks) move in opposite directions for some sample periods. For your project, this debate matters less than the shared conclusion: *some* measure of dealer balance-sheet health is a priced state variable. SecDB gives you multiple such measures simultaneously (VaR utilization, component VaR, factor concentration), so you can test which one carries the most information in your sample without taking sides in the AEM-vs.-HKM debate.

2.5 Adrian and Shin (2010): Repos and VIX

While He-Krishnamurthy and AEM/HKM focus on the cross-section of expected returns, Adrian and Shin (2010, JFI) provide direct evidence on the *time-series* forecasting power of dealer balance sheets.

Repurchase Agreements (Repos)

A **repo** (repurchase agreement) is a short-term collateralized loan. A dealer sells a security (e.g., a Treasury bond) to a counterparty and agrees to buy it back the next day (or within a few days) at a slightly higher price. The price difference is the interest rate. Repos are the primary funding mechanism for dealer inventories: a dealer with \$100 billion in bond inventory might fund \$95 billion of it through overnight repos. Changes in total repo outstanding directly reflect changes in dealer balance-sheet size—when repos grow, dealers are expanding; when repos shrink, dealers are deleveraging.

The VIX Index

The **VIX** (Chicago Board Options Exchange Volatility Index) measures the market’s expectation of 30-day S&P 500 volatility, implied from option prices. It is often called the “fear index.” A VIX of 15 means the market expects annualized volatility of 15%. The

long-run average is roughly 19–20. During crises, VIX spikes: it reached 80 in October 2008 and 66 in March 2020. A **VIX innovation** is the unexpected change in VIX—the component not predicted by its own past values.

Adrian and Shin study whether changes in dealer repo positions forecast changes in market risk. Their logic: when dealers expand their balance sheets via repos, they are absorbing more risk. When they contract, they are pulling back. The aggregate repo position is a real-time measure of dealer risk appetite.

Adrian and Shin (2010)

Growth in broker-dealer repo positions forecasts VIX innovations. Using weekly data from January 1990 to August 2007, a regression of VIX changes on lagged repo growth yields an R^2 of 11.6%, compared to 8.9% using only the lagged VIX level as a predictor. The incremental R^2 from repo growth—2.7 percentage points—may seem small, but at weekly frequency for a financial variable, this is economically meaningful. The result is stronger for the **volatility risk premium** (the gap between implied and realized volatility), which has a direct interpretation as the price of risk.

Predictor	R^2 (VIX changes)	Interpretation
Lagged VIX level only	8.9%	Mean reversion in VIX
+ Repo growth	11.6%	Dealer balance sheets add info
Incremental R^2	+2.7 pp	Dealer risk appetite signal

Data: weekly, January 1990–August 2007, U.S. primary dealers.

Why Repos Track Dealer Risk Appetite

Imagine you run a bond desk. To hold \$1 billion in Treasuries, you need funding. You repo out the bonds overnight—lending them as collateral and receiving \$990 million in cash. Tomorrow, you buy them back for \$990.01 million (the overnight interest). If you want to hold *more* bonds, you do more repos. If you want to reduce exposure, you unwind repos. At the aggregate level, total repos outstanding is therefore a near-real-time measure of how much balance sheet the dealer sector is deploying. When repos grow, dealers are leaning into risk. When repos shrink, dealers are pulling back. Adrian and Shin’s result says that this risk-appetite measure forecasts future volatility—because when dealers are aggressively expanding, they are absorbing risks that may later unwind.

This result is directly relevant to your project for two reasons:

1. It establishes that **VIX innovations** are the cleanest target variable for testing intermediary signals. The VIX represents the market price of volatility risk, and dealer balance sheets predict it. Your Phase 2 feature testing will use VIX innovations as one of four targets (Section 2.9).
2. It provides a **benchmark** for your features. Adrian and Shin achieve 2.7pp of incremental R^2 with quarterly-lagged, aggregate repo data. With daily, granular VaR and scenario data from SecDB, your features should contain at least as much information—and likely more. If your features cannot beat 2.7pp incremental R^2 for VIX forecasting, something is wrong with your feature engineering, not with the theory.

2.6 Adrian and Brunnermeier (2016): CoVaR

Value at Risk (VaR)

Value at Risk at confidence level α is the loss threshold such that the probability of exceeding it is α . Formally, $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\}$ for a return random variable X . In plain language: the 99% daily VaR of \$10 million means “we expect to lose more than \$10 million on only 1% of trading days.” Chapter 3 covers VaR computation in detail; here you only need the concept.

Adrian and Brunnermeier (2016, AER) introduce **CoVaR**—the VaR of the financial system *conditional on* a particular institution being in distress. This shifts VaR from a firm-level risk measure to a *systemic* risk measure.

CoVaR and ΔCoVaR

The **CoVaR** of the financial system conditional on institution j is:

$$P\left(R_{\text{system}} \leq \text{CoVaR}_q^{\text{system}|j} \mid R_j = \text{VaR}_q^j\right) = q \quad (2.7)$$

where R_{system} is the return on the financial system and R_j is institution j 's return.

- $\text{CoVaR}_q^{\text{system}|j}$ — the q -quantile loss of the system, given that institution j is at its own VaR (i.e., in distress).
- q — the tail quantile (typically $q = 0.05$ or 0.01).

The **ΔCoVaR** of institution j is the difference between the system's CoVaR when j is in distress versus when j is at its median state:

$$\Delta\text{CoVaR}_q^j = \text{CoVaR}_q^{\text{system}|j \text{ at VaR}} - \text{CoVaR}_q^{\text{system}|j \text{ at median}} \quad (2.8)$$

ΔCoVaR measures institution j 's *contribution* to systemic risk: how much worse the system's tail risk becomes when j is in trouble.

2.6.1 Key Findings

Adrian and Brunnermeier (2016)

Four institution-level characteristics significantly predict ΔCoVaR :

1. **Leverage** — higher leverage $\Rightarrow$ larger systemic risk contribution.
2. **Size** — larger institutions contribute more to systemic risk.
3. **Maturity mismatch** — funding long-term assets with short-term debt increases fragility.
4. **Asset price booms** — rapid growth in asset values signals building vulnerability.

Critically, ΔCoVaR is **countercyclical**: it builds up during booms (when measured VaR is *low*) and realizes during crises. A forward-looking version of ΔCoVaR measured in 2006Q4 would have predicted more than one-third of the realized systemic losses during the 2007–2009 financial crisis.

2.6.2 Why CoVaR Matters for Your Features

CoVaR and Your VaR Features

Your SecDB VaR features capture a version of the CoVaR logic at the desk level:

- **Firm-level delta VaR** (how the desk's trades change total firm VaR) is analogous to ΔCoVaR —it measures one desk's contribution to the firm's aggregate tail risk.
- **VaR utilization** (current VaR / VaR limit) captures the leverage-like constraint channel that Adrian and Brunnermeier identify as the strongest predictor.
- The countercyclical property of ΔCoVaR matches the He-Krishnamurthy prediction: risk builds when constraints are slack and realizes when they bind.

You will not compute CoVaR directly (it requires cross-institution data), but your features capture the same underlying economics at higher frequency and finer granularity.

2.7 Coval and Stafford (2007): Fire Sales and Crowding

The intermediary theories above focus on aggregate dealer capital. Coval and Stafford (2007, JFE) provide the microeconomic mechanism for *how* constrained intermediaries affect prices: forced selling.

Mutual Fund Flows

When investors withdraw money from a mutual fund (“outflows”), the fund must sell holdings to raise cash. When many investors withdraw simultaneously, the fund is a **forced seller**—it must sell regardless of price. This is distinct from an informed seller who sells because they believe the asset is overvalued. Forced selling creates temporary price pressure that reverses once the selling stops.

Coval and Stafford study mutual funds experiencing extreme outflows. When funds are forced to liquidate positions, they sell stocks mechanically. The key finding:

Coval and Stafford (2007)

Stocks heavily sold by distressed mutual funds (those experiencing outflows in the bottom 5% of the distribution) experience significant price declines that subsequently reverse. The fire-sale discount is approximately 10–15% over the selling period, with roughly half reversing within 6–12 months. The price impact is concentrated in stocks where the distressed fund holds a large share of daily volume—i.e., where the forced selling overwhelms the market's absorptive capacity.

The connection to intermediary asset pricing is direct: forced selling is the mechanism through which capital constraints transmit to prices. When a dealer hits its VaR limit, it must reduce risk—often by selling positions. If many dealers are constrained simultaneously (as in 2008), the forced selling is correlated across firms, amplifying the price impact.

Factor Concentration and Fire-Sale Risk

Your **factor concentration** feature (the Herfindahl index of factor-VaR contributions) captures fire-sale vulnerability directly. When factor concentration is high—meaning most of the desk's VaR comes from a few correlated positions—a constraint-driven forced sale will hit the same positions and the same markets. Low concentration means the desk can reduce risk by trimming many small, diverse positions with minimal market impact. High Herfindahl $\Rightarrow$ high fire-sale risk $\Rightarrow$ higher expected risk premia. This is a testable

prediction from Coval-Stafford combined with He-Krishnamurthy.

2.8 The Data Bottleneck—Why Your Project Has an Edge

Every paper in this chapter shares the same limitation: they use public, low-frequency, backward-looking data to measure intermediary constraints.

Paper	Data Source	Freq.	Lag	Granularity
He-Krishnamurthy (2013)	Calibration only	—	—	Aggregate
Adrian-Etula-Muir (2014)	Fed Z.1	Quarterly	~2 months	Sector aggregate
He-Kelly-Manela (2017)	CRSP/Compustat	Quarterly	~1 month	Firm (holding co.)
Adrian-Shin (2010)	Fed Z.1	Weekly	~1 week	Sector aggregate
Adrian-Brunnermeier (2016)	CRSP, Call Reports	Quarterly	~2 months	Firm
Your project	SecDB	Daily	T+0	Desk / position

The advantage is threefold:

1. **Frequency:** Daily vs. quarterly gives you roughly $60\times$ more observations per year. For ML methods that need data, this matters.
2. **Timeliness:** You observe today's risk state today. The academics observe last quarter's risk state two months later. Any predictive signal in intermediary constraints has likely dissipated by the time public data is released.
3. **Granularity:** You observe risk decomposed by desk, factor, scenario, and position. The academics observe a single aggregate number for the entire broker-dealer sector. Disaggregation lets you identify *which* constraints matter and *where* the pressure is building.

Better Data Does Not Guarantee Better Results

Three caveats temper the data advantage:

1. **Single firm:** You observe one dealer (Goldman Sachs), not the entire system. If the theory requires *aggregate* intermediary constraints, one firm's data may be a noisy proxy.
2. **Short history:** SecDB data may span only 5–10 years of daily observations, versus the 40+ year samples in the academic literature. Short samples increase overfitting risk and reduce the power of statistical tests. This is why the Deflated Sharpe Ratio (Chapter 11) is non-negotiable.
3. **Endogeneity:** The same firm whose risk data you observe is also a market participant. Its risk-taking affects prices, and prices affect its risk measures. Disentangling cause and effect is harder with internal data than with external data.

Acknowledge these limitations in your pitch (Phase 0) and your final report (Phase 5). The correct framing is: “better data that *should* contain more signal, pending rigorous validation”—not “we have the best data so the signal is guaranteed.”

2.9 From Theory to Features: The Mapping

Let us make the connection between the theory in this chapter and the features you will build in Phase 2 (Chapter 9) fully explicit.

Theory cept	Con-	Academic Proxy	Your Feature	SecDB	Prediction
Capital constraint (He-Krishnamurthy)	con- (He-	Aggregate capital ratio	VaR (%)	utilization	High utilization $\Rightarrow$ higher risk premia
Balance-sheet expansion (Adrian-Shin)		Repo growth	Delta VaR, com- ponent VaR		Rising VaR $\Rightarrow$ dealers absorbing risk $\Rightarrow$ future volatility
Leverage factor (AEM)	factor	Sector leverage growth	VaR change	rate-of-	Rapid VaR changes $\Rightarrow$ leverage adjustment
Cross-asset pric- ing (HKM)	pricing	Capital ratio across 7 classes	Cross-asset com- ponent VaR shifts		Risk reallocation be- tween desks $\Rightarrow$ return predictability
Crowding / fire sales (Coval- Stafford)	/ fire (Coval-	Mutual fund flow pressure	Factor concentra- tion (HHI)		High Herfindahl $\Rightarrow$ crowded risk $\Rightarrow$ frag- ile
Systemic risk (Adrian- Brunnermeier)	(Adrian-	Δ CoVaR	Firm-level delta VaR		Desk contribution to firm risk
Tail risk (He- Krishnamurthy)	(He-	Crisis vs. normal calibration	Scenario P&L dis- persion		Wide scenario spread $\Rightarrow$ tail uncertainty

Each row is a testable hypothesis. In Phase 2, you will test each one against four targets: VIX innovations, asset-class drawdowns, realized volatility, and cross-asset momentum reversals. The theory in this chapter tells you *why* each feature should predict—without that “why,” you are data-mining.

Theory-Motivated Features vs. Data-Mining

The most important methodological principle in this project is: **every feature must be motivated by theory before you look at the data**. This chapter provides that motivation. When you present results in Phase 5, the narrative is not “my ML model found a pattern.” The narrative is: “Intermediary asset pricing theory (He-Krishnamurthy 2013, AEM 2014, HKM 2017) predicts that dealer constraints drive risk premia. I tested this prediction using high-frequency proprietary data and found [result].” The first framing invites the response “that is probably overfit.” The second invites “tell me more.”

2.10 Summary

1. The **equity premium puzzle** shows that household consumption is too smooth to explain observed risk premia.
2. **Intermediary asset pricing** resolves this by making the financial intermediary—not the household—the marginal investor whose constraints drive the SDF.
3. **He and Krishnamurthy (2013)** provide the theoretical model: the intermediary’s capital ratio is the state variable; risk premia spike when capital is scarce and the equity constraint binds.
4. **Adrian, Etula, and Muir (2014)** show empirically that a single broker-dealer leverage factor prices equities, momentum, and bonds with $R^2 = 77\%$, outperforming the CAPM (24%) and matching Fama-French (73%). Sample: 1968Q1–2009Q4, quarterly.

5. **He, Kelly, and Manela (2017)** extend the result across seven asset classes (equities, government and corporate bonds, sovereign bonds, options, CDS, commodities, FX) using the primary-dealer capital ratio. The capital risk premium is approximately 9% per year. Sample: 1970Q1–2012Q4.
6. **Adrian and Shin (2010)** demonstrate that repo growth forecasts VIX innovations (R^2 rises from 8.9% to 11.6% when repos are included). This motivates VIX innovations as your cleanest target variable.
7. **Adrian and Brunnermeier (2016)** define CoVaR—systemic VaR—and show that leverage, size, and maturity mismatch predict systemic risk contributions. A forward-looking measure from 2006Q4 would have predicted more than one-third of realized crisis losses.
8. All academic work uses **quarterly, lagged, aggregate** data. Your project uses **daily, real-time, desk-level** SecDB data—a frequency and granularity advantage of roughly 60×.
9. **Coval and Stafford (2007)** show that forced selling by constrained funds creates 10–15% fire-sale discounts that partially reverse. Factor concentration (Herfindahl) measures vulnerability to this mechanism.
10. **Better data alone is not enough.** Single-firm data, short history, and endogeneity are real limitations.
11. Every Phase 2 feature maps to a specific theoretical prediction from this chapter. Theory-motivated features are your defense against overfitting and your strongest pitch to the desk.
12. **Publication decay caveat:** McLean and Pontiff (2016) show that published anomalies lose 30–50% of their magnitude post-publication. The intermediary factors here may have partially decayed. Your project’s edge is not the theory (which is public) but the proprietary data that measures it at higher resolution.

Key Results Reference

Paper	Key Claim	Headline Result
He & Krishnamurthy (2013, AER)	Intermediary capital ratio drives risk premia	Theoretical model: risk premia nonlinear in capital ratio; spike when constraint binds; calibrated to crisis dynamics
Adrian, Etula & Muir (2014, JF)	Leverage growth is a priced SDF factor	Single factor: cross-sectional $R^2 = 77\%$ on 41 portfolios; avg. pricing error $\sim 1\%/yr$; CAPM $R^2 = 24\%$, FF3 $R^2 = 73\%$
He, Kelly & Manela (2017, JFE)	One factor prices all asset classes	Capital ratio prices 7 asset classes; capital risk premium $\approx 9\%/yr$; positive price of risk in each class separately
Adrian & Shin (2010, JFI)	Repos forecast VIX	Repo growth raises VIX forecasting R^2 from 8.9% to 11.6% (weekly, 1990–2007)
Adrian & Brunnermeier (2016, AER)	CoVaR measures systemic risk	Leverage, size, maturity mismatch predict ΔCoVaR ; 2006Q4 measure predicts $> \frac{1}{3}$ of 2007–09 realized losses
Coval & Stafford (2007, JFE)	Forced selling creates fire-sale discounts	10–15% discount on stocks sold by distressed funds; $\sim 50\%$ reversal within 6–12 months

Chapter 3

Risk Systems and Value-at-Risk

Project Connection — Phase 0 & Phase 2

This chapter teaches you what every SecDB risk output actually measures. In Phase 0 you will audit which risk cube nodes are accessible; this chapter ensures you understand the data before engineering features. In Phase 2, every feature family—VaR dynamics, factor concentration, scenario P&L, VaR utilization—maps directly to concepts here. When you build features in Chapter 9, you will reference the mathematics and intuitions developed below.

3.1 What VaR Measures

Quantiles and the Normal Distribution

A **quantile** q_α of a random variable X is the value below which a fraction α of the probability mass lies: $\Pr(X \leq q_\alpha) = \alpha$. For a normal distribution $\mathcal{N}(\mu, \sigma^2)$, the α -quantile is $q_\alpha = \mu + \sigma z_\alpha$, where z_α is the standard normal quantile. Key values: $z_{0.95} = 1.645$, $z_{0.99} = 2.326$, $z_{0.975} = 1.960$.

Value-at-Risk is the single most important number produced by a bank’s risk system. Every trading desk has a VaR limit, every risk report leads with VaR, and every regulator demands it. The concept is simple: how much could we lose on a bad day?

Value-at-Risk

Let L denote the portfolio loss over a holding period h (typically 1 day or 10 days). The **Value-at-Risk** at confidence level α is the smallest loss threshold ℓ such that the probability of exceeding ℓ is at most $1 - \alpha$:

$$\text{VaR}_\alpha(L) = \inf\{ \ell \in \mathbb{R} : \Pr(L > \ell) \leq 1 - \alpha \} \quad (3.1)$$

- L — portfolio loss (positive = losing money).
- α — confidence level (e.g., 0.99 means the 99th percentile).
- h — holding period; always state it (“99% 1-day VaR”).
- $1 - \alpha$ — the tail probability you are willing to tolerate.

Equivalently, VaR_α is the α -quantile of the loss distribution.

What VaR Tells You

VaR answers: “What is the most I could lose on a normal bad day?” The 99% 1-day VaR says: “On 99 out of 100 trading days, my loss will be less than this number.” It does *not* say anything about what happens on that 1-in-100 day—you could lose \$1 more than VaR or \$1 billion more. VaR is a fence, not a cliff.

Worked Example:

Normal-Distribution VaR Suppose your portfolio’s daily P&L is normally distributed with mean $\mu = \$50,000$ and standard deviation $\sigma = \$2,000,000$.

The 99% 1-day VaR is:

$$\begin{aligned}\text{VaR}_{0.99} &= -(\mu - z_{0.99} \sigma) \\ &= -(50,000 - 2.326 \times 2,000,000) \\ &= -50,000 + 4,652,000 \\ &= \$4,602,000\end{aligned}\tag{3.2}$$

- We subtract μ because VaR is a *loss* (sign convention: loss positive).
- The $z_{0.99} = 2.326$ factor comes from the standard normal inverse CDF.
- Interpretation: on 99% of days, the portfolio loses less than \$4.6 million.

The 95% 1-day VaR uses $z_{0.95} = 1.645$:

$$\text{VaR}_{0.95} = -(50,000 - 1.645 \times 2,000,000) = \$3,240,000$$

Lower confidence $\Rightarrow$ lower VaR. This should be intuitive: a less extreme “bad day” threshold is easier to breach.

Loss Conventions

Finance has conflicting sign conventions. In risk management, **loss is positive**: if your portfolio drops \$5M, $L = +5,000,000$. In portfolio theory, **returns are positive** when you make money. VaR is reported as a positive number. Always check the convention when reading code or papers.

3.2 Expected Shortfall: VaR’s Better Sibling

VaR has a critical deficiency: it ignores what happens *beyond* the threshold. Two portfolios can have identical VaR but wildly different tail losses. Expected Shortfall fixes this.

Expected Shortfall

The **Expected Shortfall** (also called CVaR or Conditional VaR) at confidence level α is the expected loss, conditional on the loss exceeding VaR:

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]\tag{3.3}$$

For continuous distributions, equivalently:

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L) du\tag{3.4}$$

- ES_α averages *all* quantiles beyond VaR_α —it captures tail severity.

- For the normal distribution: $ES_\alpha = \mu + \sigma \frac{\phi(z_\alpha)}{1-\alpha}$, where ϕ is the standard normal PDF.
- $ES_\alpha \geq VaR_\alpha$ always (it looks deeper into the tail).

Why Expected Shortfall Matters

VaR fails a property called **subadditivity**: for some portfolios, $VaR(A+B) > VaR(A) + VaR(B)$. This means VaR can penalize diversification—combining two portfolios can *increase* measured risk, which is economically absurd.

ES is **subadditive**: $ES(A+B) \leq ES(A) + ES(B)$ always. This makes it a *coherent* risk measure in the sense of Artzner et al. (1999). The Basel Committee’s FRTB framework (Basel Committee, 2019) replaced VaR with ES at the 97.5% confidence level for internal-model capital calculations precisely for this reason.

Worked Example:

VaR Subadditivity Failure Consider two independent bonds, each with a 4% probability of defaulting and losing \$100, and a 96% probability of no loss.

Individual 95% VaR: For each bond, the 95th percentile loss is \$0 (since 96% of outcomes have zero loss, and $96\% > 95\%$). So $VaR_{0.95}(\text{Bond}_1) = VaR_{0.95}(\text{Bond}_2) = \0 .

Portfolio 95% VaR: Holding both bonds (50/50), the probability of at least one default is $1 - (0.96)^2 = 7.84\%$. Since $7.84\% > 5\%$, the 95th percentile loss is now positive: $VaR_{0.95}(\text{Portfolio}) > \0 .

Diversification *increased* measured VaR—a clear failure of subadditivity. ES would not exhibit this pathology.

Basel FRTB: ES Replaces VaR (Basel Committee, 2019)

The Fundamental Review of the Trading Book replaced 99% 10-day VaR with 97.5% Expected Shortfall for internal-model capital. The 97.5% confidence level was chosen to provide “a broadly similar level of risk capture as the existing 99th percentile VaR threshold” while yielding more stable model output and less sensitivity to extreme outliers. The Committee estimated this would increase market risk capital requirements by approximately 22% on average.

3.3 VaR Computation Methods

There are three standard approaches to computing VaR. Each makes different trade-offs between accuracy, speed, and assumptions. SecDB uses variants of all three depending on the desk and asset class.

Covariance Matrix

For a portfolio with n assets, the **covariance matrix** $\Sigma \in \mathbb{R}^{n \times n}$ has entries $\Sigma_{ij} = \text{Cov}(r_i, r_j)$ where r_i is the return on asset i . The portfolio variance is $\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$, where $\mathbf{w}$ is the vector of portfolio weights. Estimating Σ reliably is hard when n is large (Chapter 5 discusses shrinkage estimators).

3.3.1 Historical Simulation

The simplest approach: replay actual historical returns through the current portfolio.

Algorithm:

1. Collect the last T days of returns for every position in the portfolio (typically $T = 250$ or $T = 500$).
2. Revalue the current portfolio under each historical day's returns, producing T hypothetical P&L outcomes.
3. Sort the P&L from worst to best.
4. The VaR at confidence α is the $(1 - \alpha) \times T$ -th worst outcome (e.g., for 99% VaR with $T = 500$, take the 5th-worst day).

Advantages: No distributional assumptions; captures fat tails, skewness, and nonlinear positions naturally; easy to explain.

Disadvantages: Requires history; assumes the past is representative of the future; slow to adapt to regime changes; suffers from the cliff effect (below).

The Historical Simulation Cliff Effect

If the lookback window is 500 days, the 99% VaR is literally the 5th-worst day. Add or remove one extreme day from the window, and VaR can jump discontinuously. This creates **spurious features**: VaR changes driven by old observations rolling out of the window rather than by genuine changes in portfolio risk.

Adrian and Shin (2014) document that intermediary leverage is “negatively aligned with the banks’ Value-at-Risk”—VaR constrains leverage in real time. If the VaR number itself is noisy due to cliff effects, features derived from VaR rate-of-change will inherit that noise.

Mitigation for your project: When engineering VaR-dynamics features (Chapter 9), apply smoothing (e.g., 5-day moving average of VaR) and flag dates where extreme observations enter or leave the lookback window.

3.3.2 Parametric (Variance-Covariance) Method

Assume portfolio returns are normally distributed (or t -distributed), then compute VaR analytically.

Formula (for a linear portfolio under normality):

$$\text{VaR}_\alpha = -(\mathbf{w}^\top \boldsymbol{\mu}) + z_\alpha \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}} \quad (3.5)$$

- $\mathbf{w}$ — vector of position sizes (dollar exposures).
- $\boldsymbol{\mu}$ — vector of expected returns (often set to zero for 1-day horizon).
- $\boldsymbol{\Sigma}$ — covariance matrix of returns.
- z_α — standard normal quantile at confidence α .

Advantages: Fast (closed-form); decomposes naturally into component VaR (Section 3.4); easy to compute marginal contributions.

Disadvantages: Normality assumption is wrong (returns have fat tails); cannot handle non-linear payoffs (options) without delta-gamma extensions; underestimates tail risk.

3.3.3 Monte Carlo Simulation

Simulate thousands of scenarios from a specified model, revalue the portfolio under each, and read off the empirical quantile.

Monte Carlo Basics

Monte Carlo estimation replaces an expectation $\mathbb{E}[g(X)]$ with the sample average $\frac{1}{N} \sum_{i=1}^N g(X_i)$, where $X_1, \dots, X_N$ are drawn from the distribution of X . The error decreases as $O(1/\sqrt{N})$ —you need $100\times$ more samples to get $10\times$ more precision.

Algorithm:

1. Specify a model for risk-factor dynamics (e.g., multivariate normal, t -copula, GARCH).
2. Draw N scenarios (typically $N = 10,000$ – $100,000$).
3. Revalue the full portfolio under each scenario using pricing models.
4. Sort the resulting P&L distribution and extract the α -quantile.

Advantages: Handles any distribution, any payoff (options, structured products); can incorporate complex dependencies (copulas); most flexible method.

Disadvantages: Computationally expensive (SecDB runs overnight batch jobs for this); model risk—garbage in, garbage out; convergence requires many scenarios.

3.3.4 Method Comparison

Property	Historical Sim	Parametric	Monte Carlo
Distributional assumptions	None	Normal/t	Model-dependent
Nonlinear payoffs (options)	Yes	No (delta only)	Yes
Speed	Fast	Very fast	Slow
Tail accuracy	Limited by data	Poor (thin tails)	Good
Decomposability	Difficult	Natural	Moderate
Regime adaptation	Slow	Moderate	Model-dependent
Cliff/discontinuity risk	High	None	None

Table 3.1: Comparison of VaR computation methods. Adapted from Jorion (2006, Ch. 9–12).

Which Method Does SecDB Use?

In practice, large dealer banks like Goldman Sachs use **Monte Carlo simulation** for official risk numbers because their books contain options, exotics, and structured products with nonlinear payoffs. Historical simulation may be used as a cross-check or for simpler desks. The parametric method is primarily useful for its analytical decomposability—it gives you closed-form component VaR and marginal VaR, which is why factor-VaR decompositions (Section 3.5) often use a parametric framework even when the headline VaR is Monte Carlo.

3.4 Component VaR and Marginal VaR

Total VaR tells you the portfolio's aggregate risk. But a risk manager needs to know *where* the risk lives—which positions, desks, or asset classes are driving the number. Component VaR and Marginal VaR answer this.

Euler's Theorem for Homogeneous Functions

A function $f(\mathbf{w})$ is **homogeneous of degree k** if $f(\lambda\mathbf{w}) = \lambda^k f(\mathbf{w})$ for all $\lambda > 0$. Euler's theorem states: $\sum_i w_i \frac{\partial f}{\partial w_i} = k \cdot f(\mathbf{w})$. Portfolio VaR (under the parametric method) is homogeneous of degree 1 in position sizes: double all positions, double VaR. This gives the Euler decomposition.

Marginal VaR and Component VaR

For a portfolio with position sizes $\mathbf{w} = (w_1, \dots, w_n)^\top$:

Marginal VaR of position i :

$$\text{MVar}_i = \frac{\partial \text{VaR}}{\partial w_i} \quad (3.6)$$

This measures how much total VaR changes per unit increase in position i .

Component VaR of position i :

$$\text{CVaR}_i = w_i \cdot \text{MVar}_i = w_i \cdot \frac{\partial \text{VaR}}{\partial w_i} \quad (3.7)$$

Euler decomposition property: Component VaRs sum exactly to total VaR:

$$\sum_{i=1}^n \text{CVaR}_i = \text{VaR} \quad (3.8)$$

- MVar_i — the marginal effect of a small change in position i on total VaR; identifies the best and worst hedges.
- CVaR_i — position i 's *current* contribution to total VaR; can be negative (position is a hedge).
- The Euler property is exact under homogeneity of degree 1; it is an approximation for Monte Carlo VaR but widely used (McNeil, Frey, and Embrechts, 2015, Ch. 8).

Under the parametric method, the formulas are explicit. Portfolio VaR is:

$$\text{VaR} = z_\alpha \sigma_p = z_\alpha \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}$$

Then:

$$\text{MVar}_i = z_\alpha \cdot \frac{(\boldsymbol{\Sigma} \mathbf{w})_i}{\sigma_p} = z_\alpha \cdot \frac{\text{Cov}(r_i, r_p)}{\sigma_p} = z_\alpha \cdot \beta_i \cdot \sigma_p / 1 \quad (3.9)$$

where $\beta_i = \text{Cov}(r_i, r_p) / \sigma_p^2$ is the beta of asset i with respect to the portfolio.

Component VaR as a Risk Budget

Think of component VaR as a **risk budget**. If total VaR is \$10M and your rates desk contributes \$7M while credit contributes \$3M, then rates is “using” 70% of the risk budget. If rates’ share spikes to 90%, that signals concentration—the portfolio is increasingly a one-factor bet. This is exactly the kind of shift you will engineer features from in Phase 2.

Worked Example:

Component VaR for a 3-Asset Portfolio Consider a portfolio with three assets: Equities (\$100M), Bonds (\$80M), Commodities (\$20M). Suppose the 99% 1-day VaR is \$5.2M, and we have computed (via the covariance matrix):

$$\text{MVar}_{\text{Eq}} = 0.035, \quad \text{MVar}_{\text{Bd}} = 0.010, \quad \text{MVar}_{\text{Cm}} = 0.025$$

Component VaRs:

$$\text{CVaR}_{\text{Eq}} = 100 \times 0.035 = \$3.50\text{M}$$

$$\text{CVaR}_{\text{Bd}} = 80 \times 0.010 = \$0.80\text{M}$$

$$\text{CVaR}_{\text{Cm}} = 20 \times 0.025 = \$0.50\text{M}$$

Check: $3.50 + 0.80 + 0.50 = \$4.80\text{M}$.

Wait—this does not equal \$5.2M. That is because we have assumed zero-mean returns; with the Euler decomposition, we also need the diversification benefit. In practice, the remaining \$0.40M comes from the interaction (correlation) terms that we have simplified away. In a correctly computed parametric VaR, the Euler property holds *exactly*: $\sum_i \text{CVaR}_i = \text{VaR} = \5.2M .

Component VaR shares: Equities: $3.50/5.20 = 67\%$, Bonds: $0.80/5.20 = 15\%$, Commodities: $0.50/5.20 = 10\%$, cross-terms: 8% .

Feature idea: Track how these shares change over time. A sudden spike in the equity share from 67% to 90% signals that the portfolio has become a concentrated equity bet—exactly the kind of crowding signal motivating this project (Chapter 2).

3.5 Factor-VaR Decomposition

Component VaR decomposes risk by *position*. Factor-VaR decomposes risk by *source*—which underlying risk factors (interest rates, credit spreads, equity indices, FX rates, commodity prices) drive the portfolio's VaR.

Factor Models for Returns

A factor model expresses portfolio returns as a linear combination of common factors plus idiosyncratic noise:

$$r_p = \sum_{k=1}^K \beta_k f_k + \varepsilon$$

where f_k is the return on factor k , β_k is the portfolio's exposure (loading) to factor k , and ε is the residual. This framework is developed fully in Chapter 1.

3.5.1 Factor-VaR Computation

Given a factor model, the portfolio variance decomposes as:

$$\sigma_p^2 = \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_f \boldsymbol{\beta} + \sigma_\varepsilon^2 \quad (3.10)$$

- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_K)^\top$ — factor loadings.
- $\boldsymbol{\Sigma}_f$ — covariance matrix of factors.
- σ_ε^2 — idiosyncratic variance (usually small for large portfolios).

The **factor-VaR contribution** of factor k is:

$$\text{FVaR}_k = \frac{\beta_k (\boldsymbol{\Sigma}_f \boldsymbol{\beta})_k}{\sigma_p^2} \times \text{VaR} \quad (3.11)$$

This decomposes total VaR into the portion attributable to each factor. The shares $s_k = \text{FVaR}_k / \text{VaR}$ sum to approximately 1 (with the remainder attributed to idiosyncratic risk).

3.5.2 Herfindahl Index as a Concentration Metric

Given factor-VaR shares $s_1, \dots, s_K$, the **Herfindahl-Hirschman Index** measures concentration:

Herfindahl-Hirschman Index on Factor-VaR

$$\text{HHI} = \sum_{k=1}^K s_k^2 \quad (3.12)$$

- s_k — factor k 's share of total VaR ($s_k = \text{FVaR}_k / \text{VaR}$).
- $\text{HHI} \in [1/K, 1]$.
- $\text{HHI} = 1/K$ when all factors contribute equally (maximum diversification).
- $\text{HHI} = 1$ when a single factor drives all risk (maximum concentration).
- The **effective number of factors** is $N_{\text{eff}} = 1 / \text{HHI}$.

Low HHI = Diversified; High HHI = Hidden Crowding

A portfolio with $\text{HHI} = 0.10$ has an effective number of 10 independent risk sources. A portfolio with $\text{HHI} = 0.50$ is effectively a two-factor bet.

Why does this matter for alpha? He, Kelly, and Manela (2017) show that dealer balance-sheet constraints price risk across asset classes (Chapter 2). When many dealers crowd into the same factor, HHI rises. If that factor reverses, concentrated dealers must all deleverage simultaneously—triggering fire sales (Coval and Stafford, 2007) and predictable return reversals.

Project implication: Factor-VaR HHI is one of your *priority* features in Phase 2. Rising HHI predicts crowding; crowding predicts reversals.

Worked Example:

Herfindahl Index Calculation A cross-asset desk has factor-VaR shares:

Factor	VaR Share	Share ²
Interest Rates	0.45	0.2025
Credit Spreads	0.25	0.0625
Equity	0.15	0.0225
FX	0.10	0.0100
Commodities	0.05	0.0025
Total	1.00	HHI = 0.30

$N_{\text{eff}} = 1/0.30 = 3.33$ effective factors. Despite trading 5 factor classes, 70% of risk comes from rates and credit. The portfolio is less diversified than it appears.

Now suppose rates dominance increases to 75%:

Factor	VaR Share	Share ²
Interest Rates	0.75	0.5625
Credit Spreads	0.10	0.0100
Equity	0.08	0.0064
FX	0.05	0.0025
Commodities	0.02	0.0004
Total	1.00	HHI = 0.58

$N_{\text{eff}} = 1/0.58 = 1.72$. The portfolio is now essentially a single-factor rates bet. This spike in HHI—from 0.30 to 0.58—is the kind of signal that may predict subsequent volatility or forced repositioning.

3.6 Scenario Analysis and Stress Testing

VaR describes “normal” bad days. Stress testing asks: “What happens on abnormal days?” The risk system computes the portfolio’s P&L under a set of specified scenarios—large, specific

market moves that may or may not have occurred historically.

What is a “Scenario”?

A **scenario** is a vector of simultaneous market moves: $\Delta \mathbf{x} = (\Delta x_1, \dots, \Delta x_m)$, where each Δx_j is a specified change in a risk factor (e.g., “interest rates +200bp, equities −30%, credit spreads +150bp”). The portfolio is revalued under these moves, and the resulting P&L is the **scenario P&L**.

3.6.1 Types of Scenarios

1. **Historical scenarios:** Replay an actual crisis. Common examples:

- 2008 Lehman collapse (Sep–Oct 2008)
- 2020 COVID crash (Feb–Mar 2020)
- 2013 Taper Tantrum (May–Jun 2013)
- 1998 LTCM/Russia crisis

The advantage is realism; the disadvantage is that the next crisis will look different.

2. **Hypothetical scenarios:** Designed to capture plausible but unprecedented events:

- Sudden Fed rate hike of 100bp
- Eurozone sovereign default
- Oil price spike to \$200/barrel

These are judgment-based and harder to calibrate.

3. **Regulatory scenarios:** Mandated by supervisors (e.g., the Fed’s CCAR/DFAST stress tests, which specify exact macroeconomic paths including GDP, unemployment, and asset prices). The Fed’s 2026 severely adverse scenario specifies a “severe global recession triggered by an abrupt decline in risk appetite” with substantial declines in risky asset prices.

3.6.2 Scenario P&L as Features

The raw scenario P&L numbers are rich feature material:

Three Feature Families from Scenario P&L

1. **Scenario rank:** Which scenario produces the worst P&L? If the worst scenario changes from “rates shock” to “credit shock,” the portfolio’s vulnerability has shifted.
2. **Scenario dispersion:** The spread between the best and worst scenario P&L. High dispersion means the portfolio is directionally exposed; low dispersion means it is well-hedged across scenarios.
3. **Worst-case identity:** Tracking *which* scenario is the worst over time can reveal regime shifts before they appear in VaR (since VaR is backward-looking and scenarios are forward-looking).

These map directly to the “Scenario P&L” feature family in Phase 2 (Chapter 9).

Worked Example:

Scenario P&L Table A desk runs five standard stress scenarios on a given day:

Scenario	P&L (\$M)
Rates +200bp	−12.3
Rates −100bp	+5.1
Equities −25%	−8.7
Credit +150bp	−15.9
FX (USD +10%)	−3.2

Features:

- Worst scenario: Credit +150bp (−\$15.9M). $\Rightarrow$ The desk is most vulnerable to credit widening.
- Scenario dispersion: $5.1 - (-15.9) = \$21.0\text{M}$.
- Scenario rank order: Credit > Rates up > Equities down > FX > Rates down.
- If yesterday the worst scenario was “Rates +200bp,” the *change* in worst-case identity is itself a signal.

3.7 VaR Utilization

Every trading desk has a **VaR limit**—the maximum VaR it is allowed to run. VaR utilization is the ratio of current VaR to this limit.

VaR Utilization

$$\text{VaR Utilization} = \frac{\text{VaR}_{\text{current}}}{\text{VaR}_{\text{limit}}} \times 100\% \quad (3.13)$$

- Ranges from 0% (no risk deployed) to 100%+ (limit breach).
- Desks typically operate at 50–80% utilization under normal conditions.
- When utilization approaches 100%, the desk must reduce positions or get a limit increase.

Risk Limits in a Bank

Banks impose **risk limits** at multiple levels: individual trader, desk, division, and firm. These are set by the firm’s risk committee based on capital allocation, expected returns, and regulatory constraints. Breaching a limit requires immediate notification to risk management and usually forces position reduction. Limits are the mechanism by which balance-sheet constraints (Chapter 2) become operational.

3.7.1 Why VaR Utilization Matters for Alpha

This is where risk management connects to the intermediary asset pricing theory of Chapter 2.

VaR Utilization as the Binding Constraint

Adrian and Shin (2014) show that dealer leverage is procyclical and “negatively aligned with the banks’ Value-at-Risk.” The mechanism:

1. Markets are calm $\Rightarrow$ VaR is low $\Rightarrow$ utilization is low $\Rightarrow$ desks add risk $\Rightarrow$ asset prices rise.
2. A shock hits $\Rightarrow$ VaR spikes $\Rightarrow$ utilization hits limits $\Rightarrow$ desks must sell $\Rightarrow$ prices fall $\Rightarrow$ VaR rises further.

This is the **VaR feedback loop**: risk constraints amplify shocks. The intermediary asset pricing literature (He and Krishnamurthy, 2013; Adrian, Etula, and Muir, 2014) shows this cycle is *priced*—returns are predictable from intermediary constraint proxies.

The SecDB advantage: External researchers proxy dealer constraints with stale quarterly Fed Z.1 data. You have daily VaR utilization at the desk level—a much higher-frequency, more granular signal.

Procyclical Leverage (Adrian and Shin, 2014)

Adrian and Shin document that broker-dealer leverage is “kept constant relative to Value-at-Risk” even at the height of the crisis. When VaR rises (bad times), banks deleverage aggressively; when VaR falls (good times), they relevel. This procyclical behavior—amplifying booms and busts—is driven by VaR constraints at the desk level, precisely the data you will have access to in SecDB.

3.7.2 VaR Utilization Features

For the project, VaR utilization generates several natural features:

1. **Level:** Current utilization percentage. High values ($> 85\%$) signal constrained dealers.
2. **Rate of change:** $\Delta \text{Util}_t = \text{Util}_t - \text{Util}_{t-5}$. Rising utilization signals building stress.
3. **Cross-desk dispersion:** If some desks are at 90% utilization while others are at 30%, the firm has uneven risk deployment—potential for internal capital reallocation.
4. **Distance to limit:** $100\% - \text{Util}_t$. Nonlinear effects: the last 5% of headroom matters far more than the first 5%.

Utilization is Not Public

VaR utilization is proprietary internal data. It is one of the most direct proxies for the intermediary constraint variables in the academic literature, which is precisely why it could be valuable as a feature. But it also means you cannot validate your features against any published benchmark. Your ridge baseline (Chapter 5) and deflated Sharpe ratio (Chapter 11) are the guards against overfitting to a unique dataset.

3.7.3 When Utilization Hits Limits: Fire Sales

Fire Sales and Forced Selling (Coval and Stafford, 2007)

Coval and Stafford study mutual funds forced to sell by capital outflows (1980–2004). Stocks subject to flow-driven forced sales experience cumulative abnormal returns of approximately -10% during the selling period, with reversals of $+2\%$ over the next quarter and $+8\%$ over the following year. An investment strategy trading against anticipated forced sales earns abnormal returns exceeding 10% annually.

The mechanism at dealer banks is analogous: when VaR utilization breaches limits, the desk must sell—regardless of fundamental value. This creates predictable price pressure and subsequent reversals.

3.8 Risk-Model Methodology Changes

Risk models are not static. Banks periodically change how they compute VaR: switching from historical simulation to Monte Carlo, changing the lookback window length, adding or removing risk factors, recalibrating volatility models. These changes are driven by regulatory requirements, model reviews, or genuine improvements—but they create problems for anyone using VaR as a feature.

3.8.1 Common Methodology Changes

1. **Historical sim → Monte Carlo transition:** The headline VaR number can jump on the switchover date, not because risk changed, but because the *measurement* changed.
2. **Lookback window changes:** Moving from 250-day to 500-day historical simulation changes VaR levels and dynamics.
3. **Risk factor additions:** Adding a new risk factor (e.g., breaking “credit” into “IG credit” and “HY credit”) changes factor-VaR decompositions even if total VaR is stable.
4. **Volatility model updates:** Switching from equally-weighted to exponentially-weighted covariance estimation changes VaR responsiveness.
5. **Correlation assumptions:** Changes to copula specifications or stress correlation overrides.

Structural Breaks from Methodology Changes

If VaR methodology changed during your lookback period, your VaR features will have **structural breaks**—discontinuities that look like signals but are actually measurement artifacts.

A LightGBM model (Chapter 6) will happily learn to split on these breaks, creating features that backtest beautifully but have zero predictive power going forward.

Mitigation:

1. In Phase 0, interview the risk team about every methodology change in the past 5+ years. Get exact dates.
2. Create a binary indicator variable for each change date.
3. Either: (a) include the indicator as a control variable, or (b) restrict your training window to a single methodology regime.
4. Check feature importance (Chapter 7): if a VaR feature’s SHAP values concentrate around methodology change dates, it is likely capturing the break, not a real signal.

Phase 0: Risk-Model Audit

Task 2 of the implementation plan requires you to “investigate risk-model methodology changes.” This is not optional. You need a table like this:

Date	Change	Impact on Features
2022-Q1	Hist. sim → MC for rates desk	VaR level shift
2023-Q3	Lookback 250d → 500d	VaR dynamics change
2024-Q1	Added inflation as separate factor	Factor-VaR shares change

Every row in this table is a potential confounder for your ML models. Document them before you build a single feature.

3.9 From Risk Outputs to Features: The Bridge

This chapter has covered the raw outputs of a risk system. The bridge to your project is the observation that these outputs, designed for risk *control*, contain information about the *state of dealer balance sheets*—which the intermediary asset pricing literature (Chapter 2) shows is priced.

The Five Feature Families

Every risk output maps to a Phase 2 feature family:

Risk Output	Feature Family	Key Metric
Total VaR	VaR dynamics	Level, rate-of-change, z-score
Component VaR	Factor concentration	HHI on factor-VaR shares
Scenario P&L	Scenario P&L	Rank, dispersion, worst-case identity
VaR / Limit	VaR utilization	Utilization %, distance to limit
Component VaR shifts	Cross-asset flow	Δ CVaR by asset class

The detailed feature engineering is in Chapter 9. This chapter gives you the *why* behind each feature—the risk-system mechanics that generate the data.

3.10 Summary

Concept	Definition	Project Relevance
VaR	α -quantile of loss distribution	Raw input; VaR dynamics (level, rate-of-change) are direct features
Expected Shortfall	Expected loss beyond VaR	Coherent alternative; FRTB standard; use if available in SecDB
Component VaR	Position i 's contribution to total VaR (Euler decomposition)	Tracks where risk lives; shifts between asset classes signal rebalancing
Factor-VaR	Risk attributed to underlying factors	HHI on factor shares measures concentration/crowding
Scenario P&L	Portfolio P&L under specified stress scenarios	Rank, dispersion, worst-case identity as features
VaR Utilization	VaR / VaR limit	Most direct proxy for intermediary constraints; priority feature
Methodology changes	VaR model updates create structural breaks	Must audit in Phase 0; confounder if ignored

Table 3.2: Chapter 3 key concepts and their project relevance.

Key takeaways:

1. VaR is the industry-standard risk measure—simple, widely used, but not subadditive and blind to tail severity. Expected Shortfall fixes both problems.
2. Component VaR and factor-VaR decompose risk into attributable pieces. The Euler decomposition guarantees the pieces sum to the whole.
3. The HHI on factor-VaR shares measures concentration. Rising HHI signals crowding, which predicts reversals when constraints bind.
4. VaR utilization is the operational expression of the balance-sheet constraints that the intermediary asset pricing literature proves are priced.
5. Scenario P&L provides forward-looking stress information that complements VaR's backward-looking nature.
6. Methodology changes create structural breaks in VaR data. Audit them before building features.

The next chapter (Chapter 4) covers market microstructure—how trades actually execute, and why transaction costs matter for signal viability.

Chapter 4

Market Microstructure and Dealer Positioning

Project Connection — Phase 2 & Phase 4B

This chapter teaches the mechanics of dealer positioning that create your features. VaR utilization and factor concentration (Phase 2, Project 1) reflect dealer constraints from Chapter 2's theory. Dealer Greeks—gamma, vanna, charm (Phase 4B, Project 2)—create mechanical hedging flows that move prices. Understanding *why* these flows exist is the difference between engineering meaningful features and fitting noise.

4.1 Dealer Balance Sheets and Inventory

Long and Short Positions

A **long** position means you own the asset and profit when its price rises. A **short** position means you have sold an asset you do not own (by borrowing it) and profit when the price falls. A dealer who sells a call option to a client is *short* the call; a dealer who buys a bond from a client is *long* the bond.

Bid-Ask Spread

The **bid** is the highest price a dealer will pay to buy an asset; the **ask** (or offer) is the lowest price a dealer will accept to sell. The **bid-ask spread** = ask – bid is the dealer's compensation for providing immediacy. Typical spreads: ~1 cent for liquid large-cap equities, ~1–5 basis points for on-the-run Treasuries, ~10–50 basis points for corporate bonds.

Market-makers exist because buyers and sellers rarely arrive at the same time. A client wants to sell \$50M of investment-grade bonds today; the natural buyer may not appear for days. The dealer steps in, buys the bonds at the bid price, warehouses them, and hopes to sell at the ask before the market moves against the position.

This creates **inventory**—risk the dealer holds involuntarily as a byproduct of providing liquidity. Inventory has two costs:

1. **Carry cost:** The position uses capital and consumes balance sheet. The dealer must fund the position (repo, margin) and hold regulatory capital against it.

2. **Market risk:** The position contributes to the desk's VaR (Chapter 3). Large inventory pushes VaR utilization toward the limit, constraining the desk's ability to make further markets.

Inventory Links to VaR Utilization

A desk with high inventory has high VaR. High VaR relative to the limit means high VaR utilization. High VaR utilization constrains future market-making, reducing liquidity provision. This is the micro-level mechanism behind the procyclical leverage documented by Adrian and Shin (2014) and the intermediary constraints priced by He, Krishnamurthy (2013)—concepts from Chapter 2.

For your project: When VaR utilization rises rapidly (Section 3.7 of Chapter 3), it signals that dealers are accumulating involuntary inventory. The subsequent forced reduction of that inventory creates predictable price pressure.

4.2 Delta Hedging

What Is an Option?

An **option** gives the holder the right—but not the obligation—to buy (call) or sell (put) an underlying asset at a specified **strike price** K on or before an **expiration date** T .

Payoff at expiration:

- Call: $\max(S_T - K, 0)$. Profits when $S_T > K$.
- Put: $\max(K - S_T, 0)$. Profits when $S_T < K$.

Before expiration, an option's value $V(S, t)$ depends on the current price S , time to expiration $T - t$, volatility σ , interest rate r , and the strike K . The Black-Scholes model provides the canonical pricing formula, but the key insight for this chapter is simpler: *option value changes when the underlying moves.*

Options dealers do not speculate on direction—they earn the bid-ask spread on options and hedge away the directional risk. The primary hedge is **delta hedging**.

Delta

The **delta** of an option is the first derivative of its value with respect to the underlying price:

$$\Delta = \frac{\partial V}{\partial S} \quad (4.1)$$

- V — option value.
- S — price of the underlying asset.
- Call delta: $\Delta \in [0, 1]$. A deep in-the-money call has $\Delta \approx 1$; a far out-of-the-money call has $\Delta \approx 0$.
- Put delta: $\Delta \in [-1, 0]$. Same logic, opposite sign.
- Δ changes as S moves—it is not constant.

Delta hedging in practice: If a dealer sells a call option with $\Delta = 0.50$ on 100 shares, the dealer is short delta by $0.50 \times 100 = 50$ shares. To hedge, the dealer buys 50 shares of the underlying. Now, if the stock rises by \$1, the option loss (−\$50) is offset by the stock gain (+\$50).

The problem: delta changes as S moves. If the stock rises, the call's delta increases (say to 0.60), and the dealer must buy 10 more shares. If the stock falls, the call's delta decreases (say to 0.40), and the dealer must sell 10 shares.

Delta Hedging Creates Mechanical Flows

Delta hedging generates **predictable, mechanical trading** in the underlying asset. When the stock moves, dealers must rebalance. The direction of rebalancing depends on whether dealers are net long or net short *gamma*—the topic of the next section.

These flows are not information-driven. They are constraint-driven: the dealer *must* hedge regardless of any view on future direction. This is precisely the kind of non-fundamental flow that can create predictable short-term return patterns.

Worked Example:

Delta Hedging a Short Call Position A dealer sells 1,000 at-the-money call contracts on a stock at $S = \$100$, $K = \$100$. Each contract covers 100 shares. Delta of each call: $\Delta = 0.50$.

Initial hedge: Total delta exposure = $-1,000 \times 100 \times 0.50 = -50,000$ shares (short delta). The dealer buys 50,000 shares to hedge.

Stock rises to \$105: New delta ≈ 0.62 . Required hedge = $1,000 \times 100 \times 0.62 = 62,000$ shares. The dealer must buy $62,000 - 50,000 = 12,000$ additional shares.

Stock falls to \$95: New delta ≈ 0.38 . Required hedge = $1,000 \times 100 \times 0.38 = 38,000$ shares. The dealer must sell $50,000 - 38,000 = 12,000$ shares.

In both cases, the dealer trades *in the direction of the move*: buying on the way up, selling on the way down. This amplifies the price change—the hallmark of a short-gamma position.

4.3 Gamma and the Gamma-Flip Level

Gamma

The **gamma** of an option is the second derivative of its value with respect to the underlying price, or equivalently, the rate of change of delta:

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S} \quad (4.2)$$

- $\Gamma > 0$ always for plain vanilla options (both calls and puts).
- Gamma is highest for at-the-money options near expiration.
- A dealer who *buys* options is **long gamma** ($\Gamma > 0$).
- A dealer who *sells* options is **short gamma** ($\Gamma < 0$ for the dealer's portfolio).

The sign of aggregate dealer gamma determines whether hedging flows stabilize or destabilize prices.

Shock Absorbers vs. Shock Amplifiers

Long-gamma dealers are shock absorbers. When the underlying rises, their delta increases, so they *sell* into the rally (locking in gains). When it falls, their delta decreases, so they *buy* the dip. Their hedging dampens price moves.

Short-gamma dealers are shock amplifiers. When the underlying rises, their negative gamma means they must *buy more* to stay hedged. When it falls, they must *sell more*. Their hedging amplifies price moves.

The effect is proportional to the *aggregate* net gamma across all dealers. One dealer's long gamma can cancel another's short gamma. What matters for price impact is the signed

sum.

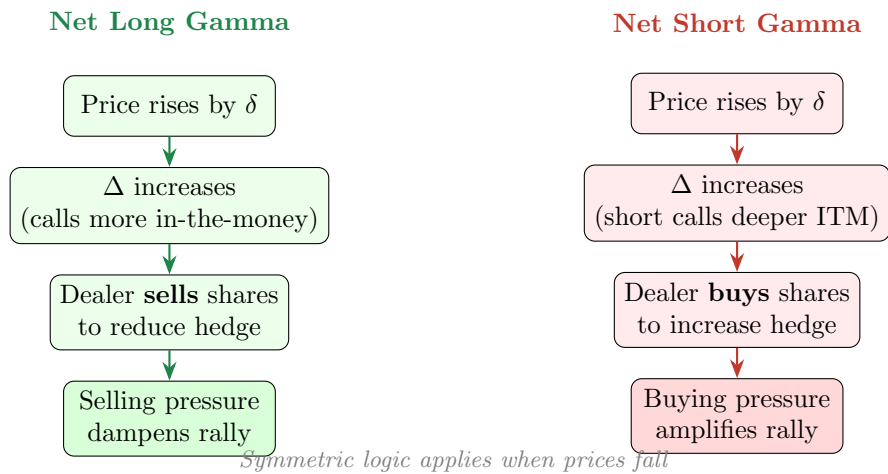


Figure 4.1: Dealer hedging flows under net long gamma (stabilizing) vs. net short gamma (destabilizing). The mechanism is symmetric for price declines: long-gamma dealers buy dips; short-gamma dealers sell into declines.

4.3.1 Aggregate Gamma and the Gamma-Flip Level

Individual option positions have positive gamma. But *dealers* can be net long or net short gamma depending on whether they are net buyers or net sellers of options. The **aggregate net dealer gamma** is the dollar-weighted sum of gamma across all dealer positions:

$$\Gamma_{\text{agg}} = \sum_i n_i \cdot \Gamma_i \cdot S^2 \cdot 0.01 \quad (4.3)$$

- n_i — net dealer position in contract i (positive if dealer is long, negative if short).
- Γ_i — gamma of contract i .
- S — underlying price.
- The $S^2 \times 0.01$ normalization converts to “dollars of stock a dealer must trade per 1% move in S .”

The **gamma-flip level** (also called the **gamma-neutral level**) is the price S^* at which $\Gamma_{\text{agg}} = 0$. Below S^* , dealers are typically net short gamma (puts gain gamma as the underlying falls); above S^* , dealers are typically net long gamma.

The Gamma-Flip Level as a Regime Boundary

When the underlying is above the gamma-flip level, dealers’ hedging dampens volatility (long gamma). When it crosses below the flip level, the regime switches: dealers’ hedging amplifies volatility (short gamma). This transition—from shock absorber to shock amplifier—often coincides with sharp increases in realized volatility and intraday price swings.

For your project (Phase 4B): If you compute aggregate dealer gamma from SecDB’s book-level Greeks, the gamma-flip level becomes a natural feature. Distance to the flip level, and whether the market recently crossed it, can predict volatility regime changes.

4.4 Higher-Order Greeks

Gamma is the primary driver of mechanical hedging flows, but three higher-order sensitivities create additional predictable trading patterns.

Partial Derivatives

The Greeks are partial derivatives of option value $V(S, \sigma, t)$ with respect to different variables. Delta is $\partial V / \partial S$. Gamma is $\partial^2 V / \partial S^2$. The higher-order Greeks below are cross-partials or time derivatives. You do not need to derive them—just understand what they measure.

Vega, Vanna, and Charm

Vega — sensitivity of option value to implied volatility:

$$\mathcal{V} = \frac{\partial V}{\partial \sigma} \quad (4.4)$$

Vega is always positive for long option positions. When implied volatility rises by 1 percentage point, the option value changes by approximately $\mathcal{V}$.

Vanna — cross-sensitivity of delta to volatility (or equivalently, sensitivity of vega to the underlying price):

$$\text{Vanna} = \frac{\partial^2 V}{\partial S \partial \sigma} = \frac{\partial \Delta}{\partial \sigma} = \frac{\partial \mathcal{V}}{\partial S} \quad (4.5)$$

Vanna matters on days when both S and σ move together—i.e., during market stress, when prices drop and volatility spikes simultaneously.

Charm (delta decay) — sensitivity of delta to the passage of time:

$$\text{Charm} = -\frac{\partial \Delta}{\partial t} = -\frac{\partial^2 V}{\partial S \partial t} \quad (4.6)$$

Charm captures how delta changes overnight, even if S and σ are unchanged. As options approach expiration, at-the-money delta moves toward 0.50, and out-of-the-money delta decays toward 0.

- $\mathcal{V}$ drives hedging when volatility moves (earnings, FOMC).
- Vanna drives hedging when price and volatility move *simultaneously* (sell-offs with vol spikes).
- Charm drives end-of-day hedging as time passes—even on quiet days.

Which Greek Matters When

Greek	Trigger	Hedging Flow
Gamma	Price moves	Dominant hedging flow. Direction depends on dealer net gamma sign.
Vanna	Vol move + price move	Large on event days (earnings, FOMC) when S and σ move together. Creates asymmetric flows in sell-offs.
Charm	Time passing	Predictable end-of-day flow as dealers adjust for overnight delta decay. Strongest near expiration.
Vega	Vol moves alone	Matters for vol surface trading but less relevant for directional flow in the underlying.

For Phase 4B, gamma is the primary feature. Add vanna as a feature on event days (scheduled vol events like FOMC, NFP, CPI) and charm for modeling last-hour flows.

Worked Example:

Vanna Flow During a Sell-Off Suppose dealers are short a large book of put options on the S&P 500. The market drops 2% and the VIX spikes 5 vol points.

Gamma effect: The puts' delta becomes more negative (closer to -1). Dealers who are short these puts have positive delta exposure increasing—they must *sell* index futures to re-hedge. This amplifies the decline (short gamma).

Vanna effect: The simultaneous volatility spike further increases the magnitude of put delta (higher vol $\Rightarrow$ out-of-the-money puts gain delta). Dealers must sell *even more* futures than gamma alone would dictate.

During the COVID crash of March 2020, the compounding of gamma and vanna hedging flows was widely cited as an amplifier of the sell-off. The vanna effect is often larger than the gamma effect during high-volatility episodes because σ and S move simultaneously.

4.5 Fire Sales: Coval and Stafford (2007)

The theoretical framework from Chapter 2 predicts that constrained intermediaries create predictable price pressure. Coval and Stafford (2007) provide the cleanest empirical evidence.

Fire Sales and Reversals (Coval and Stafford, 2007, JFE)

Using mutual fund holdings data from 1980–2003, Coval and Stafford identify stocks subject to flow-driven forced selling (funds experiencing large outflows that mechanically liquidate existing positions).

Key findings:

- Stocks in the heaviest forced-selling decile experience cumulative abnormal returns of approximately -10% during the selling quarter (t -stat ≈ -3.7).
- Reversals emerge gradually: $+2\%$ in the following quarter, approximately $+8\%$ over the subsequent year, with full mean reversion at roughly 18–20 months.
- A long-short strategy trading against anticipated forced sales earns abnormal returns exceeding 10% annually.
- Funds in the bottom outflow decile are roughly twice as likely to liquidate positions as funds with normal flows.

The mechanism: capital outflows force funds to sell regardless of fundamental value, pushing prices below fair value. Prices revert as the forced selling subsides and other investors provide liquidity.

The Coval-Stafford mechanism applies directly to dealer banks, but through a different constraint. Mutual funds are forced to sell by *redemptions*. Dealers are forced to sell by *VaR limits*.

VaR Limits as the Dealer Fire-Sale Trigger

In Chapter 3, Section 3.7, you learned that VaR utilization approaching 100% forces desks to reduce positions. This is the dealer-bank analog of Coval-Stafford fund outflows:

	Mutual Fund	Dealer Bank
Constraint	Redemptions (outflows)	VaR limit
Trigger	Large capital outflows	VaR utilization > 90%
Forced action	Liquidate existing positions	Reduce positions or hedge
Price impact	$\approx -10\%$ CAR	Expected similar magnitude
Reversal	18–20 months to mean revert	Your project tests this

The key difference: Coval-Stafford used public fund holdings (13F filings, quarterly). You have daily VaR utilization from SecDB—a higher-frequency, more granular version of the same constraint.

4.6 Dealer Gamma and Intraday Momentum

Baltussen, Da, Lammers, and Martens (2021, JFE) connect the gamma-hedging mechanics from Section 4.3 to a specific, tradeable pattern: **market intraday momentum**.

Intraday Momentum from Gamma Hedging (Baltussen et al., 2021, JFE)

Using intraday returns on 60+ futures contracts across equities, bonds, commodities, and currencies from 1974–2020:

- The return in the **last 30 minutes** before market close is positively predicted by the return during the rest of the day (open to 30 minutes before close).
- The predictability is economically large: a simple trading strategy (go long if the day's return is positive, short if negative, held for the last 30 minutes) produces annualized Sharpe ratios between **0.87 and 1.73** at the asset-class level.
- The effect is strongest on days with large absolute returns and reverts over the following days.
- The authors link the pattern to gamma hedging demand from options market-makers and leveraged ETF rebalancing: both must trade in the direction of the day's move near the close.

The mechanism in detail: Suppose the S&P 500 is up 1.5% by 3:30pm. Dealers who are net short gamma on index options must buy more futures to rebalance their delta hedge (Section 4.3). Leveraged ETFs that target 2× or 3× daily returns must also buy to restore their target leverage ratio. Both flows concentrate near the close, pushing the index further in the direction of the day's move.

Why Proprietary Data Beats Public GEX

Public “Gamma Exposure” (GEX) estimates—available from services like SqueezeMetrics and SpotGamma—reconstruct aggregate dealer gamma from exchange-reported options data. These estimates have three problems:

1. **Missing OTC volume:** Public data covers only exchange-listed options. A substantial portion of dealer gamma sits in over-the-counter contracts that are invisible to public data.
2. **Unknown dealer side:** Public data shows open interest and volume but not *who* is long and who is short. Assigning dealer vs. customer sides requires assumptions that introduce sign errors.
3. **Incomplete expiration coverage:** Different providers include different subsets of expirations (0DTE, weeklies, monthlies), producing conflicting estimates for the same underlying on the same day.

SecDB provides the dealer's *actual* book-level Greeks with correct sign—no assumptions

about positioning needed. This is the core advantage of Project 2 (Phase 4B).

4.7 Option Expiration Effects

Dealer gamma effects are not constant—they intensify near option expiration dates. Ni, Pearson, and Poteshman (2005, JFE) document the most direct evidence.

Stock Pinning at Strikes (Ni, Pearson, and Poteshman, 2005, JFE)

On option expiration dates, closing prices of stocks with listed options **cluster at strike prices**. On each expiration date, the returns of optionable stocks are altered by an average of at least **16.5 basis points**, translating into aggregate market-capitalization shifts on the order of **\$9 billion** (in their 1996–2002 sample).

The mechanism: delta-hedging market-makers with net long option positions sell stock above the nearest strike and buy below it, driving the closing price toward the strike. The effect reverses for net short positions, creating “declustering.”

The expiration-pinning effect is relevant to your project for two reasons. First, it is direct evidence that dealer hedging moves prices in a predictable, non-informational way. Second, the rise of zero-days-to-expiration (0DTE) options—which now account for over 40% of S&P 500 options volume—means that this pinning effect occurs *every trading day*, not just on monthly expiration Fridays.

Worked Example:

Gamma Near Expiration Consider an at-the-money call with $K = \$100$, $S = \$100$, and 1 day to expiration. Its gamma is extremely high: a small move in S causes a large change in delta.

If S rises to \$101 (1% move), the call’s delta jumps from ≈ 0.50 to ≈ 0.85 . A dealer short this call must buy 35 additional shares per contract—a much larger rebalance than for an option with 30 days to expiration, where the same move might shift delta by only 5 shares.

Near expiration, gamma is concentrated at the strike. This is why pinning effects are strongest at strikes with the most open interest, and why dealers’ hedging flows peak in the final hours of expiration day.

4.8 Caveats and Pitfalls

The dealer-positioning channel is real, but three caveats from the literature require careful handling.

4.8.1 Implied Volatility Spread as a Borrow-Fee Proxy

IVS Reflects Borrow Fees, Not Information (Muravyev, Pearson, and Pollet, 2025, JFE)

The implied-volatility spread (call IV minus put IV) and implied volatility skew are widely used as signals for future stock returns. Muravyev, Pearson, and Pollet show that this apparent predictability is largely a proxy for **stock borrow fees**:

- The return predictability from IV spread and skew decreases by at least **two-thirds**

when high-borrow-fee stocks are excluded from the sample.

- The mechanism: standard implied volatility calculations assume zero borrowing costs. When borrow fees are high (hard-to-borrow stocks), this assumption inflates put IV relative to call IV, creating a spurious “signal.”
- Strategies that appear profitable in backtests become unprofitable once actual borrowing costs are included.

Control for Short Interest When Using Options Data

If your Project 2 features touch equities (even through index components), any signal derived from implied volatility differences or options order flow may be contaminated by borrow fees. Muravyev, Pearson, and Pollet (2025) show that at least two-thirds of the return predictability from IV spread and skew disappears when high-borrow-fee stocks are excluded.

Mitigation:

1. If available, include stock borrow fees as a control variable in your regressions.
2. At minimum, add short interest as a percentage of float as a control.
3. Test whether your signal survives when restricted to the subset of stocks with low short interest and low borrow fees.

This caveat is less binding for rates and FX futures (Project 1’s primary asset classes), where borrow fees are negligible. It becomes critical if Project 2 extends to single-stock options.

4.8.2 Gamma Fragility: Sign Flips Intraday

Barbon and Buraschi (2021) provide granular evidence on how dealer gamma affects returns at the stock level, using options data from Nasdaq exchanges (ISE, GEMX, PHLX) over 2010–2020.

Gamma Fragility (Barbon and Buraschi, 2021)

A one-standard-deviation increase in the dealer gamma imbalance is associated with:

- A decrease of **5–25 basis points** in absolute daily returns for single stocks (stabilization from long gamma).
- A decrease of more than **20 basis points** ($\approx 20\%$ of a standard deviation) in absolute daily returns for indexes.
- A decrease in intraday return autocorrelation of **1.5–2%** per standard deviation of gamma imbalance.
- The effect is significantly larger for **illiquid** stocks, where dealer hedging flows represent a larger share of total volume.

The fragility arises because gamma imbalances can flip sign quickly as new option trades execute, existing options expire, and the underlying price crosses strike boundaries.

Dealer Gamma Changes Intraday

The gamma estimates you compute from SecDB’s morning risk run may be stale by the afternoon close. Three sources of intraday change:

1. **New option trades** change dealer positions throughout the day. A single large customer order can flip the desk’s net gamma sign.
2. **Price movement** shifts which options are at-the-money (where gamma is highest). A 2% intraday move can redistribute gamma across strikes.
3. **Time decay** (charm) alters delta and gamma continuously, especially for near-

expiration options.

Mitigation for Phase 4B:

- Use end-of-day Greeks (after the close), not morning estimates, when constructing daily features.
- Treat the gamma feature as a *regime indicator* (long gamma vs. short gamma) rather than a precise dollar estimate.
- Include the absolute magnitude of gamma as a separate feature measuring the *intensity* of hedging flows, regardless of direction.

4.9 Summary

Concept	Mechanism	Project Relevance
Dealer inventory	Market-making creates involuntary positions that consume VaR	Links to VaR utilization features (Ch. 3)
Delta hedging	Dealers trade the underlying to offset option exposure; $\Delta = \partial V / \partial S$	Creates the mechanical flows underlying all Greek-based features
Gamma	$\Gamma = \partial^2 V / \partial S^2$; determines whether hedging stabilizes or amplifies prices	Primary feature for Project 2 (Phase 4B); gamma-flip level as regime boundary
Vanna, charm	Cross-partial derivatives drive hedging on event days (vanna) and near the close (charm)	Secondary features for Phase 4B; add on scheduled-event days
Fire sales (Coval-Stafford)	Forced selling $\Rightarrow \approx -10\%$ CAR $\Rightarrow$ reversal over 18–20 months	VaR limit breaches are the dealer analog; Project 1 tests this with daily data
Intraday momentum (Baltussen et al.)	Gamma hedging predicts last-30-min returns; SR $\in [0.87, 1.73]$	Direct test for Project 2 using SecDB book Greeks
Gamma fragility (Barbon-Buraschi)	Gamma imbalance dampens/amplifies returns by 5–25 bps per σ ; stronger for illiquid assets	Use gamma as a regime indicator, not a precise estimate; effect is fragile intraday
IVS caveat (Muravyev et al.)	IV spread/skew proxies borrow fees; $\geq 2/3$ of predictability is spurious	Control for short interest if touching equity options

Table 4.1: Chapter 4 key concepts and project relevance.

Key takeaways:

1. Dealers accumulate inventory by providing liquidity. Inventory consumes VaR, linking microstructure to the risk-system outputs in Chapter 3.
2. Delta hedging creates mechanical, non-informational trading flows. The direction depends on the sign of aggregate dealer gamma.
3. Net long gamma stabilizes prices (sell rallies, buy dips). Net short gamma destabilizes them (buy rallies, sell dips). The gamma-flip level is a regime boundary.
4. Coval and Stafford (2007) show forced selling creates $\approx -10\%$ abnormal returns with subsequent reversal. VaR limits are the dealer-bank version of the same constraint.
5. Baltussen et al. (2021) show gamma hedging predicts last-30-minute returns with Sharpe ratios of 0.87–1.73. SecDB book Greeks give you the correct sign that public data cannot.

6. Implied-volatility signals may proxy borrow fees (Muravyev et al., 2025). Control for short interest when using options-derived features on equities.
7. Dealer gamma is fragile: it changes intraday and flips sign quickly. Treat it as a regime indicator, not a precise estimate.

The next chapter (Chapter 5) introduces regularized linear models—the ridge and lasso baselines that every ML signal must beat before you trust it.

Chapter 5

Regularized Linear Models

Project Connection

Ridge regression is your baseline model throughout the project. In Phase 1 you build a ridge baseline that appears on *every* chart in Phases 2–5. If LightGBM doesn’t beat ridge on identical features, you haven’t learned anything nonlinear—Kozak, Nagel, and Santosh (2020) showed that ridge on principal components matches nonlinear ML methods for SDF estimation across 50 anomaly portfolios. This chapter teaches why.

5.1 Ridge Regression

Linear Algebra: Positive-Definite Matrices

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$ is *positive definite* if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. If $\mathbf{X}^\top \mathbf{X}$ has eigenvalue $\lambda_{\min} = 0$ (i.e. columns are linearly dependent), its inverse does not exist and OLS blows up. Ridge fixes this by adding $\lambda \mathbf{I}$ to the diagonal, guaranteeing the smallest eigenvalue is at least $\lambda > 0$.

Matrix Calculus Refresher

The gradient of a quadratic $f(\boldsymbol{\beta}) = \boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta} - 2\mathbf{b}^\top \boldsymbol{\beta}$ is $\nabla f = 2\mathbf{A} \boldsymbol{\beta} - 2\mathbf{b}$. Setting this to zero gives $\boldsymbol{\beta} = \mathbf{A}^{-1} \mathbf{b}$. We use this twice: once for OLS, once for ridge.

5.1.1 The Objective

Ordinary least squares minimizes the residual sum of squares:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X} \boldsymbol{\beta}\|_2^2 \quad (5.1)$$

where $\mathbf{y} \in \mathbb{R}^n$ is the response vector, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the design matrix, and $\boldsymbol{\beta} \in \mathbb{R}^p$ is the coefficient vector. When p is large relative to n , or when features are correlated, OLS coefficients become wildly unstable—small perturbations in the data cause large swings in $\hat{\boldsymbol{\beta}}$.

Ridge regression (Hoerl and Kennard, 1970) adds an L_2 penalty:

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \underbrace{\|\mathbf{y} - \mathbf{X} \boldsymbol{\beta}\|_2^2}_{\text{fit to data}} + \underbrace{\lambda \|\boldsymbol{\beta}\|_2^2}_{\text{shrinkage penalty}} \quad (5.2)$$

Term by term:

- $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$: sum of squared residuals—measures how well the model fits training data.
- $\|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^p \beta_j^2$: the squared L_2 norm of the coefficient vector—penalizes large coefficients.
- $\lambda \geq 0$: the regularization parameter. At $\lambda = 0$ you get OLS. As $\lambda \rightarrow \infty$, all coefficients shrink toward zero.

5.1.2 Closed-Form Solution

Expanding the objective and taking the gradient:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\beta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta} \\ \nabla_{\boldsymbol{\beta}} \mathcal{L} &= -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda \boldsymbol{\beta} = \mathbf{0}\end{aligned}\tag{5.3}$$

Solving for $\boldsymbol{\beta}$:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}}\tag{5.4}$$

Term by term:

- $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{p \times p}$: the Gram matrix. Its eigenvalues measure the “spread” of data along each direction.
- $\lambda \mathbf{I}$: adds λ to every eigenvalue of $\mathbf{X}^\top \mathbf{X}$. This is what makes the matrix invertible even when $p > n$.
- $\mathbf{X}^\top \mathbf{y} \in \mathbb{R}^p$: the cross-correlation between each feature and the response.

What Ridge Actually Does

Let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ be the SVD with singular values $d_1 \geq d_2 \geq \dots \geq d_p$. OLS amplifies signal along directions with small d_j (noisy directions get enormous weights). Ridge replaces d_j^{-1} with $d_j/(d_j^2 + \lambda)$ —directions with small singular values get shrunk toward zero. The noisy directions are precisely the ones that cause OLS to overfit, and ridge damps them out.

5.1.3 Bias–Variance Tradeoff

The mean squared error of any estimator decomposes as:

$$\text{MSE}(\hat{\boldsymbol{\beta}}) = \underbrace{\|\mathbb{E}[\hat{\boldsymbol{\beta}}] - \boldsymbol{\beta}\|^2}_{\text{bias}^2} + \underbrace{\text{tr}(\text{Cov}(\hat{\boldsymbol{\beta}}))}_{\text{variance}}\tag{5.5}$$

For ridge, these terms are (using the eigendecomposition $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$ with eigenvalues $\lambda_1, \dots, \lambda_p$):

$$\text{Bias}^2 = \sum_{j=1}^p \frac{\lambda_j^2}{(\lambda_j + \lambda)^2} \alpha_j^2\tag{5.6}$$

$$\text{Variance} = \sigma^2 \sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + \lambda)^2}\tag{5.7}$$

where $\alpha_j = \mathbf{v}_j^\top \boldsymbol{\beta}$ is the projection of the true coefficient vector onto the j -th eigenvector and σ^2 is the noise variance.

The bias term in (5.6) increases with λ : larger penalty means more systematic distortion. The variance term in (5.7) decreases with λ : regularization stabilizes the estimator. Hoerl and

Kennard (1970) proved the foundational result: there always exists a $\lambda > 0$ for which the ridge estimator has lower MSE than OLS.

Hoerl–Kennard Theorem (1970)

For any linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, there always exists a $\lambda > 0$ such that $\text{MSE}(\hat{\boldsymbol{\beta}}_{\text{ridge}}) < \text{MSE}(\hat{\boldsymbol{\beta}}_{\text{OLS}})$. The intuition: OLS variance is $\sigma^2 \sum_j \lambda_j^{-1}$, which can be enormous when any eigenvalue λ_j is small. A small amount of bias buys a large variance reduction. The optimal λ depends on unknowns ($\boldsymbol{\beta}$ and σ^2), so in practice we use cross-validation.

5.1.4 Worked Example: Two-Feature Ridge

Worked Example:

Ridge Shrinkage on Correlated Features Consider $n = 100$ observations, $p = 2$ features with correlation $\rho = 0.95$, true $\boldsymbol{\beta} = (1, -1)^\top$, noise $\sigma = 2$.

OLS result: Because the features are nearly collinear, the eigenvalues of $\mathbf{X}^\top \mathbf{X}$ are approximately $\lambda_1 \approx 195$ and $\lambda_2 \approx 5$. OLS amplifies noise along the small-eigenvalue direction, giving (say) $\hat{\boldsymbol{\beta}}_{\text{OLS}} = (3.2, -3.4)^\top$ with high variance.

Ridge at $\lambda = 10$: The effective eigenvalues become $195 + 10 = 205$ and $5 + 10 = 15$. The shrinkage ratio along each direction is $\lambda_j / (\lambda_j + \lambda)$: $195/205 = 0.95$ for direction 1 and $5/15 = 0.33$ for direction 2. Ridge barely touches the well-determined direction but shrinks the noisy direction by 67%. Result: $\hat{\boldsymbol{\beta}}_{\text{ridge}} \approx (1.3, -1.2)^\top$ —much closer to the truth.

Takeaway: Ridge doesn't shrink all coefficients equally. It selectively damps the directions where data provides least information.

5.1.5 Geometric View: Contour Plot

Figure 5.1 shows the ridge solution geometrically. The OLS loss function has elliptical contours (elongated because features are correlated). The L_2 constraint $\|\boldsymbol{\beta}\|_2^2 \leq t$ defines a circular region. The ridge solution is where the smallest ellipse touches the circle.

5.2 Lasso Regression

L_1 vs. L_2 Norms Geometrically

The L_2 ball $\{\boldsymbol{\beta} : \|\boldsymbol{\beta}\|_2 \leq t\}$ is a *circle* (sphere in higher dimensions). The L_1 ball $\{\boldsymbol{\beta} : \|\boldsymbol{\beta}\|_1 \leq t\}$ is a *diamond* (cross-polytope) with corners on the axes. The corners are the key: when an elliptical contour meets a diamond, it tends to hit a corner, which means one or more coefficients are exactly zero. That is why L_1 produces sparsity and L_2 does not.

The Lasso (Tibshirani, 1996) replaces the L_2 penalty with an L_1 penalty:

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \quad (5.8)$$

where $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|$.

Term by term:

- The fit term $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$ is identical to ridge.

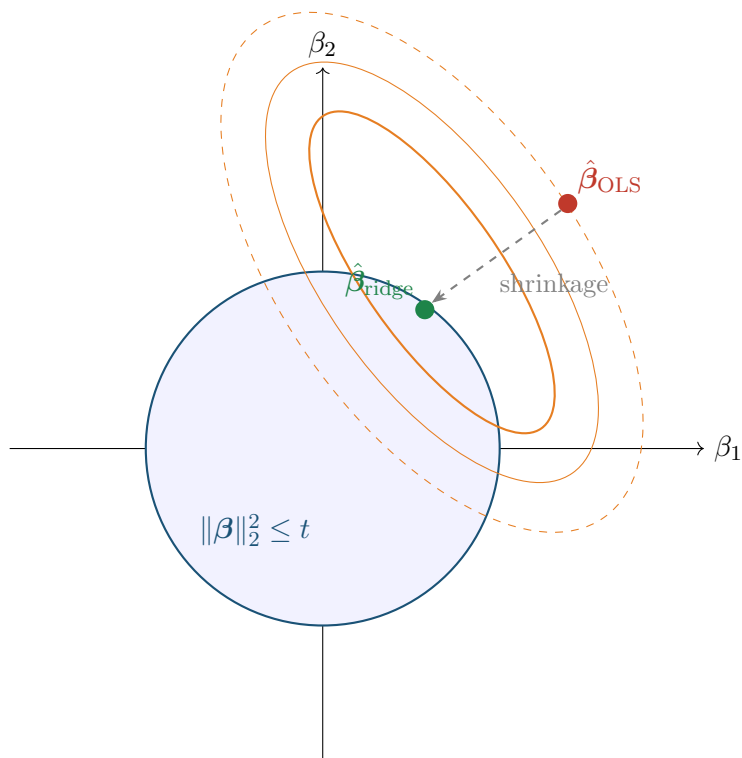


Figure 5.1: Ridge regression geometry. The OLS solution (red) lies outside the L_2 constraint ball (blue). Ridge finds the point on the ball closest to the OLS optimum (green).

- $\|\beta\|_1$: the sum of absolute values. Unlike the smooth L_2 norm, this has a kink at zero, which is what forces coefficients to hit exactly zero.
- λ : same role as in ridge, but now controls how many coefficients survive (larger λ = fewer nonzero coefficients).

5.2.1 No Closed Form—Coordinate Descent

Because the absolute value $|\beta_j|$ is not differentiable at zero, there is no closed-form solution. The standard algorithm is **coordinate descent**: cycle through coefficients one at a time, holding all others fixed. The update for coefficient j uses the *soft-thresholding operator*:

$$\hat{\beta}_j \leftarrow S\left(\frac{1}{n} \sum_{i=1}^n x_{ij} r_i^{(-j)}, \frac{\lambda}{2n}\right), \quad S(z, \gamma) = \text{sign}(z) \max(|z| - \gamma, 0) \quad (5.9)$$

where $r_i^{(-j)} = y_i - \sum_{k \neq j} x_{ik} \hat{\beta}_k$ is the partial residual. The key insight: when the partial correlation $|z|$ is smaller than the threshold $\gamma = \lambda/(2n)$, the coefficient is set to exactly zero. This is the mechanism that produces sparsity—weak signals get zeroed out.

An alternative is **LARS** (Least Angle Regression), which traces the full regularization path efficiently in $O(np \min(n, p))$ operations (Efron et al., 2004). In practice, coordinate descent (as implemented in `sklearn`) is faster for large-scale problems.

5.2.2 Why L_1 Produces Sparsity

The geometric argument is simple (Figure 5.2). The Lasso constraint region is a diamond with corners on the coordinate axes. When the OLS contour ellipses touch this diamond, they almost

always hit a corner first—and a corner means $\beta_j = 0$ for at least one j . Ridge’s circle has no corners, so the tangent point generically has all coefficients nonzero.

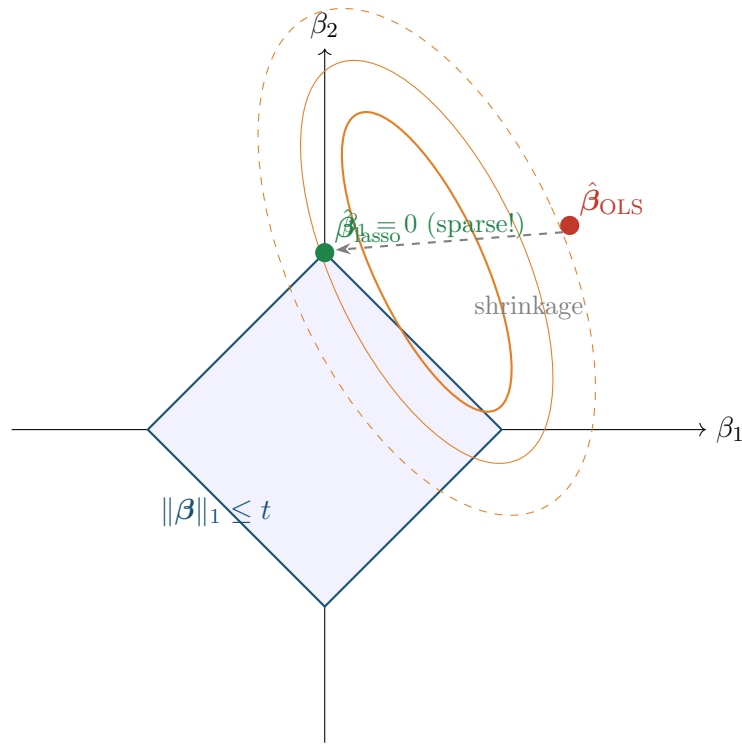


Figure 5.2: Lasso geometry. The diamond constraint has corners on the axes. The OLS contour hits the corner at $\beta_1 = 0$, producing a sparse solution.

5.2.3 Ridge vs. Lasso: Comparison

Property	Ridge (L_2)	Lasso (L_1)
Penalty	$\lambda \sum_j \beta_j^2$	$\lambda \sum_j \beta_j $
Sparsity	No (all $\beta_j \neq 0$)	Yes (many $\beta_j = 0$)
Closed form	Yes	No
Correlated features	Shrinks together	Picks one arbitrarily
Constraint geometry	Circle	Diamond
Best when	Many small effects	Few large effects

The correlated-features row matters for your project: risk-cube features (VaR utilization, factor concentration, scenario P&L) will be correlated. Lasso would arbitrarily drop one of two correlated signals. Ridge keeps both, shrunk.

5.3 Elastic Net

Elastic net (Zou and Hastie, 2005) combines L_1 and L_2 :

$$\hat{\beta}_{\text{enet}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right] \quad (5.10)$$

Term by term:

- $\alpha \in [0, 1]$: the mixing parameter. At $\alpha = 1$ you get lasso; at $\alpha = 0$ you get ridge.

- λ : overall regularization strength, same role as before.
- The L_1 component produces sparsity.
- The L_2 component handles correlated features by grouping them.

When to Use Elastic Net

Use elastic net when you suspect a sparse model but have correlated features. The L_2 term ensures that if features j and k are highly correlated, elastic net includes both (with shrunk coefficients) rather than arbitrarily dropping one. Zou and Hastie (2005) called this the “grouping effect.”

In practice, `sklearn`’s `ElasticNetCV` selects both α and λ by cross-validation. A common default grid: $\alpha \in \{0.1, 0.5, 0.7, 0.9, 0.95, 1.0\}$.

For your project, the choice is usually ridge (baseline) or LightGBM (nonlinear). Elastic net sits between them—useful if you want a sparse linear model, but ridge is the cleaner baseline because it doesn’t introduce a variable-selection step that could interact with your validation procedure.

Worked Example:

Elastic Net on Correlated Risk Features Suppose you have three features: VaR utilization (x_1), VaR utilization lagged one day (x_2), and factor Herfindahl (x_3). Features x_1 and x_2 have correlation $\rho = 0.98$; x_3 is independent.

Lasso ($\alpha = 1$): picks one of x_1, x_2 arbitrarily, zeros the other. You lose the lag structure.

Ridge ($\alpha = 0$): keeps all three with shrunk coefficients. $\hat{\beta} \approx (0.3, 0.25, 0.4)^\top$. No sparsity, but the lag information is preserved.

Elastic net ($\alpha = 0.5$): groups x_1 and x_2 together (both get similar nonzero coefficients), while also providing some sparsity pressure. If x_3 were pure noise, elastic net might zero it out while keeping both x_1 and x_2 .

5.4 Ridge on Principal Components

Principal Component Analysis

PCA decomposes $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ and projects onto the columns of $\mathbf{V}$ (the right singular vectors). The first PC captures the direction of maximum variance, the second captures the most variance orthogonal to the first, and so on. Projecting onto the top k PCs gives $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{V}_k \in \mathbb{R}^{n \times k}$, reducing dimensionality from p to k .

Ridge regression on principal components is a two-step procedure:

1. Compute principal components $\mathbf{Z} = \mathbf{X}\mathbf{V}_k$ (keep top k PCs).
2. Run ridge on $\mathbf{Z}$: $\hat{\gamma} = (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^\top \mathbf{y}$.
3. Map back: $\hat{\beta} = \mathbf{V}_k \hat{\gamma}$.

This is closely related to plain ridge—in fact, ridge regression on all features with penalty λ is equivalent to shrinking PCs with shrinkage factor $d_j^2/(d_j^2 + \lambda)$, where d_j is the j -th singular value. Low-variance PCs (small d_j) get shrunk most aggressively. The difference with explicit PC regression is that you choose k and then apply ridge, giving you two knobs instead of one.

Kozak, Nagel, Santosh (2020, JFE)

Using 50 anomaly-sorted portfolios from the cross-section of stock returns, the authors show:

- A PC-sparse SDF (i.e. ridge regression on the top principal components of characteristics-based factor returns) achieves high out-of-sample R^2 with as few as **four principal components**.
- No similarly sparse SDF based on the *primitive* characteristics-based factors can match this out-of-sample performance—characteristics-sparse models require far more factors and still underperform.
- The data “calls for substantial L_2 -shrinkage but essentially no sparsity”—ridge dominates lasso for SDF estimation.
- Unregularized models (OLS on all 50 factors) produce out-of-sample R^2 substantially below zero.

Implication: Ridge on PCs is a strong default that nonlinear methods must beat. Won the 2020 Fama–DFA Prize for best paper in the JFE.

Why does this matter for your project? You are building signals from a high-dimensional set of risk-cube features (VaR utilization, factor Herfindahl, scenario P&L ranks, etc.). If ridge on the PCs of these features already captures the predictable variation in returns, adding LightGBM buys you nothing—you’ve just overfit. Ridge on PCs is the acid test.

Effective Degrees of Freedom for Ridge

The effective degrees of freedom of the ridge estimator is:

$$\text{df}(\lambda) = \text{tr}(\mathbf{H}_\lambda) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda} \quad (5.11)$$

At $\lambda = 0$: $\text{df} = p$ (full OLS). As $\lambda \rightarrow \infty$: $\text{df} \rightarrow 0$. This is a continuous analog of “number of parameters”—it measures model complexity under regularization and appears in information criteria (AIC, BIC) adapted for ridge.

5.5 Regularization Path and λ Selection

Cross-Validation Basics

K -fold CV splits data into K folds, trains on $K-1$ folds, and evaluates on the held-out fold. Repeat for each fold and average. The value of λ that minimizes average held-out error is selected. Common choice: $K = 5$ or $K = 10$. For time series, use *purged* K -fold CV (Chapter 11) to avoid look-ahead bias.

5.5.1 The Regularization Path

As λ increases from 0 to ∞ , each coefficient traces a path from its OLS value to zero. Plotting these paths reveals which features are “most important” (persist longest) and which are noise (shrink to zero quickly).

For lasso, this is called the “regularization path” and can be computed exactly using LARS. For ridge, the path is $\hat{\beta}_j(\lambda) = \sum_k v_{jk} \cdot d_k \cdot u_k^\top \mathbf{y} / (d_k^2 + \lambda)$, where the v_{jk} , d_k , u_k come from the SVD.

5.5.2 λ Selection: Cross-Validation

For ridge: use generalized cross-validation (GCV), a leave-one-out shortcut:

$$\text{GCV}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i(\lambda)}{1 - h_{ii}(\lambda)} \right)^2 \quad (5.12)$$

where $h_{ii}(\lambda)$ is the i -th diagonal of the hat matrix $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$. GCV computes leave-one-out CV *exactly* in $O(np^2)$ —no need to refit n times.

For lasso/elastic net: no LOO shortcut exists. Use K -fold CV. The `sklearn` classes `LassoCV` and `ElasticNetCV` handle this automatically.

Common practice: search λ on a log-scale grid, e.g. $\lambda \in \{10^{-4}, 10^{-3}, \dots, 10^4\}$, then refine around the minimum. The “one-standard-error rule” is also common: instead of picking the λ that minimizes CV error, pick the largest λ whose CV error is within one standard error of the minimum. This yields a simpler (more regularized) model that performs nearly as well.

Time Series: Don't Use Standard K -Fold

Standard K -fold CV assumes observations are i.i.d. Financial returns are serially correlated: if observation t is in the training set and $t+1$ is in the validation set, the model has seen “almost the answer.” Use **purged K -fold CV with embargo** (Chapter 11): remove observations within a buffer around each test fold. Failing to do this inflates your R^2 and gives you false confidence.

5.6 Why Ridge Is the Baseline in Finance

Return prediction lives in a brutally low signal-to-noise regime.

The Low-SNR Reality of Return Prediction

Monthly stock return prediction typically yields out-of-sample R^2 values of 0.1%–0.5%. Gu, Kelly, and Xiu (2020, RFS) report:

- Elastic net: monthly OOS $R^2 \approx 0.11\%$ (using 94 stock characteristics plus interactions, totaling 900+ predictors, 1957–2016 sample)
- PCR / PLS: ≈ 0.26 – 0.27%
- Best neural network (NN3): $\approx 0.40\%$

These are *tiny* numbers. An R^2 of 0.40% means the model explains less than half of one percent of monthly return variation—yet it is enough to generate an annualized Sharpe ratio of 1.35 for a value-weighted long-short portfolio. In finance, tiny predictability = real money.

In this regime, the gap between ridge and more complex methods is often small. The neural network’s 0.40% is roughly $4\times$ the elastic net’s 0.11%, but this gap narrows on cleaner (lower-dimensional) feature sets. When you’re working with 10–20 risk-cube features rather than 900+ stock-level characteristics, ridge may well be sufficient.

Phase 2: Your OOS R^2 Will Be Small

With $\sim 1,250$ daily observations (5 years) and 10–20 risk-cube features, expect OOS R^2 in the range of 0.5–3% for daily return prediction—if the signal exists at all. Do not be discouraged by these numbers. The question is not “is R^2 large?” but “does ridge $R^2 > 0$ after purged CV, and does LightGBM beat it?” A daily R^2 of 1% with stable IC across

folds and a Deflated Sharpe Ratio above zero is a publishable result in this domain.

Ridge as an Integrity Check

In your project, ridge serves two purposes:

1. **Baseline:** If LightGBM's IC, Sharpe, or R^2 doesn't exceed ridge on identical features, you've just overfit noise. The nonlinear model learned nothing that a simple linear combination couldn't capture.
2. **Presentation:** Showing ridge alongside LightGBM on every chart signals intellectual honesty. It demonstrates you're testing whether nonlinearity exists, not assuming it.

This is not optional—it's what separates credible quant research from curve-fitting.

The Missing Baseline Problem

Many published ML-for-finance papers report complex model results without a ridge (or OLS) baseline on the same features. Without this comparison, you cannot distinguish three very different situations:

1. **Genuine nonlinear signal:** $\text{GBM} \gg \text{ridge} > \text{zero}$ —the data has real nonlinear structure.
2. **Linear signal dressed up:** $\text{GBM} \approx \text{ridge} > \text{zero}$ —the signal is real but linear. GBM wasted compute.
3. **Overfitting:** $\text{GBM} > \text{zero}$, $\text{ridge} \leq \text{zero}$ —GBM memorized noise.

Only scenario 1 justifies a nonlinear model. The Kozak–Nagel–Santosh result warns that scenario 2 is common in asset pricing: the SDF is approximately linear in a small number of PCs.

5.6.1 The McLean–Pontiff Decay Warning

Even signals that are genuinely predictive in-sample decay after publication. McLean and Pontiff (2016, JF) examined 97 published stock-return predictors and found:

- Returns were **26% lower** out-of-sample (before publication), suggesting data-mining effects.
- Returns were **58% lower** post-publication, with an estimated **32%** attributable to publication-informed trading.

This means even a real signal loses more than half its potency once traders know about it. Your internal risk-cube signals have an advantage here: they are proprietary and not subject to publication-informed arbitrage. But the 26% out-of-sample decay is a data-mining effect that applies everywhere—which is why Deflated Sharpe Ratio (Chapter 11) is non-negotiable.

5.7 Implementation Notes

Practical points for your codebase:

1. **Standardize features first.** Ridge penalizes $\|\beta\|_2^2$, which is scale-dependent. If feature 1 is in basis points and feature 2 is a ratio, the penalty is asymmetric. Always standardize to zero mean and unit variance before fitting.
2. **sklearn API:**
 - `Ridge(alpha=lambda)`: single λ .
 - `RidgeCV(alphas=[...])`: built-in GCV for λ selection.
 - `Lasso`, `LassoCV`, `ElasticNet`, `ElasticNetCV`: analogous.

Note: `sklearn` uses `alpha` where this chapter uses λ .

3. **Intercept handling.** By default, `sklearn` fits an unpenalized intercept (mean of $\mathbf{y}$). This is correct—you don’t want to shrink the intercept toward zero.
4. **λ grid.** Start with `np.logspace(-4, 4, 50)`. If the optimal λ hits the boundary, extend the grid.
5. **For your project** (`src/validation/baseline.py`): wrap `RidgeCV` with purged K -fold CV from Chapter 11, not `sklearn`’s default GCV (which assumes i.i.d. data).

5.8 Summary

Concept	Key Takeaway
Ridge (L_2)	Closed-form: $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$. Shrinks all coefficients, never zeros them out. Always exists a λ that beats OLS in MSE (Hoerl–Kennard, 1970).
Lasso (L_1)	Produces sparse models ($\beta_j = 0$). No closed form. Arbitrarily picks one of correlated features.
Elastic net	Combines $L_1 + L_2$. Grouping effect handles correlated features. Two hyperparameters (λ, α).
Ridge on PCs	4 PCs suffice for SDF estimation; ridge on PCs matches nonlinear ML (Kozak–Nagel–Santosh, 2020).
λ selection	GCV for ridge (LOO shortcut), K -fold CV for lasso/elastic net. Use purged CV for time series.
Baseline role	Ridge is the bar GBM must clear. Three scenarios: genuine nonlinear signal, linear signal dressed up, or overfitting. Only scenario 1 justifies nonlinear models.
Low-SNR reality	Monthly OOS $R^2 \sim 0.1$ – 0.4% for stock returns (Gu–Kelly–Xiu, 2020). Published signals decay 58% post-publication (McLean–Pontiff, 2016).

What’s next: Chapter 6 covers tree-based methods (random forests, gradient boosting, LightGBM)—the models that must beat the ridge baseline you’ve just learned to build.

Chapter 6

Tree-Based Methods for Finance

Phase 2: Your Primary Model

LightGBM is your primary model in Phase 2. This chapter teaches the algorithm, the hyperparameter choices specific to financial data, and the Gu–Kelly–Xiu (2020) evidence that tree methods are competitive with—and often beat—deep learning for tabular financial data. The key insight: aggressive regularization (low depth, high `min_child_samples`) is critical in finance’s low signal-to-noise regime. Every LightGBM result you report must appear alongside the ridge baseline from Chapter 5.

6.1 Decision Trees

Information Gain and Impurity Measures

A *splitting criterion* measures how “mixed” a node is. For regression, the standard measure is variance (or equivalently, MSE). For classification, the two workhorses are:

- **Gini impurity:** $G = \sum_{k=1}^K \hat{p}_k(1 - \hat{p}_k)$, where $\hat{p}_k$ is the fraction of class k in the node. Gini is zero when all samples belong to one class (pure node) and maximized when classes are equally represented.
- **Entropy:** $H = -\sum_{k=1}^K \hat{p}_k \log_2 \hat{p}_k$. Same idea, different scale.

Information gain from a split is the parent impurity minus the weighted average of child impurities. The algorithm picks the split that maximizes this gain. In practice, Gini and entropy produce nearly identical trees; the choice rarely matters.

6.1.1 CART: Recursive Binary Splitting

Classification and Regression Tree (CART)

A decision tree partitions the feature space $\mathbb{R}^p$ into J non-overlapping rectangular regions $R_1, R_2, \dots, R_J$ by a sequence of binary splits. For regression, the prediction in region R_j is the mean of the training responses in that region:

$$\hat{f}(\mathbf{x}) = \sum_{j=1}^J c_j \cdot \mathbf{1}(\mathbf{x} \in R_j), \quad c_j = \frac{1}{|R_j|} \sum_{i: \mathbf{x}_i \in R_j} y_i \quad (6.1)$$

where c_j is the leaf value (the average target in leaf j) and $\mathbf{1}(\mathbf{x} \in R_j)$ is the indicator function.

The tree is built top-down by **greedy recursive binary splitting**. At each node, the algorithm considers every feature $j \in \{1, \dots, p\}$ and every possible split point s , and selects the pair (j, s) that minimizes the sum of squared residuals in the two child nodes:

$$\min_{j,s} \left[\sum_{i: x_{ij} \leq s} (y_i - c_L)^2 + \sum_{i: x_{ij} > s} (y_i - c_R)^2 \right] \quad (6.2)$$

where c_L and c_R are the means of the left and right child nodes. This process repeats recursively until a stopping criterion is met (e.g., minimum samples per leaf, maximum depth, or minimum impurity decrease).

6.1.2 Why Single Trees Overfit

A fully grown tree (no stopping criterion) creates one leaf per training observation—it memorizes the training data perfectly. The training MSE is zero, but the test MSE is terrible. The fundamental problem: a single tree is a high-variance estimator. Small perturbations in the training data produce completely different tree structures.

This is especially catastrophic in finance. With monthly stock returns, the signal-to-noise ratio is roughly 0.05 (the true signal explains $\sim 0.25\%$ of return variation; see the Gu–Kelly–Xiu numbers in Section 6.8). A single tree will partition on noise, find spurious splits, and produce worthless out-of-sample predictions. No one uses single decision trees for return prediction.

6.2 Random Forests

Bootstrap Aggregation (Bagging)

Given a training set of n observations, a *bootstrap sample* draws n observations *with replacement*. On average, each bootstrap sample contains $\sim 63.2\%$ of the unique training observations (the rest are duplicates). **Bagging** fits a model to each of B bootstrap samples and averages the predictions: $\hat{f}_{\text{bag}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(\mathbf{x})$. Averaging reduces variance without increasing bias—the variance of the mean of B i.i.d. estimates is σ^2/B . The catch: if individual trees are correlated (because they all split on the same dominant feature), the variance reduction is limited.

Random forests (Breiman, 2001) combine bagging with a simple but powerful modification: at each split, only a random subset of m features (out of p total) is considered as split candidates.

Random Feature Subsets Decorrelate Trees

If one feature is very strong, every bagged tree will split on it first, making the trees highly correlated. Averaging correlated estimators provides limited variance reduction. By restricting each split to a random $m \ll p$ features, the forest forces trees to explore different parts of the feature space, producing *decorrelated* trees. The variance of the forest average is:

$$\text{Var}(\hat{f}_{\text{RF}}) = \rho \sigma^2 + \frac{1 - \rho}{B} \sigma^2 \quad (6.3)$$

where ρ is the average pairwise correlation between trees and σ^2 is the variance of a single tree. As $B \rightarrow \infty$, the second term vanishes but the first does not—the irreducible variance floor is $\rho \sigma^2$. Smaller m reduces ρ (more decorrelated trees) at the cost of slightly higher individual-tree bias. The standard default is $m = \lfloor p/3 \rfloor$ for regression and $m = \lfloor \sqrt{p} \rfloor$ for classification.

6.2.1 Out-of-Bag Error

Because each bootstrap sample omits $\sim 36.8\%$ of observations, each tree has a built-in holdout set. The **out-of-bag (OOB) error** evaluates each observation using only the trees for which it was *not* in the bootstrap sample:

$$\hat{f}_{\text{OOB}}(\mathbf{x}_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} \hat{f}_b(\mathbf{x}_i) \quad (6.4)$$

where $\mathcal{B}_i = \{b : \text{observation } i \text{ not in bootstrap sample } b\}$. OOB error is approximately equivalent to leave-one-out cross-validation and comes for free—no separate validation split needed.

OOB Is Not Sufficient for Time Series

OOB error assumes observations are exchangeable (i.i.d.). In financial time series, this assumption fails: if observation $t+1$ is in a tree's training set and observation t is OOB, the tree has seen the future. For your project, use purged K -fold CV (Chapter 11) instead of OOB for model selection. OOB is still useful as a quick sanity check during development, but never as your primary validation metric.

6.2.2 Why Forests Are Better—but Still Limited

Random forests reduce variance dramatically compared to single trees. They are robust to hyperparameters (forests of 500+ trees with default m generally work well) and handle mixed feature types, missing values, and interactions naturally.

But forests have two limitations for financial prediction:

1. **No sequential correction:** each tree is fit independently. If 100 trees all make the same systematic error, the 101st tree will too. There is no mechanism to focus on the ensemble's mistakes.
2. **Equal weighting:** each tree contributes equally to the average, regardless of its accuracy.

Gradient boosting addresses both limitations.

6.3 Gradient Boosting

Gradient Descent in Parameter Space

Standard gradient descent updates parameters in the direction of the negative gradient: $\boldsymbol{\theta}_{m+1} = \boldsymbol{\theta}_m - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_m)$. Gradient boosting applies the same idea, but in *function space*: instead of updating parameters, it adds a new function (a small tree) that points in the direction of the steepest descent of the loss.

6.3.1 The Algorithm

Gradient boosting builds an additive ensemble sequentially. Starting from an initial constant prediction $F_0(\mathbf{x}) = \bar{y}$, each iteration m fits a small tree $h_m(\mathbf{x})$ to the *negative gradient* of the loss function evaluated at the current ensemble's predictions:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \quad (6.5)$$

Term by term:

- $F_{m-1}(\mathbf{x})$: the ensemble’s prediction after $m-1$ trees. This is the running “best guess.”
- $h_m(\mathbf{x})$: a shallow decision tree (typically depth 3–6) fit to the *pseudo-residuals* $r_{im} = -\frac{\partial \mathcal{L}(y_i, F_{m-1}(\mathbf{x}_i))}{\partial F_{m-1}(\mathbf{x}_i)}$. For squared-error loss, the pseudo-residuals are simply $r_{im} = y_i - F_{m-1}(\mathbf{x}_i)$ —the ordinary residuals.
- $\eta \in (0, 1]$: the **learning rate** (or shrinkage parameter). Smaller η means each tree contributes less, requiring more trees but producing a smoother, more regularized ensemble. Typical values: $\eta \in [0.01, 0.1]$.

Boosting as Iterative Refinement

Think of boosting as iterative refinement. The first tree captures the biggest, most obvious patterns in the data. The second tree looks at what the first tree got wrong (the residuals) and corrects those errors. The third tree corrects what the first two together still get wrong. Each subsequent tree focuses on progressively smaller, harder-to-find patterns. The learning rate η controls how aggressively each correction is applied. A small η (say 0.01) means each tree barely nudges the prediction—you need thousands of trees, but the ensemble generalizes better because no single tree dominates. A large η (say 1.0) means each tree makes a full-strength correction—fast convergence but high risk of overfitting. In finance, always err toward smaller η .

6.3.2 Why Boosting Beats Bagging

The key difference from random forests:

- **Forests**: each tree is fit independently on a random subsample. Trees don’t communicate. The ensemble reduces variance by averaging.
- **Boosting**: each tree is fit *sequentially* on the residuals of the current ensemble. Trees explicitly correct each other’s errors. The ensemble reduces *both* bias and variance.

This sequential error correction is why gradient boosting typically outperforms random forests for structured/tabular data. Gu, Kelly, and Xiu (2020) found that gradient-boosted trees (GBRT) achieved a monthly OOS R^2 of 0.34% on individual stock returns, substantially above random forests.

6.4 XGBoost

XGBoost (Chen and Guestrin, 2016) is gradient boosting with three key enhancements: a regularized objective, a second-order Taylor expansion for faster optimization, and systems-level engineering for speed.

6.4.1 Regularized Objective

XGBoost adds explicit regularization to the tree complexity. The objective at iteration m is:

$$\mathcal{L}^{(m)} = \sum_{i=1}^n \ell(y_i, F_{m-1}(\mathbf{x}_i) + h_m(\mathbf{x}_i)) + \Omega(h_m) \quad (6.6)$$

where the complexity penalty on tree h_m is:

$$\Omega(h_m) = \gamma J + \frac{1}{2} \lambda \sum_{j=1}^J w_j^2 \quad (6.7)$$

Term by term:

- $\ell(y_i, \hat{y}_i)$: the per-sample loss (e.g., squared error, log-loss).
- $\Omega(h_m)$: the regularization term penalizing tree complexity.
- γ : penalty per additional leaf. Larger γ prunes the tree more aggressively. Acts like a minimum-gain threshold: a split must improve the loss by at least γ to be accepted.
- J : the number of leaves in tree h_m .
- λ : L_2 penalty on the leaf weights w_j . Same idea as ridge regression (Chapter 5), but applied to leaf values rather than regression coefficients.
- w_j : the output value (prediction) in leaf j .

6.4.2 Second-Order Approximation and Optimal Leaf Weights

XGBoost applies a second-order Taylor expansion to the loss around the current prediction $F_{m-1}(\mathbf{x}_i)$:

$$\mathcal{L}^{(m)} \approx \sum_{i=1}^n \left[g_i h_m(\mathbf{x}_i) + \frac{1}{2} h_i h_m(\mathbf{x}_i)^2 \right] + \Omega(h_m) + \text{const.} \quad (6.8)$$

where $g_i = \partial \ell / \partial F_{m-1}(\mathbf{x}_i)$ is the first-order gradient and $h_i = \partial^2 \ell / \partial F_{m-1}(\mathbf{x}_i)^2$ is the second-order gradient (Hessian) of the loss with respect to the current prediction.

For a given tree structure, the optimal leaf weight for leaf j is:

$$w_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (6.9)$$

where I_j is the set of training indices assigned to leaf j . The denominator $\sum h_i + \lambda$ is the analog of $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})$ from ridge—the Hessians measure curvature and λ adds regularization.

6.4.3 Split Gain

XGBoost evaluates candidate splits using the gain formula:

$$\text{Gain} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I_P} g_i)^2}{\sum_{i \in I_P} h_i + \lambda} \right] - \gamma \quad (6.10)$$

Term by term:

- The first two fractions: the objective value of the left and right children.
- The third fraction: the objective value of the parent (no split).
- $-\gamma$: the cost of adding one more leaf. If the gain from splitting is less than γ , the split is rejected—built-in pruning.

6.4.4 Growth Strategy: Level-Wise

XGBoost grows trees **level-wise** (breadth-first): it splits all nodes at the current depth before moving to the next depth level. This produces balanced trees—every branch has the same depth. Level-wise growth is safe (it doesn't chase isolated high-gain nodes) but can be wasteful: it may split nodes that contribute little, spending capacity on unimportant regions of the feature space.

6.5 LightGBM

LightGBM (Ke et al., 2017) is an alternative gradient boosting implementation designed for speed and memory efficiency. It introduces four key innovations over XGBoost: histogram-based splitting, leaf-wise growth, Gradient-based One-Side Sampling (GOSS), and Exclusive Feature Bundling (EFB).

6.5.1 Histogram-Based Splitting

Standard gradient boosting evaluates every unique feature value as a candidate split point. With n observations and p features, this requires $O(n \cdot p)$ operations per level—expensive when n is large.

LightGBM bins each continuous feature into k discrete bins (default $k = 255$) using equal-frequency histograms. Split finding then scans k bins instead of n unique values:

- **Complexity:** $O(k \cdot p)$ per level instead of $O(n \cdot p)$. Since $k = 255 \ll n$, this is much faster.
- **Memory:** storing bin indices (1 byte per feature per sample) instead of floating-point values (4–8 bytes) drastically reduces memory footprint.
- **Accuracy trade-off:** binning introduces a small discretization error, but Ke et al. (2017) show this has negligible effect on model quality. The regularization effect of binning can even help prevent overfitting.

6.5.2 Leaf-Wise Growth

Leaf-Wise vs. Level-Wise Growth

XGBoost grows trees **level-wise**: split all nodes at the current depth, then proceed to the next depth. This produces balanced trees but wastes splits on low-gain nodes.

LightGBM grows trees **leaf-wise**: at each step, find the leaf with the highest gain *across the entire tree* and split it. This concentrates the tree's limited capacity on the most informative regions of the feature space.

Leaf-wise growth typically produces lower training loss with the same number of leaves. The risk: it can create very deep, unbalanced trees that overfit. This is why `max_depth` and `num_leaves` are critical hyperparameters for LightGBM—you must constrain depth explicitly.

6.5.3 GOSS: Gradient-Based One-Side Sampling

Gradient-based One-Side Sampling exploits a key observation: samples with large gradients (i.e., large residuals) are the ones the model gets most wrong and therefore carry the most information about which splits will reduce loss.

The GOSS algorithm:

1. Sort all samples by the absolute value of their gradient $|g_i|$.
2. Keep the top $a \times 100\%$ of high-gradient samples (default: $a = 0.2$, keep 20%).
3. Randomly sample $b \times 100\%$ of the remaining low-gradient samples (default: $b = 0.1$).
4. Up-weight the low-gradient subsample by a factor of $(1 - a)/b$ to maintain the correct gradient distribution.

This reduces the effective sample size for split finding to $(a + b) \times n$ while preserving the gradient distribution. Ke et al. (2017) show that GOSS achieves almost the same split quality as full-data scanning with a fraction of the computation.

6.5.4 EFB: Exclusive Feature Bundling

Many real-world feature sets contain mutually exclusive features—features that rarely take nonzero values simultaneously (e.g., one-hot encoded categoricals, sparse indicator features). EFB identifies such features and bundles them into a single composite feature, reducing the effective number of features.

The algorithm:

1. Build a conflict graph: two features conflict if they frequently take nonzero values for the same sample.
2. Greedily bundle features with few conflicts (graph coloring heuristic).
3. Merge bundled features by offsetting their bin ranges so no information is lost.

Finding the optimal bundling is NP-hard, but the greedy approximation works well in practice. EFB is most impactful on high-dimensional sparse datasets. For your project’s 10–20 risk-cube features, EFB matters less—but it is part of what makes LightGBM fast by default.

6.5.5 Speed: LightGBM vs. XGBoost

Ke et al. (2017) report that LightGBM achieves up to $20\times$ speedup over XGBoost on large datasets while reaching the same accuracy. The speedup comes from the combination of histogram binning ($O(k)$ vs. $O(n)$ per feature), leaf-wise growth (fewer wasted splits), and GOSS (fewer samples scanned).

For your project, the speed difference is less dramatic ($\sim 1,250$ daily observations is tiny by ML standards), but LightGBM remains the preferred choice because:

1. The Python API (`lightgbm`) is mature and well-documented.
2. Built-in handling of categorical features (no need for one-hot encoding).
3. Native integration with SHAP for feature importance (Chapter 7).
4. It is the industry default for tabular data at quant desks and Kaggle competitions alike.

6.6 LightGBM vs. XGBoost: Comparison

Feature	XGBoost	LightGBM
Growth strategy	Level-wise (balanced)	Leaf-wise (greedy, deeper)
Split finding	Exact or approximate (quantile sketch)	Histogram-based ($k=255$ bins)
Data sampling	Row subsampling	GOSS (gradient-based)
Feature reduction	Column subsampling	Column subsampling + EFB
Regularization	$L_1 + L_2$ on leaf weights, γ pruning	Same, plus <code>min_child_samples</code>
Categorical features	Requires encoding	Native handling
Speed (large n)	Slower	Up to $20\times$ faster
Overfitting risk	Lower (balanced trees)	Higher (must tune <code>max_depth</code>)
Default choice	Mature, well-tested	Industry standard for tabular data

Bottom line: For your project, use LightGBM as the primary model. XGBoost is a fine alternative if you encounter LightGBM-specific issues (e.g., package installation problems on the GS compute environment). The two produce similar accuracy when tuned well; the choice is primarily about speed and API convenience.

6.7 Hyperparameters for Financial Data

Never Use Default LightGBM Parameters on Financial Data

LightGBM's defaults are tuned for Kaggle-style competitions with high signal-to-noise ratios and 10^5 – 10^7 samples. Financial return prediction has $\text{SNR} \sim 0.05$ and $\sim 10^3$ samples. Default parameters (`num_leaves=31`, `max_depth=-1`, `min_child_samples=20`) will overfit catastrophically: you will see excellent in-sample R^2 and negative out-of-sample R^2 . Always start from the conservative grid below and tune from there.

The table below gives recommended starting ranges for financial return prediction:

Parameter	Finance range	Default	Why
<code>max_depth</code>	3–5	–1 (no limit)	Shallow trees = less overfitting. Depth 4 allows up to 3-way interactions.
<code>num_leaves</code>	8–31	31	Should be $\leq 2^{\text{max_depth}}$.
<code>min_child_samples</code>	50–200	20	Forces each leaf to contain enough samples for statistical reliability. With $n = 1,250$, a leaf of 20 is 1.6% of data—noise.
<code>learning_rate</code>	0.01–0.05	0.1	Slow learning + early stopping. Lower η with more trees.
<code>n_estimators</code>	500–5000	100	Paired with low <code>learning_rate</code> . Use early stopping on validation loss.
<code>subsample</code>	0.6–0.8	1.0	Row subsampling (per tree). Reduces variance.
<code>colsample_bytree</code>	0.6–0.8	1.0	Feature subsampling (per tree). Decorrelates trees, like random forests.
<code>reg_alpha</code> (L_1)	0–0.1	0	Mild L_1 on leaf weights. Usually less important than depth/samples.
<code>reg_lambda</code> (L_2)	1–10	0	L_2 on leaf weights. Shrinks leaf predictions toward zero.

The Core Principle: Constrain Capacity, Not Just Trees

In finance, regularization must be multi-layered:

1. **Small trees:** `max_depth` 3–5 limits the complexity of each tree.
2. **Slow learning:** `learning_rate` 0.01–0.05 means each tree barely nudges the ensemble.
3. **Subsampling:** `subsample` and `colsample_bytree` 0.6–0.8 inject randomness.
4. **Leaf regularization:** `min_child_samples` 50–200 prevents tiny, noisy leaves.
5. **Early stopping:** monitor validation loss, stop when it plateaus or increases.

No single knob is sufficient. You need all five simultaneously because the signal is so weak that overfitting can occur through any ungarded channel.

Worked Example:

Tuning LightGBM for VaR Utilization Signals Suppose you have $n = 1,250$ daily observations and $p = 15$ risk-cube features. You want to predict next-day VIX innovations using VaR utilization and factor concentration features.

Step 1: Set a conservative base.

```
params = {
    'max_depth': 4,
    'num_leaves': 15,
    'min_child_samples': 100,
    'learning_rate': 0.02,
    'n_estimators': 3000,
    'subsample': 0.7,
    'colsample_bytree': 0.7,
    'reg_lambda': 5.0,
    'early_stopping_rounds': 50,
}
```

Step 2: Early stopping. Use the purged validation fold (Chapter 11) as the early-stopping monitor. Training will stop when validation MSE has not improved for 50 rounds. With `learning_rate=0.02`, you might stop at ~ 800 trees out of the 3,000 maximum.

Step 3: Compare to ridge. If this LightGBM's purged-CV R^2 does not exceed the ridge baseline from Chapter 5 on the same features, reduce `max_depth` to 3 or increase `min_child_samples` to 150. If it still doesn't beat ridge, the signal may be approximately linear—and that is a valid finding.

6.8 The GKX Horse Race

Gu, Kelly, and Xiu (2020, RFS) ran the most comprehensive ML horse race in empirical asset pricing: 30,000 individual US stocks, 60 years (1957–2016), 94 stock characteristics plus 8 macro variables plus 74 industry dummies (~ 900 predictors after interactions), with monthly return prediction evaluated on 30 years of out-of-sample data.

Gu–Kelly–Xiu (2020): The ML Horse Race

Monthly out-of-sample R^2 for individual stock return prediction (value-weighted):

Model	Monthly OOS R^2
OLS (all 900+ predictors)	< 0 (negative)
Elastic net (ENet+H)	0.11%
PLS	0.27%
PCR	0.26%
Random forest	0.28%
Gradient-boosted trees (GBRT)	0.34%
Neural network (NN3, 3 hidden layers)	0.40%

Portfolio-level results: the NN3 long-short decile portfolio earned an annualized Sharpe ratio of 1.35 (value-weighted). Trees and neural nets achieve their edge through nonlinear predictor interactions that linear methods miss.

6.8.1 What the Numbers Mean

1. **OLS collapses:** with 900+ predictors and $\sim 30,000$ stocks/month, OLS massively overfits. The OOS R^2 is *negative*—worse than predicting the mean. This is the curse of

- dimensionality in action.
2. **Regularization rescues linear models:** elastic net shrinks the 900+ coefficients, pulling OOS R^2 to 0.11%. PLS and PCR do even better (0.26–0.27%) by projecting onto a few linear combinations.
 3. **Trees beat linear methods:** GBRT at 0.34% is $3\times$ elastic net's 0.11%. The gain comes from capturing nonlinear interactions between characteristics that linear methods miss.
 4. **Deep learning offers a modest further gain:** NN3's 0.40% is $\sim 18\%$ above GBRT's 0.34%. The margin is real but not enormous—and neural networks are harder to tune, interpret, and deploy.
 5. **Tiny R^2 , real money:** 0.40% monthly R^2 means the model explains less than half a percent of return variation. Yet it produces a long-short Sharpe ratio of 1.35—excellent by asset management standards. In finance, tiny predictability translates to economically meaningful portfolio performance.

Why Trees Are Your Default

For your project, GBRT (via LightGBM) is the right default model for three reasons:

1. **Competitive accuracy:** GBRT's 0.34% monthly R^2 is close to the best neural network (0.40%). On your smaller, lower-dimensional feature set (10–20 risk-cube features vs. GKK's 900+), the gap will likely shrink further.
2. **Interpretability:** LightGBM integrates natively with SHAP (Chapter 7), giving you feature importance that you can present to the desk. Neural networks require more work to interpret.
3. **Robustness:** trees handle mixed feature types, missing data, and feature interactions without the extensive preprocessing that neural networks require. With $\sim 1,250$ observations, you cannot afford the hyperparameter search that deep learning demands.

If LightGBM does not beat ridge, the signal is linear (a valid finding). Only if both ridge and LightGBM produce $IC > 0$ should you consider whether neural networks add value—and even then, the GKK evidence suggests the incremental gain is modest.

6.9 When Trees Fail

Trees are not a universal solution. They fail in specific, predictable ways that matter for financial applications.

6.9.1 Extrapolation

Trees cannot extrapolate beyond the range of training data. A tree's prediction is always a weighted average of training-set target values—it can only output values it has seen. If VaR utilization hits 98% but the training data never exceeded 85%, the tree will predict the same value as for 85%.

This matters in finance because extreme events are precisely when you most need accurate predictions. Linear models (including ridge) can extrapolate along the learned coefficient direction, which is at least directionally correct. Trees are capped at the training-set boundary.

Mitigation: include recent extreme events in the training window when possible, and supplement tree predictions with linear models for tail scenarios.

6.9.2 Non-Stationarity and Regime Changes

Trees learn fixed split thresholds. If the split “VaR utilization > 75%” was informative in 2019–2023, the tree will apply it forever. But if the risk limit structure changes in 2024 (e.g., the desk is restructured), 75% may no longer be the relevant threshold.

Mitigation: rolling-window retraining with an expanding or sliding window (see backtesting in Chapter 12). Re-estimate the model every month or quarter using only recent data. Regime conditioning (Chapter 13) can further help by allowing the model to adapt to changing market environments.

6.9.3 Small Samples

With $n = 1,250$ daily observations and depth-4 trees, the deepest leaves partition the data into at most $2^4 = 16$ bins. If `min_child_samples=100`, you have at most $1,250/100 = 12$ effective leaves—the tree is heavily constrained. This is by design (overfitting prevention), but it limits the complexity of interactions the tree can capture.

Mitigation: panel structure (Chapter 8). If you pool multiple asset classes, your effective sample size grows. With 5 asset classes $\times$ 1,250 days = 6,250 observations, you can afford slightly deeper trees or lower `min_child_samples`—but always validate with purged CV.

6.9.4 Comparison: Where Trees vs. Linear Models Win

Scenario	Trees win	Linear models win
Feature interactions	Trees capture them automatically	Must engineer interactions by hand
Nonlinear effects	Natural (via recursive splits)	Must add polynomial terms
Extrapolation	Fail (bounded by training range)	Extrapolate along coefficients
Small n , low SNR	Overfit risk; need heavy tuning	More stable (fewer parameters)
Interpretability	SHAP, but complex	Coefficients are directly readable
Missing data	Handle natively	Require imputation

6.10 Summary

Concept	Key Takeaway
Decision trees (CART)	Recursive binary splitting on features. Single trees overfit catastrophically in low-SNR regimes. Never use alone.
Random forests	Bagging + random feature subsets decorrelate trees, reducing variance. OOB error is free but invalid for time series.
Gradient boosting	Sequential: each tree corrects the ensemble's residuals. Reduces both bias and variance. Learning rate η controls regularization.
XGBoost	Level-wise growth, built-in L_1/L_2 on leaf weights, second-order Taylor expansion for split gain.
LightGBM	Leaf-wise growth, histogram binning ($k=255$), GOSS (gradient-based sampling), EFB (feature bundling). Up to $20\times$ faster than XGBoost.
Financial hyperparameters	<code>max_depth</code> 3–5, <code>min_child_samples</code> 50–200, <code>subsample</code> 0.6–0.8, <code>learning_rate</code> 0.01–0.05. Never use defaults.
GKX horse race	GBRT $R^2 = 0.34\%$, best NN $R^2 = 0.40\%$ monthly OOS (Gu–Kelly–Xiu, 2020). Trees are the right default for tabular financial data.
When trees fail	Cannot extrapolate, brittle to regime changes, need rolling retraining. Supplement with linear models for tail scenarios.

What's next: Chapter 7 covers model interpretation—SHAP values, feature importance stability (MDA across folds), and how to present LightGBM results to a desk that thinks in terms of economic intuition, not tree splits.

Chapter 7

Model Interpretation

Phase 2–5: Interpretation Is Not Optional

SHAP analysis appears throughout your project: in Phase 2 for identifying which risk-cube feature families drive predictions, in Phase 3 for the checkpoint (is feature importance stable across CV folds?), and in Phase 5 for the final presentation (SHAP waterfalls convert “trust me” into “here is what drives this prediction”). MDI/MDA stability across CV folds is your test for whether importance rankings are reliable—if the top feature flips between folds, you are fitting noise. Every LightGBM result you present must include SHAP output alongside the ridge baseline from Chapter 5.

7.1 Why Interpretation Matters in Finance

Model interpretation in quantitative finance is not a nice-to-have—it is a gating requirement for deployment. Four reasons, in order of career relevance:

1. **Regulatory and compliance scrutiny.** Financial regulators (Fed SR 11-7, OCC 2011-12) require model risk management for any model that influences trading or risk limits. A black box that cannot explain *why* it issued a signal will not survive compliance review. Phase 4A of your project initiates compliance review at weeks 16–17; you need interpretable output ready before that.
2. **Sponsor trust.** Your desk sponsor thinks in terms of economic mechanisms: “VaR utilization spikes precede forced selling, which causes drawdowns.” If you show a SHAP waterfall decomposing a prediction into the contribution of VaR utilization (+12 bps), factor concentration (+8 bps), and scenario P&L dispersion (−3 bps), the sponsor can evaluate whether the model’s reasoning matches the theory from Chapter 2. If you show only an aggregate R^2 , the sponsor has no way to assess whether the model learned something real or memorized noise.
3. **Debugging and feature auditing.** SHAP dependence plots reveal when a model exploits spurious features. If the top SHAP feature is “day of week” rather than VaR utilization, you have a data leak or a calendar artifact—not alpha. Interpretation catches these problems before the OOS test wastes your one-shot holdout (Chapter 11).
4. **Presentation.** At the final desk presentation (Phase 5), the single most convincing chart is the SHAP waterfall plot. Chapter 14 covers presentation strategy in detail; this chapter gives you the tools to produce the charts.

7.2 SHAP from Game Theory

Cooperative Game Theory Basics

A *cooperative game* is a pair (N, v) , where $N = \{1, 2, \dots, p\}$ is a set of players and $v : 2^N \rightarrow \mathbb{R}$ is a *value function* mapping every coalition (subset) $S \subseteq N$ to a real-valued payoff $v(S)$, with $v(\emptyset) = 0$. The central question: given that the grand coalition N achieves payoff $v(N)$, how should the total payoff be divided *fairly* among the p players? A player's *marginal contribution* to coalition S is $v(S \cup \{i\}) - v(S)$ —the additional payoff player i brings when joining S . The challenge is that player i 's marginal contribution depends on which coalition i joins, so there is no single number that captures i 's worth without averaging over coalitions.

7.2.1 Shapley Values

Lloyd Shapley (1953) proved that there is exactly one allocation rule satisfying four fairness axioms. That allocation is the *Shapley value*:

Shapley Value

For a cooperative game (N, v) with $p = |N|$ players, the Shapley value of player i is:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{i\}) - v(S)] \quad (7.1)$$

Term by term:

- $S \subseteq N \setminus \{i\}$: every possible coalition that does not include player i . There are 2^{p-1} such coalitions.
- $v(S \cup \{i\}) - v(S)$: the marginal contribution of player i to coalition S .
- $\frac{|S|! (p - |S| - 1)!}{p!}$: the weight, equal to the fraction of all $p!$ orderings in which the $|S|$ players in S arrive before i and the remaining $p - |S| - 1$ arrive after. This ensures every ordering of players is counted equally.

The Shapley value is the unique allocation satisfying four axioms:

The Four Shapley Axioms

1. **Efficiency**: contributions sum to the total payoff: $\sum_{i=1}^p \phi_i(v) = v(N) - v(\emptyset)$. No value is created or lost in the allocation.
 2. **Symmetry**: if players i and j contribute identically to every coalition ($v(S \cup \{i\}) = v(S \cup \{j\})$ for all S not containing i or j), then $\phi_i = \phi_j$. Equal contributors receive equal credit.
 3. **Dummy**: if player i adds nothing to any coalition ($v(S \cup \{i\}) = v(S)$ for all S), then $\phi_i = 0$. Irrelevant features get zero attribution.
 4. **Additivity** (linearity): for two games v and w on the same player set, $\phi_i(v + w) = \phi_i(v) + \phi_i(w)$. This lets us decompose complex games into simpler components.
- Shapley (1953) proved that *no other allocation rule* satisfies all four simultaneously.

7.2.2 From Game Theory to ML

Lundberg and Lee (2017, NeurIPS) connected Shapley values to model interpretation by defining the game as follows:

- **Players:** the p features $\{x_1, x_2, \dots, x_p\}$.
- **Value function:** $v(S) = \mathbb{E}[f(\mathbf{x}) \mid x_S]$ —the expected model prediction when only the features in S are observed, marginalizing over the features not in S .
- **Grand coalition payoff:** $v(N) = f(\mathbf{x})$ —the full model prediction for this specific observation.
- **Empty coalition:** $v(\emptyset) = \mathbb{E}[f(\mathbf{x})]$ —the average prediction across the dataset.

The SHAP value of feature i for observation $\mathbf{x}$ is then the Shapley value ϕ_i under this game. By the efficiency axiom:

$$f(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})] + \sum_{i=1}^p \phi_i(\mathbf{x}) \quad (7.2)$$

Term by term:

- $f(\mathbf{x})$: the model's prediction for this observation.
- $\mathbb{E}[f(\mathbf{x})]$: the baseline—the average prediction across the dataset.
- $\phi_i(\mathbf{x})$: feature i 's contribution to pushing this prediction away from the baseline. Positive means feature i pushed the prediction up; negative means down.

This is the decomposition that SHAP waterfall plots visualize: starting from the baseline $\mathbb{E}[f(\mathbf{x})]$, each feature's SHAP value is a step that walks the prediction to its final value $f(\mathbf{x})$.

SHAP as Fair Credit Assignment

Imagine three analysts jointly produce a trading signal. Analyst A specializes in rates, Analyst B in credit, Analyst C in FX. The signal earns \$10M. How much credit does each analyst deserve?

You cannot just look at what each analyst produces alone—their value depends on who else is on the team. Analyst A might add \$6M when working solo but only \$2M when B is already on the team (because A and B have overlapping rates/credit expertise). The Shapley value computes A's average marginal contribution across *all* possible team compositions: A alone, A with B, A with C, A with B and C. This is exactly what SHAP does for features: it averages each feature's marginal contribution across all possible subsets of other features. No double-counting, no leftover value.

7.2.3 Computational Cost

The naive computation of Equation (7.1) requires evaluating $v(S)$ for all 2^p coalitions. With $p = 15$ risk-cube features, that is $2^{15} = 32,768$ coalition evaluations *per observation*. For $n = 1,250$ observations, this totals ~ 41 million evaluations—expensive for a generic model. This exponential cost is why Lundberg and Lee (2017) proposed model-specific approximations, the most important of which is TreeSHAP.

Worked Example:

SHAP by Enumeration for 3 Features Consider a model $f(x_1, x_2, x_3)$ predicting next-day VIX innovation, with $\mathbb{E}[f] = 0$ (the baseline). For one specific observation, $f(\mathbf{x}) = 0.12$ (predicting a 12 bps VIX increase).

We want ϕ_1 , the SHAP value of feature x_1 (VaR utilization). There are $2^{3-1} = 4$ coalitions $S \subseteq \{2, 3\}$ not containing player 1:

S	$ S $	Weight $\frac{ S !(3- S -1)!}{3!}$	$v(S \cup \{1\}) - v(S)$
$\emptyset$	0	$\frac{0!2!}{3!} = \frac{2}{6} = \frac{1}{3}$	$v(\{1\}) - v(\emptyset) = 0.08 - 0 = 0.08$
$\{2\}$	1	$\frac{1!1!}{3!} = \frac{1}{6}$	$v(\{1, 2\}) - v(\{2\}) = 0.11 - 0.03 = 0.08$
$\{3\}$	1	$\frac{1!1!}{3!} = \frac{1}{6}$	$v(\{1, 3\}) - v(\{3\}) = 0.10 - 0.02 = 0.08$
$\{2, 3\}$	2	$\frac{2!0!}{3!} = \frac{2}{6} = \frac{1}{3}$	$v(\{1, 2, 3\}) - v(\{2, 3\}) = 0.12 - 0.05 = 0.07$

$$\phi_1 = \frac{1}{3}(0.08) + \frac{1}{6}(0.08) + \frac{1}{6}(0.08) + \frac{1}{3}(0.07) = 0.0267 + 0.0133 + 0.0133 + 0.0233 = 0.077$$

Similarly, suppose $\phi_2 = 0.025$ and $\phi_3 = 0.018$. Check efficiency: $\phi_1 + \phi_2 + \phi_3 = 0.077 + 0.025 + 0.018 = 0.12 = f(\mathbf{x}) - \mathbb{E}[f]$. ✓

VaR utilization (x_1) gets the most credit: it contributed 7.7 bps of the 12 bps prediction. This is the number that appears in the SHAP waterfall.

7.3 TreeSHAP

Tree Structure Notation

A single decision tree t has L_t leaves and maximum depth D_t . Each internal node stores a feature index, a split threshold, and pointers to left/right children. Each leaf stores a prediction value w_j . An observation $\mathbf{x}$ follows a unique root-to-leaf path, landing in exactly one leaf. The key insight for TreeSHAP: the tree structure determines which features are “used” (split on) along each path, and this structure can be exploited to compute exact Shapley values efficiently.

For generic models, computing exact Shapley values costs $O(2^p)$ —exponential in the number of features. Lundberg et al. (2020, *Nature Machine Intelligence*) introduced TreeSHAP, an algorithm that exploits the recursive structure of decision trees to compute *exact* Shapley values in polynomial time.

TreeSHAP: Exact Shapley in Polynomial Time

For an ensemble of T trees, each with at most L leaves and maximum depth D , TreeSHAP computes exact SHAP values in:

$$O(T \cdot L \cdot D^2) \text{ time,} \quad O(D^2 + p) \text{ memory} \quad (7.3)$$

where p is the number of features. Compare this to the naive $O(T \cdot L \cdot 2^p)$: with $p = 15$ features, TreeSHAP is roughly $2^{15}/D^2 \approx 32,768/16 \approx 2,000\times$ faster (at depth $D = 4$). The algorithm produces **exact** Shapley values—not a sampling approximation—because it recursively pushes probability mass through tree paths, accounting for all feature subsets simultaneously.

The key algorithmic insight is the *additivity* axiom applied to tree ensembles. By additivity, the SHAP value for the full ensemble is the sum of SHAP values for each individual tree: $\phi_i^{\text{ensemble}} = \sum_{t=1}^T \phi_i^{(t)}$. So the problem reduces to computing exact SHAP values for a single tree, then summing across trees.

For a single tree, the algorithm traverses from root to leaf along the observation’s decision path. At each internal node, it tracks a set of “path fractions” that encode how the prediction would change if each subset of features were observed or unobserved. Because a tree of depth D splits

on at most D features along any root-to-leaf path, the number of relevant feature subsets is bounded by $O(2^D)$ —not $O(2^p)$. With $D = 4$ and $p = 15$, this is 16 vs. 32,768. The D^2 factor in the complexity arises from maintaining and updating these path fractions at each node.

Three properties make TreeSHAP the gold standard for LightGBM/XGBoost interpretation:

1. **Exact, not approximate.** Kernel SHAP (also from Lundberg and Lee, 2017) uses weighted regression to approximate Shapley values for any model, but introduces sampling noise. TreeSHAP produces the exact answer for tree ensembles.
2. **Fast enough for production.** With your LightGBM configuration from Chapter 6 ($T \approx 800$ trees after early stopping, $D = 4$, $L \leq 15$), computing SHAP values for all $n = 1,250$ observations takes seconds, not hours.
3. **Native integration.** The `shap` Python library (by Lundberg) calls LightGBM’s internal C++ API to extract tree structures directly, avoiding serialization overhead. The call is a single line: `shap.TreeExplainer(model).shap_values(X)`.

TreeSHAP Interventional vs. Path-Dependent

TreeSHAP has two modes: “interventional” and “tree_path_dependent” (the default). The path-dependent method is faster but can assign nonzero SHAP values to features the model never split on, because it uses the tree’s internal node-cover statistics to marginalize. The interventional method marginalizes using background data, which is theoretically cleaner but slower. For your project, use the default path-dependent mode for exploratory analysis (fast, directionally correct) and switch to interventional for final presentation charts where precision matters. Set the mode via `shap.TreeExplainer(model, feature_perturbation="interventional", data=X_background)`.

7.4 SHAP Plots

The `shap` library produces four core plot types. Each answers a different question.

7.4.1 Summary Plot (Beeswarm)

```
shap.summary_plot(shap_values, X)
```

The beeswarm plot shows, for *every observation*, the SHAP value of each feature. Each dot is one observation; the x -axis is the SHAP value (contribution to prediction), and color indicates the feature value (red = high, blue = low). Features are sorted by mean absolute SHAP value (global importance).

What to look for:

- Is VaR utilization consistently at the top? (If not, re-examine the thesis from Chapter 2.)
- Do high feature values (red dots) appear on the right (positive SHAP)? If VaR utilization is high and SHAP is positive, the model learned “high VaR utilization predicts higher VIX innovation”—consistent with intermediary theory.
- Are there features with wide SHAP spread but no clear color pattern? These may be capturing nonlinear or interaction effects.

7.4.2 Dependence Plot

```
shap.dependence_plot("var_utilization", shap_values, X)
```

The dependence plot shows one feature’s value (x -axis) against its SHAP value (y -axis), colored by the feature that most strongly interacts with it. It is the SHAP analog of a partial dependence

plot but is observation-specific (not averaged).

What to look for:

- Nonlinearity: is the relationship between VaR utilization and its SHAP value linear, or does it exhibit a threshold effect (e.g., SHAP jumps sharply above 80% utilization)? A threshold would support the Coval–Stafford (2007) fire-sale mechanism (Chapter 2).
- Interactions: if the coloring feature is factor concentration (HHI), the plot reveals whether VaR utilization’s impact is amplified when factor concentration is also high—a two-way interaction that LightGBM captures but ridge misses.

7.4.3 Waterfall Plot

```
shap.plots.waterfall(shap_values[i])
```

The waterfall shows, for a *single observation*, how each feature pushes the prediction away from the baseline $\mathbb{E}[f(\mathbf{x})]$. Starting at the baseline, features are stacked as positive (red) or negative (blue) bars, arriving at the final prediction.

The Money Chart for Phase 5

The waterfall plot is the single most effective chart in your final presentation. It converts a LightGBM prediction into a narrative: “For this date, VaR utilization pushed the predicted VIX innovation up by 8 bps, factor concentration added 5 bps, but low scenario dispersion pulled it back by 3 bps, giving a net prediction of +12 bps above the mean.” Sponsors, portfolio managers, and risk committees understand waterfall decompositions because they mirror the additive P&L attribution they already use. Pick 2–3 observations for your presentation: one where the model was right, one where it was wrong, and one during a stress event.

7.4.4 Force Plot

```
shap.plots.force(shap_values[i])
```

The force plot shows the same information as the waterfall but in a horizontal bar format. Features pushing the prediction higher appear on the left (red); features pushing it lower appear on the right (blue). The force plot is more compact than the waterfall and useful for dashboards, but less readable for presentations. Use waterfall for slides, force plots for interactive monitoring.

7.5 MDI vs. MDA

Feature Importance Before SHAP

Before SHAP existed, the standard feature importance measures for tree ensembles were MDI and MDA, both introduced by Breiman (2001). These are aggregate (global) measures—they assign a single importance score to each feature across the entire dataset, unlike SHAP which provides per-observation attributions. They are still widely reported and you should understand their strengths and weaknesses.

7.5.1 MDI: Mean Decrease in Impurity

Mean Decrease in Impurity (MDI)

For a tree ensemble, the MDI of feature j is the sum of the impurity decreases at all nodes that split on feature j , weighted by the number of samples reaching each node, averaged across all trees:

$$\text{MDI}_j = \frac{1}{T} \sum_{t=1}^T \sum_{\substack{k \in \text{nodes of tree } t \\ \text{splitting on feature } j}} p_{tk} \cdot \Delta I_{tk} \quad (7.4)$$

where T is the number of trees, p_{tk} is the fraction of samples reaching node k in tree t , and ΔI_{tk} is the impurity decrease (information gain) at that node.

MDI is computed as a byproduct of tree training—it comes for free and requires no additional data evaluation. In LightGBM, this is the default `feature_importances_` attribute (with `importance_type='split'` counting split frequency, or `importance_type='gain'` using total gain, which is closer to Equation (7.4)).

7.5.2 MDA: Mean Decrease in Accuracy (Permutation Importance)

Mean Decrease in Accuracy (MDA)

The MDA of feature j is the decrease in model accuracy when feature j 's values are randomly permuted across observations, breaking the association between x_j and the target:

$$\text{MDA}_j = \text{Score}(\mathbf{X}) - \text{Score}(\mathbf{X}_\pi^{(j)}) \quad (7.5)$$

where $\text{Score}(\mathbf{X})$ is the model's accuracy (or R^2 , or negative MSE) on the original data, $\mathbf{X}_\pi^{(j)}$ is the dataset with column j randomly permuted, and $\text{Score}(\mathbf{X}_\pi^{(j)})$ is the accuracy after permutation.

If feature j is important, permuting it destroys the signal and accuracy drops substantially; MDA_j is large. If feature j is irrelevant, permuting it changes nothing; $\text{MDA}_j \approx 0$.

MDA is model-agnostic—it works for any model, not just trees. The cost is that it requires p additional passes through the dataset (one permutation per feature), each evaluating the full model.

Practical note: always compute MDA on held-out data, not training data. In-sample MDA can be misleading because the model has already memorized the training set; permuting a feature may appear to cause a large accuracy drop even if the feature is just noise that the model overfit to. In your project, compute MDA on the purged-CV test fold for each split.

Worked Example:

MDA for VaR Utilization You have a trained LightGBM model evaluated on the test fold ($n_{\text{test}} = 250$ observations). The model achieves $R^2 = 0.018$ on the original test data (i.e., it explains 1.8% of variance in VIX innovations—tiny, but potentially economically meaningful).

To compute MDA for VaR utilization (x_1):

1. Randomly shuffle the VaR utilization column across the 250 test observations, leaving all other features intact.
2. Re-evaluate the model on the permuted data: $R_\pi^2 = 0.004$.

3. $MDA_1 = 0.018 - 0.004 = 0.014$.

Repeat for factor concentration (x_2): $R_\pi^2 = 0.011$, so $MDA_2 = 0.018 - 0.011 = 0.007$.

Repeat for scenario P&L rank (x_3): $R_\pi^2 = 0.017$, so $MDA_3 = 0.018 - 0.017 = 0.001$.

Ranking: VaR utilization (0.014) > factor concentration (0.007) > scenario P&L rank (0.001). This ranking should be consistent with the SHAP-based ranking from the beeswarm plot. If it is not, investigate whether the discrepancy comes from nonlinear interaction effects (which SHAP captures per-observation but MDA averages out).

To get confidence intervals, repeat the permutation 10–30 times with different random seeds and report the mean $\pm$ standard deviation.

7.5.3 MDI vs. MDA: Comparison

Criterion	MDI (Impurity-Based)	MDA (Permutation-Based)
Computation	Free (byproduct of training)	Requires p extra model evaluations
Bias	Biased toward high-cardinality and continuous features (Strobl et al., 2007)	Unbiased under independence
Correlated features	Splits importance across correlated features arbitrarily	Can over-estimate importance of correlated features (permuting one leaves the correlated partner intact)
Evaluation data	Uses training data (in-sample)	Should use held-out data (OOS)
Scale	Relative (sums to total impurity decrease)	Absolute (in units of the accuracy metric)
Model-agnostic?	Trees only	Any model
Recommended?	Quick sanity check only	Primary importance measure for your project

MDI Is Biased—Do Not Use as Your Primary Importance Measure

MDI systematically inflates the importance of features with more unique values. A continuous feature (e.g., VaR utilization, which can take any value in $[0, 1]$) has hundreds of possible split points, giving it more chances to appear in a split by random chance alone. A binary feature (e.g., “VaR limit breach yes/no”) has only one split point. Even if the binary feature is more predictive, MDI may rank the continuous feature higher simply because the tree had more opportunities to split on it.

Strobl et al. (2007) demonstrated this bias rigorously: in simulations with uninformative features, MDI assigned nonzero importance to continuous noise features while correctly zeroing out categorical noise features. Use MDI only as a fast sanity check during development. For any result you report—in the checkpoint memo, the research report, or the presentation—use MDA or SHAP.

7.5.4 SHAP vs. MDI/MDA: When to Use Which

All three measures answer “which features matter?” but at different levels of granularity:

- **SHAP**: local (per-observation) attributions that sum to the prediction. The richest information. Use for presentation charts, debugging individual predictions, and detecting

nonlinear effects.

- **MDA**: global importance with a clear unit (accuracy drop). Use as the primary aggregate importance measure for checkpoint reports and stability analysis. Complements SHAP because it is computed on a different principle (perturbation vs. game-theoretic allocation).
- **MDI**: global importance, free, but biased. Use only as a quick sanity check during inter-active development. Never report MDI as a final result.

In practice, you will compute all three. If they agree on the top features, you have robust evidence. If SHAP and MDA agree but MDI disagrees, the MDI bias is the likely explanation—ignore MDI. If SHAP and MDA disagree, the signal may be driven by a small number of extreme observations (high-leverage points); investigate with SHAP dependence plots.

7.6 Feature Importance Stability

A feature importance ranking that changes every time you retrain the model is not a reliable signal—it is an artifact of noise. Stability testing is how you distinguish between the two.

7.6.1 The Problem

Compute MDA on CV fold 1: VaR utilization is the top feature, factor concentration is second. Compute MDA on CV fold 2: factor concentration is top, scenario P&L dispersion is second, VaR utilization drops to fourth. Which ranking do you trust? Neither—the instability suggests the model is not reliably identifying the same signal across different slices of the data.

This matters especially for your project because $n \approx 1,250$ is small. With small samples, the particular observations in each fold have outsized influence on the tree structure, which directly affects importance rankings. Unstable importance rankings are a leading indicator of poor out-of-sample performance.

7.6.2 Method: MDA Across CV Folds

The protocol:

1. Run purged K -fold CV (Chapter 11) with $K = 5$.
2. For each fold k , train the LightGBM model on the training split and compute $\text{MDA}_j^{(k)}$ for every feature j on the test split.
3. Collect the K importance vectors: $\{\text{MDA}^{(1)}, \text{MDA}^{(2)}, \dots, \text{MDA}^{(K)}\}$.
4. Rank features within each fold. Compute Spearman rank correlation between all $\binom{K}{2}$ pairs of fold rankings.

7.6.3 Stability Metric: Rank Correlation

Feature Importance Stability

Given K fold-level importance rankings $r^{(1)}, r^{(2)}, \dots, r^{(K)}$ over p features, the *importance stability* is the average pairwise Spearman rank correlation:

$$\rho_{\text{stability}} = \frac{2}{K(K-1)} \sum_{k=1}^{K-1} \sum_{l=k+1}^K \rho_S(r^{(k)}, r^{(l)}) \quad (7.6)$$

where ρ_S denotes the Spearman rank correlation coefficient.

Interpretation:

- $\rho_{\text{stability}} > 0.8$: strong stability. The same features dominate across folds. Proceed with confidence.
- $\rho_{\text{stability}} \in [0.5, 0.8]$: moderate stability. The top 2–3 features are consistent, but mid-ranked features shuffle. Report the stable features; be cautious about the unstable ones.
- $\rho_{\text{stability}} < 0.5$: weak stability. Importance rankings are essentially random across folds. This is a red flag—the model may be fitting noise. At the Phase 3 checkpoint, this is grounds for concern about the signal.

7.6.4 Complementary Check: SHAP Stability

Apply the same protocol using mean absolute SHAP values instead of MDA:

1. For each fold k , compute SHAP values for the test split.
2. Compute mean $|\phi_j|$ per feature, yielding a SHAP-based importance ranking.
3. Calculate $\rho_{\text{stability}}$ as above.

If both MDA-based and SHAP-based stability metrics agree (both high or both low), you have a clear signal. If they disagree, investigate: the discrepancy may reveal that a feature’s importance is driven by a few extreme observations (which SHAP captures but MDA averages out).

7.6.5 What to Do When Importance Is Unstable

If $\rho_{\text{stability}} < 0.5$, do not simply report the average ranking and move on. Unstable importance indicates a structural problem that must be diagnosed:

1. **Check the ridge baseline.** If ridge coefficient rankings are stable across folds but LightGBM’s MDA rankings are not, the instability is specific to the tree model—likely because the sample is too small for the tree to reliably identify interaction effects. In this case, the linear signal exists but the nonlinear signal does not.
2. **Reduce model complexity.** Decrease `max_depth` from 4 to 3, increase `min_child_samples` from 100 to 200. Simpler trees are more stable because they have fewer degrees of freedom to fit different noise patterns in each fold.
3. **Increase effective sample size.** Pool across asset classes using the panel structure from Chapter 8. If pooling stabilizes importance, the signal exists but requires cross-asset information to estimate reliably.
4. **Accept the finding.** If none of the above works, the honest conclusion is that the feature’s importance is not distinguishable from noise at the current sample size. This is a valid finding—report it.

Phase 3 Checkpoint: Stability Criterion

At the week 13 checkpoint, one of the pass/fail criteria is “MDA stable across folds.” Concretely: if the top two features (by MDA) are the same across all 5 purged-CV folds and $\rho_{\text{stability}} > 0.6$, the importance is stable enough to proceed. If the top feature changes in every fold, the model is chasing different noise patterns in each data slice—this is the “GBM $\leq$ ridge” scenario from the design spec, and you should consider the pivot to Project 2.

7.7 Putting It Together: Interpretation Workflow

The full interpretation workflow for each signal-testing notebook in Phase 2:

1. **Train LightGBM** with the hyperparameters from Chapter 6 (depth 3–5, learning rate 0.01–0.05, purged CV).

2. **Compute SHAP values** via TreeSHAP for the test-fold observations: `shap.TreeExplainer(model).shap_values(X_test)`.
3. **Generate four plots**: summary (beeswarm), dependence (for the top 2 features), waterfall (for 2–3 representative observations).
4. **Compute MDA** on each CV fold. Calculate $\rho_{\text{stability}}$ and report it.
5. **Sanity-check against theory**: does the SHAP-implied feature ranking match the economic hypothesis? If VaR utilization and factor concentration are the top SHAP features (consistent with He–Kelly–Manela, Coval–Stafford), the model’s reasoning aligns with intermediary theory. If “day of month” ranks first, investigate a data issue.
6. **Compare to ridge coefficients**: ridge from Chapter 5 gives direct coefficient magnitudes. If the ridge coefficient ranking roughly matches the SHAP ranking, the signal is approximately linear. If they diverge, LightGBM is capturing nonlinearity or interactions that ridge misses—and the SHAP dependence plot should reveal where.

7.8 Summary

Concept	Key Takeaway
SHAP values	Per-observation, per-feature attribution grounded in Shapley’s uniqueness theorem. Satisfy efficiency, symmetry, dummy, and additivity.
TreeSHAP	Computes <i>exact</i> Shapley values for tree ensembles in $O(TLD^2)$ time (Lundberg et al., 2020). The gold standard for LightGBM interpretation.
Waterfall plot	Decomposes a single prediction into feature contributions. The most effective chart for desk presentations.
Summary (beeswarm) plot	Shows SHAP values for all observations, revealing global feature importance and the direction of effects.
MDI (impurity-based)	Free but biased toward high-cardinality features (Strobl et al., 2007). Use only as a quick sanity check.
MDA (permutation)	Model-agnostic, unbiased under independence. Your primary non-SHAP importance measure. Compute on held-out data.
Importance stability	Average pairwise Spearman rank correlation of MDA rankings across CV folds. $\rho > 0.8$: stable; $\rho < 0.5$: red flag.
Interpretation workflow	TreeSHAP → four plots → MDA stability → theory sanity check → ridge comparison. Every Phase 2 notebook follows this.

What’s next: Chapter 8 covers panel econometrics—pooling observations across asset classes to increase effective sample size, which directly affects the stability metrics introduced here.

Chapter 8

Panel Econometrics

Project Connection — Phase 4A Panel Extension

In Phase 4A you move from firm-level aggregates to asset-class-level risk outputs and test within-class vs. cross-class prediction using panel data. This chapter teaches the econometric methods you need: fixed effects, clustered standard errors, and Fama–MacBeth regressions. The panel structure is also how you increase your effective sample size—a critical concern when firm-level VaR gives only $\sim 1,250$ daily observations over five years. Stacking four asset classes doubles your observations; stacking seven (the He–Kelly–Manela setup) nearly sextuples them.

8.1 Why Panel Data

Panel Data

A **panel** (or **longitudinal**) data set tracks multiple entities $i = 1, \dots, N$ over multiple time periods $t = 1, \dots, T$. Each observation is indexed by the pair (i, t) , giving a total of up to $N \times T$ rows. When every entity is observed at every date, the panel is *balanced*; when some (i, t) pairs are missing, it is *unbalanced*.

Three data structures appear throughout empirical finance:

1. **Cross-section.** Many entities, one date. Example: returns of 500 stocks on a single day.
2. **Time series.** One entity, many dates. Example: daily VIX over five years ($T \approx 1,250$).
3. **Panel.** Many entities, many dates. Example: daily VaR utilisation for each of $N = 7$ asset classes over $T = 1,250$ trading days.

Panels give you three advantages a pure time series cannot:

- **More observations.** Stacking N entities multiplies your sample size. With $N = 7$ asset classes and $T = 1,250$ days you have up to 8,750 rows instead of 1,250.
- **Control for unobserved heterogeneity.** Each entity may have a different average level of the dependent variable for reasons you cannot measure. Fixed effects absorb these permanent differences (Section 8.2).
- **Cross-sectional variation in exposure.** At each date, some asset classes have high VaR utilisation and others have low. This within-date variation identifies the coefficient on VaR utilisation even if aggregate risk is trending over time.

Your Panel

In this project your panel has:

- **Entities** (i): asset classes—equities, rates, credit, FX, commodities, and potentially finer subdivisions.
- **Time** (t): trading days, typically $T \approx 1,000$ – $1,500$ depending on history available in SecDB.
- **Features** ($\mathbf{x}_{it}$): asset-class-specific VaR utilisation, factor concentration (HHI), scenario P&L rank, component VaR share.
- **Targets** (y_{it}): next-period return, realised volatility, drawdown indicator, or VIX innovation.

The key question is whether a risk signal that predicts returns in *one* asset class also predicts returns in *other* asset classes—the “single pricing kernel” test of He–Kelly–Manela (2017).

8.2 Fixed Effects

Dummy Variables and Degrees of Freedom

A **dummy variable** (indicator variable) equals 1 for one category and 0 otherwise. Including N dummies for N groups, plus an intercept, creates perfect collinearity; drop one dummy or suppress the intercept. Each dummy costs one degree of freedom, so adding 7 asset-class dummies to a regression with 8,750 observations reduces df by 6—negligible. With thousands of firm-level dummies, the cost is material.

Fixed Effects Model

The **entity fixed-effects** (FE) model is:

$$y_{it} = \alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it} \quad (8.1)$$

where:

- y_{it} — outcome for entity i at time t (e.g., next-day return for asset class i),
- α_i — entity-specific intercept, absorbing all *time-invariant* differences across entities,
- $\mathbf{x}_{it}$ — $K \times 1$ vector of time-varying regressors (e.g., VaR utilisation, factor HHI),
- $\boldsymbol{\beta}$ — $K \times 1$ coefficient vector, common across entities,
- ε_{it} — idiosyncratic error, assumed $\mathbb{E}[\varepsilon_{it} \mid \alpha_i, \mathbf{x}_{it}] = 0$.

The intercept α_i captures everything that differs across entities but does not change over time: equities have structurally higher average returns than FX carry; credit VaR operates at different levels than rates VaR. Fixed effects absorb these level differences so that $\boldsymbol{\beta}$ is identified purely from *within-entity* variation over time.

8.2.1 The Within Estimator

Demeaning Removes Entity Effects

To estimate $\boldsymbol{\beta}$ in the FE model, subtract each entity’s time-series mean from both sides:

$$y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'\boldsymbol{\beta} + (\varepsilon_{it} - \bar{\varepsilon}_i) \quad (8.2)$$

where $\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$ and similarly for $\bar{\mathbf{x}}_i$ and $\bar{\varepsilon}_i$. The entity effects α_i cancel. OLS on the demeaned data yields the **within estimator**, also called the **LSDV** (Least Squares

Dummy Variable) estimator.

You can equivalently include N dummy variables—one per entity—and run OLS on the full model. The coefficient $\hat{\beta}$ is identical; demeaning is just computationally cheaper when N is large (Wooldridge, 2010, Chapter 10).

Between vs. Within Variation

Panel data contains two sources of variation:

- **Between variation:** differences *across* entities in their time-series averages. Example: equities have higher average VaR utilisation than FX.
- **Within variation:** fluctuations *over time within* a single entity. Example: equity VaR utilisation rises from 60% to 85% during a stress event, then falls back.

The FE estimator discards between variation entirely—it uses only within variation. This is a feature, not a bug: between variation is contaminated by omitted time-invariant confounders. The cost is that you cannot estimate the effect of any time-invariant regressor (e.g., “is this asset class a derivative?”). If you need those effects, consider Mundlak’s (1978) compromise: include entity means $\bar{\mathbf{x}}_i$ as additional regressors in a RE model, which is algebraically equivalent to FE for the time-varying coefficients while also estimating between-entity effects.

Worked Example:

Fixed Effects with 4 Asset Classes Suppose you have $N = 4$ asset classes (equities, rates, credit, FX) observed over $T = 252$ trading days, giving $4 \times 252 = 1,008$ panel observations. You regress next-day return y_{it} on VaR utilisation x_{it} with entity fixed effects:

$$y_{it} = \alpha_i + \beta \cdot x_{it} + \varepsilon_{it}$$

Step 1: Compute entity means. For each asset class i , compute $\bar{y}_i$ and $\bar{x}_i$ across its 252 days.

Step 2: Demean. Create $\tilde{y}_{it} = y_{it} - \bar{y}_i$ and $\tilde{x}_{it} = x_{it} - \bar{x}_i$.

Step 3: OLS on demeaned data.

$$\hat{\beta}_{\text{FE}} = \frac{\sum_{i=1}^4 \sum_{t=1}^{252} \tilde{x}_{it} \tilde{y}_{it}}{\sum_{i=1}^4 \sum_{t=1}^{252} \tilde{x}_{it}^2}$$

If $\hat{\beta}_{\text{FE}} = -0.03$ with $\text{SE} = 0.01$, the interpretation is: within a given asset class, a one-standard-deviation increase in VaR utilisation predicts a 3 bp decrease in next-day return. The negative sign is consistent with the Coval–Stafford (2007) forced-selling channel (see Chapter 2).

Degrees of freedom: $N \times T - N - K = 1,008 - 4 - 1 = 1,003$. The 4 dummies consume only 4 degrees of freedom.

8.2.2 Time Fixed Effects

You can also include **time fixed effects** δ_t :

$$y_{it} = \alpha_i + \delta_t + \mathbf{x}'_{it}\beta + \varepsilon_{it} \quad (8.3)$$

Time effects absorb market-wide shocks common to all entities on a given day—aggregate volatility spikes, macro announcements, or Fed meetings. This is a **two-way fixed effects** model. After adding time effects, β is identified only from variation that differs *across entities within the same date*, net of the entity average.

When to Use Time Fixed Effects

Include time fixed effects when you want to control for aggregate shocks that hit all asset classes simultaneously. This is appropriate when you care about the *relative* effect of VaR utilisation across asset classes, not whether VaR utilisation predicts returns in general. If your goal is to predict absolute returns, omit time effects—they absorb the common signal.

8.3 Random Effects and the Hausman Test

The fixed-effects model treats α_i as parameters to estimate. The **random effects** (RE) model treats them as draws from a distribution:

$$y_{it} = \mu + \mathbf{x}'_{it}\boldsymbol{\beta} + \alpha_i + \varepsilon_{it}, \quad \alpha_i \sim (0, \sigma_\alpha^2), \quad \varepsilon_{it} \sim (0, \sigma_\varepsilon^2) \quad (8.4)$$

The critical assumption is $\mathbb{E}[\alpha_i | \mathbf{x}_{it}] = 0$: the entity effect is *uncorrelated* with the regressors.

If the RE assumption holds, the RE estimator is more efficient than FE because it uses both within-entity and between-entity variation. If the assumption fails— α_i correlates with $\mathbf{x}_{it}$ —the RE estimator is inconsistent, and FE is the safe choice.

Fixed Effects vs. Random Effects

Fixed effects says: “I do not care *why* asset classes differ in their average return levels. Maybe equities have a higher equity premium, maybe credit has illiquidity compensation—I just want to control for it.” Random effects says: “The differences across asset classes are random draws from a common distribution, and they are unrelated to VaR utilisation.” In finance, the RE assumption almost always fails. Asset classes with structurally higher returns (equities) also tend to have structurally different risk profiles (higher VaR utilisation limits). That correlation between α_i and $\mathbf{x}_{it}$ biases the RE estimator. Default to fixed effects.

8.3.1 The Hausman Test

The **Hausman test** (1978) formalises the FE-vs-RE choice:

$$H = (\hat{\boldsymbol{\beta}}_{\text{FE}} - \hat{\boldsymbol{\beta}}_{\text{RE}})' [\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{\text{FE}}) - \widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{\text{RE}})]^{-1} (\hat{\boldsymbol{\beta}}_{\text{FE}} - \hat{\boldsymbol{\beta}}_{\text{RE}}) \sim \chi_K^2 \quad (8.5)$$

where K is the number of regressors. Under H_0 : RE is consistent (both estimators converge to the same $\boldsymbol{\beta}$). Under H_1 : RE is inconsistent (FE and RE diverge). If you reject H_0 , use FE.

In practice, for finance panels, the Hausman test almost always rejects. Use fixed effects as your default.

8.4 Clustered Standard Errors

Fixed effects handle the *point estimate* correctly, but the *standard errors* require separate treatment. The problem: panel residuals are not independent. Returns of the same asset class are correlated over time (autocorrelation), and returns of different asset classes on the same date are correlated in the cross section (common shocks). Ignoring this correlation produces standard errors that are too small, t -statistics that are too large, and false discoveries.

Petersen (2009) — Standard Errors in Finance Panels

Petersen (2009, *Review of Financial Studies*) uses Monte Carlo simulations to show that OLS standard errors are severely downward biased in finance panels. Specific findings:

- With a firm effect ($\rho_X = \rho_\varepsilon = 0.50$), OLS reports $\widehat{SE} = 0.0283$ while the true standard error is 0.0506—understating by 44%.
- The bias grows with panel length: from $\sim 30\%$ understatement with 5 years per entity to $\sim 73\%$ with 50 years.
- Using unclustered OLS standard errors, 15.3% of t -statistics are significant at the 1% level—over 15 times the nominal rate.
- Only clustering by the correct dimension (firm, time, or both) produces unbiased standard errors.

Never Report Unclustered Standard Errors from a Finance Panel

Unclustered standard errors in finance panels produce t -statistics that are inflated by a factor that grows with the panel's length and the strength of within-cluster correlation. In Petersen's simulations, a nominal 1% test rejects 15.3% of the time. A “significant” result with unclustered SEs is not evidence of anything. Always cluster—the question is only *by what dimension*.

8.4.1 One-Way Clustering

Clustered Standard Errors

Let the panel regression be $y_{it} = \mathbf{x}'_{it}\boldsymbol{\beta} + \varepsilon_{it}$. The **cluster-robust** variance estimator, clustering by entity i , is:

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N \mathbf{X}'_i \hat{\boldsymbol{\varepsilon}}_i \hat{\boldsymbol{\varepsilon}}'_i \mathbf{X}_i \right) (\mathbf{X}'\mathbf{X})^{-1} \quad (8.6)$$

where $\mathbf{X}_i$ is the $T \times K$ submatrix of regressors for entity i and $\hat{\boldsymbol{\varepsilon}}_i$ is the $T \times 1$ vector of residuals for entity i .

Cluster by entity when residuals are correlated within the same entity over time (autocorrelation). This is the standard in corporate finance panels where firms have persistent unobservables.

Cluster by time when residuals are correlated across entities within the same period (cross-sectional dependence). This is common in asset pricing when a market shock hits all assets on the same date.

8.4.2 Two-Way (Double) Clustering

Finance panels typically exhibit *both* entity-level and time-level correlation. The two-way cluster-robust variance estimator (Cameron, Gelbach, and Miller, 2011) is:

$$\widehat{\text{Var}}_{\text{double}} = \widehat{\text{Var}}_{\text{entity}} + \widehat{\text{Var}}_{\text{time}} - \widehat{\text{Var}}_{\text{entity} \times \text{time}} \quad (8.7)$$

where $\widehat{\text{Var}}_{\text{entity} \times \text{time}}$ clusters by the intersection (each unique (i, t) pair—equivalent to heteroskedasticity-robust SEs). Petersen (2009) shows this is the only approach that produces correct standard errors when both dimensions of correlation are present.

Worked Example:

Unclustered vs. Clustered Standard Errors You run a panel regression of next-day return on VaR utilisation with entity fixed effects, using $N = 7$ asset classes and $T = 1,000$ days (7,000 observations). The coefficient estimate is $\hat{\beta} = -0.025$.

SE Method	SE($\hat{\beta}$)	t -statistic	p -value
OLS (no clustering)	0.004	−6.25	< 0.001
Clustered by entity	0.008	−3.13	0.002
Clustered by time	0.007	−3.57	< 0.001
Double-clustered	0.010	−2.50	0.013

The unclustered t -statistic (−6.25) overstates significance by a factor of 2.5 relative to the double-clustered t -statistic (−2.50). In this example all methods still reject at 5%, but in a marginal case the unclustered SE would produce a false discovery. With $N = 7$ clusters on the entity dimension, entity-clustered SEs may themselves be unreliable—the asymptotic approximation requires N large. When N is small, report double-clustered SEs and note the small N caveat, or use the wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008).

How Many Clusters Is Enough?

Clustered standard errors rely on an asymptotic argument in the number of clusters. With $N = 7$ asset classes, you have only 7 entity-level clusters—below the common rule-of-thumb of ~ 50 for reliable asymptotics (Cameron and Miller, 2015). Solutions: (a) use finer entities (sub-desks or individual products rather than broad asset classes); (b) use the wild cluster bootstrap; (c) cluster by time (where $T \approx 1,000$ gives many clusters) and note the limitation.

8.5 Fama–MacBeth Regressions

Fama and MacBeth (1973) introduced an alternative to pooled panel regression that has become the workhorse of cross-sectional asset pricing.

Fama–MacBeth Two-Pass Procedure

Pass 1 (time-series). For each asset $i = 1, \dots, N$, regress returns on factor realisations to estimate factor loadings:

$$R_{it} = \alpha_i + \beta_{i,1}F_{1t} + \dots + \beta_{i,K}F_{Kt} + \varepsilon_{it}, \quad t = 1, \dots, T \quad (8.8)$$

This gives $\hat{\beta}_{i,k}$ for each asset and factor.

Pass 2 (cross-sectional). At each date t , regress returns on the estimated betas:

$$R_{it} = \gamma_{0t} + \gamma_{1t}\hat{\beta}_{i,1} + \dots + \gamma_{Kt}\hat{\beta}_{i,K} + u_{it}, \quad i = 1, \dots, N \quad (8.9)$$

This gives a time series of coefficient estimates $\{\hat{\gamma}_{kt}\}_{t=1}^T$ for each factor k .

Estimate and inference. The risk premium for factor k is the time-series average:

$$\hat{\lambda}_k = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{kt} \quad (8.10)$$

The standard error uses the time-series variation of the cross-sectional estimates:

$$\widehat{\text{SE}}(\hat{\lambda}_k) = \frac{1}{\sqrt{T}} \sqrt{\frac{1}{T-1} \sum_{t=1}^T (\hat{\gamma}_{kt} - \hat{\lambda}_k)^2} \quad (8.11)$$

The key insight is that by running separate cross-sectional regressions at each date and then averaging, Fama–MacBeth *automatically accounts for cross-sectional correlation*. If all assets move together on a given date, the cross-sectional coefficient for that date reflects it, and the time-series averaging smooths it out.

However, Fama–MacBeth does *not* correct for time-series autocorrelation. If $\hat{\gamma}_{kt}$ is persistent over time, the standard error in Equation (8.11) is too small. The fix: apply Newey–West (1987) correction to the time series of $\hat{\gamma}_{kt}$ estimates.

Fama–MacBeth vs. Clustered Panel Regression

Petersen (2009) proves:

- Fama–MacBeth SEs $\approx$ clustering by time. Both handle cross-sectional correlation; neither handles time-series correlation unless adjusted.
- Clustering by entity handles time-series correlation but not cross-sectional correlation.
- Double-clustering handles both.

Use Fama–MacBeth for **pricing tests** (“is this factor priced in the cross section?”). Use panel FE with double-clustered SEs for **prediction** (“does VaR utilisation predict returns, controlling for entity effects?”).

8.5.1 Fama–MacBeth in Practice: He–Kelly–Manela

He, Kelly, and Manela (2017, *JFE*) use Fama–MacBeth regressions to test whether intermediary capital risk is priced across asset classes. Their setup:

- **Factor:** Shocks to the equity capital ratio of New York Fed primary dealers.
- **Test assets:** Portfolios spanning seven asset classes—equities, US government bonds, corporate bonds, sovereign bonds, options, CDS, commodities, and FX.
- **Pass 1:** Estimate each portfolio’s beta to the intermediary capital factor using time-series regression.
- **Pass 2:** Cross-sectional regression of average excess returns on estimated betas.

He–Kelly–Manela (2017) — Single Pricing Kernel

A single intermediary capital risk factor prices the cross section of returns across seven asset classes with a cross-sectional R^2 of 77%. The estimated capital risk premium is approximately 9% per year, with a range of 7–11% for five of the seven asset classes. The price of risk is consistently positive and of similar magnitude when estimated separately for individual asset classes—evidence for a single stochastic discount factor driven by dealer balance-sheet constraints.

This is the test you replicate in Phase 4A: if SecDB risk outputs capture the same dealer-constraint channel, your risk signals should price assets across classes in a similar Fama–MacBeth framework.

Worked Example:

Fama–MacBeth with 3 Asset Classes Suppose you have $N = 20$ portfolios across 3 asset classes and $T = 60$ monthly observations. You want to test whether exposure to a VaR-utilisation factor earns a risk premium.

Pass 1. For each portfolio i , run a time-series regression:

$$R_{it} = \alpha_i + \beta_i \cdot \Delta \text{VaRUtil}_t + \varepsilon_{it}, \quad t = 1, \dots, 60$$

This gives 20 estimated betas $\hat{\beta}_1, \dots, \hat{\beta}_{20}$.

Pass 2. At each month t , regress the cross section of 20 portfolio returns on their betas:

$$R_{it} = \gamma_{0t} + \gamma_{1t} \hat{\beta}_i + u_{it}, \quad i = 1, \dots, 20$$

This gives 60 estimates of $\hat{\gamma}_{1t}$ —the risk premium for VaR-utilisation exposure in month t .

Estimate. Average: $\hat{\lambda} = \frac{1}{60} \sum_{t=1}^{60} \hat{\gamma}_{1t} = 0.42\%$ per month.

Standard error. $\widehat{\text{SE}}(\hat{\lambda}) = \frac{1}{\sqrt{60}} \sqrt{\frac{1}{59} \sum_t (\hat{\gamma}_{1t} - 0.42\%)^2} = 0.18\%$.

t -statistic. $t = 0.42/0.18 = 2.33$ ($p = 0.023$).

Interpretation: portfolios with higher exposure to VaR-utilisation shocks earn 42 bp/month (5.0% annualised) more than low-exposure portfolios, significant at 5%.

8.6 Panel ML vs. Panel Regression

Panel regression and ML are complements, not substitutes. They answer different questions:

	Panel FE Regression	Panel ML (LightGBM)
Goal	Causal interpretation, pricing tests	Out-of-sample prediction
Inference	Clustered SEs, t -statistics	Purged CV, information coefficient
Flexibility	Linear in features	Nonlinear, interactions
Entity effects	Absorbed via demeaning	Dummy features for asset class
Interpretability	Coefficient table	SHAP values
Weakness	Misses nonlinearities	No valid p -values

8.6.1 Using Both in Your Project

Two-Track Panel Analysis

Your project uses panel data in two complementary ways:

1. **Panel FE regression** for the He–Kelly–Manela pricing test. Run Fama–MacBeth or panel FE with double-clustered SEs. Report $\hat{\lambda}$ (price of risk), t -statistics, and cross-sectional R^2 . This tells you whether SecDB risk signals are *priced*—whether bearing exposure to VaR-utilisation shocks earns a premium.
2. **LightGBM with asset-class indicators** for prediction. Include one-hot encoded asset-class dummies as features alongside risk signals. The tree can learn asset-class-specific nonlinear relationships (e.g., VaR utilisation matters more for credit than for FX). Evaluate with purged cross-validation (Chapter 11) and IC (Chapter 12).

The regression answers “is the signal priced?” with proper statistical inference. The ML model answers “can the signal predict?” with proper out-of-sample evaluation. Present both in your deliverable.

8.6.2 Entity Features in Tree Models

When you include asset-class dummies in LightGBM, the tree can split on them. A split on “is_credit = 1” followed by a split on VaR utilisation is equivalent to estimating a different VaR-utilisation slope for credit vs. other classes—an interaction effect that panel FE with a single β cannot capture without explicit interaction terms.

However, with only $N = 7$ asset classes, the dummies are low-cardinality. LightGBM handles this natively. Do *not* target-encode the asset-class variable—with 7 categories and time-series data, target encoding will leak future information unless you are extremely careful with temporal ordering.

Panel ML Validation Pitfall

When cross-validating a panel model, you must respect the time dimension. A random K -fold split will place observations from the same date into both train and test folds, leaking cross-sectional correlation. Use the purged K -fold CV from Chapter 11: split by *time*, not by row. For panel data, this means entire dates go into train or test, never split across folds.

8.7 Choosing the Right Panel Method

With multiple methods available, the decision tree is:

1. **Is your question “is this factor priced?” or “can I predict returns?”**
 - Pricing test $\Rightarrow$ Fama–MacBeth (Section 8.5). Report $\hat{\lambda}$, t -stat, cross-sectional R^2 .
 - Prediction $\Rightarrow$ Panel FE regression or panel ML (Sections 8.2, 8.6).
2. **Do you need interpretable coefficients with valid inference?**
 - Yes $\Rightarrow$ Panel FE with double-clustered SEs. Report $\hat{\beta}$, t -stats.
 - No, you want the best forecast $\Rightarrow$ LightGBM with asset-class dummies and purged CV.
3. **How many entity-level clusters do you have?**
 - $N \geq 50 \Rightarrow$ entity clustering is reliable.
 - $N < 50 \Rightarrow$ cluster by time (T is large), or use wild cluster bootstrap, and note the caveat.
4. **Present both.** In your deliverable, the panel FE regression gives the desk a causal story (“VaR utilisation predicts returns because forced selling depletes liquidity”), while the ML model gives a tradeable signal.

8.8 Practical Considerations

8.8.1 Unbalanced Panels

Your panel will likely be unbalanced: CDS data starts later than equity data, some products were introduced or discontinued, and SecDB history depth varies by desk. Fixed effects handle unbalanced panels naturally—the within estimator only requires variation over whatever time periods exist for each entity. Document which entities have shorter histories and check that results are not driven solely by the longest series.

8.8.2 Stationarity

Panel regressions assume stationarity within each entity’s time series. If VaR utilisation has a unit root (it drifts without mean reversion), the regression is spurious. Pre-test each feature: compute the augmented Dickey–Fuller statistic within each entity. If a feature is non-stationary, take first differences or use the z -score (deviation from rolling mean divided by rolling standard deviation), which you already construct in your feature engineering pipeline (Chapter 9).

8.8.3 Survivorship Bias

If your panel includes only asset classes that currently exist in SecDB, you miss any that were discontinued. This is less of a concern for broad asset classes (equities and rates are permanent) than for individual products. Note the limitation; do not overstate generalisability.

8.8.4 Implementation in Python

Python Packages for Panel Econometrics

The `linearmodels` package (by Kevin Sheppard) implements panel FE, RE, Fama–MacBeth, and clustered SEs directly:

- `PanelOLS` — entity/time fixed effects with clustered SEs.
- `FamaMacBeth` — two-pass Fama–MacBeth with Newey–West adjustment.
- `RandomEffects` — RE estimator with Hausman test via `compare()`.

Set your `DataFrame` to a `MultiIndex` with levels (`entity`, `time`) before fitting. Specify clustering with the `cluster_entity` and `cluster_time` arguments. Always check that the index is sorted by time within each entity to avoid silent misalignment.

For the ML track, `LightGBM` does not know about panel structure—it treats each row independently. You enforce temporal ordering through your validation strategy (purged K -fold from Chapter 11), not through the model itself.

8.8.5 Nickell Bias

When T is small relative to the persistence of the dependent variable, the within estimator is biased. This is the **Nickell (1981) bias**: in a dynamic panel where y_{it} depends on $y_{i,t-1}$, demeaning creates correlation between the demeaned lagged dependent variable and the demeaned error. The bias is $O(1/T)$, so it vanishes as $T \rightarrow \infty$. With $T \approx 1,000$ daily observations, Nickell bias is negligible for your panel. It becomes a concern only if you aggregate to monthly frequency ($T \approx 60$) and include a lagged dependent variable.

8.9 Summary

Concept	Key Takeaway
Panel data	Multiple entities $\times$ multiple time periods; increases sample size and controls for unobserved heterogeneity.
Fixed effects	Absorb entity-level intercepts via demeaning; identify β from within-entity variation only.
Random effects	More efficient <i>if</i> $\alpha_i \perp \mathbf{x}_{it}$; use Hausman test, but default to FE in finance.
Clustered SEs	OLS SEs are 30–73% too small in Petersen’s (2009) simulations; cluster by entity, time, or both.
Fama–MacBeth	Two-pass: time-series betas, then cross-sectional risk premia; accounts for cross-sectional but not time-series correlation.
He–Kelly–Manela	Single intermediary capital factor prices 7 asset classes with $R^2 = 77\%$ and $\sim 9\%$ annual premium.
Panel ML	Complement regression with LightGBM using asset-class dummies; validate with time-based purged CV.

Part II

Applied Methodology

Chapter 9

Feature Engineering from Risk Systems

Project Connection — Phase 0 & Phase 2

This chapter maps directly to Phase 0 (data pipeline) and Phase 2 (feature engineering). Every feature family here corresponds to a module in `src/features/`. The critical lesson: **point-in-time discipline**. A single lookahead bug—using VaR from the overnight risk run before it was actually available—invalidates your entire backtest.

9.1 Point-in-Time Discipline

Feature Matrix Notation

A feature matrix $\mathbf{X} \in \mathbb{R}^{T \times p}$ has T rows (one per time step) and p columns (one per feature). Row t contains only information that was *known and available* at time t . If any entry $X_{t,j}$ uses data that arrived after t , the matrix is contaminated by lookahead bias.

Every feature must be stamped with *when it was known*, not when it applied. Lopez de Prado (2018, Ch. 2) calls this **point-in-time** (PIT) stamping and identifies lookahead bias as the single most common and most damaging error in financial ML. The principle is simple: a feature available only at time $t + \delta$ cannot appear in any row earlier than $t + \delta$ in your feature matrix.

9.1.1 Knowledge-Date Lagging

SecDB runs its nightly risk engine in a batch process. The VaR number for business date T is computed overnight between T close and $T + 1$ open. Therefore:

- The **as-of date** of VaR is T —it measures the portfolio’s risk as of the close on date T .
- The **knowledge date** is $T + 1$ morning—this is when the number becomes available to traders and models.
- Any model using VaR_T as a feature may only do so for predictions dated $T + 1$ or later.

The gap between as-of date and knowledge date varies by data source:

Risk Output	As-of Date	Knowledge Date
Nightly VaR / ES	T close	$T + 1$ pre-open
Scenario P&L	T close	$T + 1$ pre-open
Intraday Greeks	Real-time	Real-time
VaR limit (set by risk)	Set date	Set date
Factor-VaR decomposition	T close	$T + 1$ pre-open

Intraday Greeks are the exception: they are computed in real-time by SecDB’s pricing engine and available immediately. This is why Project 2 (Chapter 4, Phase 4B) can use same-day Greek features.

Point-in-Time Stamping

For a risk output x with as-of date T and knowledge date $T + \delta$, the PIT-stamped feature is:

$$\tilde{x}_t = \begin{cases} x_T & \text{if } t \geq T + \delta \\ \text{NaN} & \text{if } t < T + \delta \end{cases} \quad (9.1)$$

where δ is the **publication lag**—typically 1 business day for overnight batch outputs. The tilde notation $\tilde{x}$ denotes the PIT-lagged version.

Worked Example:

VaR Point-in-Time Timeline Consider the following sequence for a Monday–Wednesday window:

Event	Wall-Clock Time	As-of	Knowledge
Monday close	Mon 4:00pm	—	—
Risk batch starts	Mon 6:00pm	Monday	—
Risk batch ends	Tue 5:00am	Monday	Tuesday
Tuesday open	Tue 9:30am	—	—
Model predicts	Tue 9:30am	—	Uses VaR_{Mon}
Tuesday close	Tue 4:00pm	—	—
Risk batch ends	Wed 5:00am	Tuesday	Wednesday

The model running at Tuesday open may use Monday’s VaR (batch completed at 5am Tuesday) but *not* Tuesday’s VaR (batch has not run yet). In code, this means:

$$\tilde{\text{VaR}}_{t=\text{Tue}} = \text{VaR}_{\text{Mon}}, \quad \tilde{\text{VaR}}_{t=\text{Wed}} = \text{VaR}_{\text{Tue}} \quad (9.2)$$

Using VaR_{Tue} at Tuesday close is cheating—it was not available until Wednesday morning.

A Single Lookahead Bug Invalidates Your Entire Backtest

VaR for date T , computed in the overnight batch, is available at $T + 1$ open. Using it at T close is cheating. The effect is subtle: your backtest will show inflated Sharpe ratios because the model “knows” tomorrow’s risk state today. Lopez de Prado (2018, Ch. 2) emphasizes that lookahead bias is the most frequent source of false discoveries in financial ML—more common than overfitting and harder to detect because the backtest looks clean. In your pipeline code (`src/data/point.in.time.py`), enforce PIT lagging at the data ingestion layer so no downstream code can accidentally skip it.

9.1.2 Implementation Discipline

Three rules for your `src/data/pipeline.py`:

1. **Lag at ingestion, not at modeling.** Apply the knowledge-date shift when data enters

- the pipeline. Every downstream consumer sees only PIT-valid data.
2. **Store both timestamps.** Keep the as-of date and knowledge date as separate columns. This allows auditing.
 3. **Unit test with synthetic data.** Create a test where a feature's knowledge date is deliberately set to $T + 1$. Verify that the feature at time T is **NaN**, not the T value.

9.2 Feature Family 1: VaR Utilization

VaR Utilization Recap

VaR utilization = $\text{VaR}_{\text{current}} / \text{VaR}_{\text{limit}} \times 100\%$ measures how much of the desk's risk budget is deployed. Desks typically operate at 50–80%; approaching 100% forces position reduction. See Section 3.7 of Chapter 3 for the full definition and the procyclical leverage mechanism (Adrian and Shin, 2014).

VaR utilization is one of two **priority** feature families (design spec, Phase 2). The theoretical basis is strong: VaR utilization is the most direct proxy for the binding constraint in intermediary asset pricing models (He and Krishnamurthy, 2013; Chapter 2).

9.2.1 Feature Definitions

Let $U_t = \text{VaR}_t / \text{VaR}_{\text{limit},t}$ denote VaR utilization at time t (PIT-lagged per Section 9.1). Three features:

VaR Utilization Features

(1) Level:

$$f_t^{\text{util-level}} = \tilde{U}_t \quad (9.3)$$

Current utilization percentage. Values above 85% signal constrained dealers.

(2) Rate of change over horizon $h \in \{1, 5, 21\}$ days:

$$f_t^{\text{util-}\Delta h} = \tilde{U}_t - \tilde{U}_{t-h} \quad (9.4)$$

Rising utilization signals building stress; the 5-day change captures weekly dynamics; the 21-day change captures monthly trends.

(3) Z-score over a rolling window of length w (default $w = 63$ business days, approximately 3 months):

$$f_t^{\text{util-z}} = \frac{\tilde{U}_t - \bar{U}_{t,w}}{\hat{\sigma}_{U,t,w}} \quad (9.5)$$

where $\bar{U}_{t,w}$ and $\hat{\sigma}_{U,t,w}$ are the rolling mean and standard deviation of $\tilde{U}$ over the window $[t - w + 1, t]$.

- $f_t^{\text{util-level}}$ — raw utilization; captures the constraint *level*.
- $f_t^{\text{util-}\Delta h}$ — first difference; captures the *direction* of constraint tightening or loosening.
- $f_t^{\text{util-z}}$ — standardized anomaly; captures how *unusual* today's utilization is relative to recent history.

Worked Example:

VaR Utilization Features from a 10-Day Series Given PIT-lagged utilization data:

t	$\tilde{U}_t$ (%)	$\Delta 1d$	$\Delta 5d$	Z-score ($w=5$)
1	62	—	—	—
2	64	+2	—	—
3	68	+4	—	—
4	71	+3	—	—
5	73	+2	+11	—
6	78	+5	+14	+1.63
7	76	−2	+12	+0.99
8	80	+4	+12	+1.38
9	83	+3	+12	+1.55
10	88	+5	+15	+1.82

- **Level:** At $t=10$, utilization is 88%—above the 85% warning threshold.
- **$\Delta 5d$:** The 5-day change of +15 percentage points signals rapid constraint tightening.
- **Z-score:** At +1.82, today's utilization is nearly 2 standard deviations above its recent 5-day rolling mean.

In practice, use $w = 63$ (3 months) for the z-score to have a meaningful rolling baseline. The $w = 5$ window here is for illustration only.

Why Three Features, Not Just One?

The level, rate of change, and z-score capture fundamentally different information. A desk at 85% utilization that has been there for 6 months (z-score near 0) is in a different state than a desk that jumped from 60% to 85% in a week (z-score near +2). Both have the same level, but the second desk is under acute stress and more likely to be forced to sell. The z-score distinguishes chronic from acute constraint tightening.

9.3 Feature Family 2: Factor Concentration

Factor-VaR Decomposition and HHI

Factor-VaR decomposes total VaR into contributions from underlying risk factors (rates, credit, equity, FX, commodities). The Herfindahl-Hirschman Index $HHI = \sum_k s_k^2$ measures concentration, where s_k is factor k 's share of total VaR. $HHI \in [1/K, 1]$; higher values mean more concentrated risk. See Section 3.5 and specifically equation (3.12) of Chapter 3.

Factor concentration is the second **priority** feature family. The theory: when many dealers crowd into the same factor, HHI rises. If that factor reverses, concentrated dealers must all deleverage simultaneously—triggering fire sales (Coval and Stafford, 2007) and predictable return reversals (He, Kelly, and Manela, 2017; Chapter 2).

9.3.1 Feature Definitions

Let $s_{k,t}$ denote the PIT-lagged factor-VaR share of factor k at time t .

Factor Concentration Features

(1) Herfindahl Index:

$$f_t^{\text{HHI}} = \sum_{k=1}^K \tilde{s}_{k,t}^2 \quad (9.6)$$

Measures overall concentration. K is the number of risk factors (typically 5–10 for a cross-asset desk).

(2) **Top-3 factor share:**

$$f_t^{\text{top3}} = \sum_{k \in \mathcal{T}_t} \tilde{s}_{k,t} \quad (9.7)$$

where $\mathcal{T}_t$ is the set of three factors with the largest VaR shares at time t . This is a simpler, more interpretable concentration metric.

(3) **Rolling HHI change:**

$$f_t^{\Delta \text{HHI}} = f_t^{\text{HHI}} - f_{t-h}^{\text{HHI}} \quad (9.8)$$

for $h \in \{5, 21\}$ days. Rising HHI means concentration is increasing.

- f^{HHI} — level of concentration across all factors.
- f^{top3} — concentration in the dominant factors; robust to noise in small factor shares.
- $f^{\Delta \text{HHI}}$ — direction and speed of concentration changes.

High HHI = Concentrated Risk = Fragility

High HHI means one factor dominates the risk profile. This signals crowding and fragility. He, Kelly, and Manela (2017) show that a single intermediary-capital factor prices risk across asset classes—when dealers are concentrated in one factor, the cross-asset pricing kernel is exposed to a single point of failure.

Concrete numbers: For a 5-factor desk, $\text{HHI} = 0.20$ means all factors contribute equally ($N_{\text{eff}} = 5$). $\text{HHI} = 0.50$ means the desk is effectively a 2-factor bet. $\text{HHI} > 0.60$ should raise concern about crowding.

9.4 Feature Family 3: VaR Dynamics

VaR Computation Methods and the Cliff Effect

VaR under historical simulation suffers from the cliff effect: an extreme observation entering or leaving the lookback window can cause a discontinuous jump (Section 3.3.1 of Chapter 3). VaR-dynamics features should be smoothed (e.g., 5-day moving average) to reduce this noise.

VaR dynamics capture the *time-series behavior* of the aggregate risk number—not where risk lives (that is factor concentration) or how close the desk is to its limit (that is utilization), but how risk itself is evolving.

9.4.1 Feature Definitions

Let V_t denote the desk's total VaR at time t , PIT-lagged.

VaR Dynamics Features

(1) **Delta VaR:**

$$f_t^{\Delta \text{VaR}} = \tilde{V}_t - \tilde{V}_{t-1} \quad (9.9)$$

The daily change in VaR. Large positive values indicate a rapid increase in measured risk.

(2) **Component VaR by asset class:**

$$f_t^{\text{CVaR-c}} = \widetilde{\text{CVaR}}_{c,t} \quad (9.10)$$

for each asset class $c \in \{\text{rates, credit, equity, FX, commodities}\}$. These track the risk contribution of each asset class over time (see equation (3.7) of Chapter 3 for the component VaR definition).

(3) VaR momentum:

$$f_t^{\text{VaR-mom}} = \text{sign}\left(\sum_{i=0}^4 (\tilde{V}_{t-i} - \tilde{V}_{t-i-1})\right) \cdot \left|\sum_{i=0}^4 (\tilde{V}_{t-i} - \tilde{V}_{t-i-1})\right|^{1/2} \quad (9.11)$$

Sign and square-root magnitude of the 5-day rolling VaR change. The square root compresses extreme values while preserving sign.

- $f^{\Delta\text{VaR}}$ — instantaneous risk change.
- $f^{\text{CVaR-}c}$ — risk attribution by asset class.
- $f^{\text{VaR-mom}}$ — direction and persistence of risk trends.

Methodology Breaks Masquerade as Signals

Section 3.8 of Chapter 3 warns that VaR model updates create structural breaks. A switch from historical simulation to Monte Carlo can produce a one-day ΔVaR that dwarfs any genuine portfolio change. LightGBM (Chapter 6) will happily learn to split on these artifacts. Before engineering VaR-dynamics features, obtain the methodology change dates from Phase 0's risk-model audit and either include them as binary control variables or restrict training to a single methodology regime.

9.5 Feature Family 4: Scenario P&L

Scenario Analysis Recap

A scenario specifies simultaneous moves in multiple risk factors (e.g., rates +200bp, equities −25%). The risk system revalues the portfolio under each scenario to produce a scenario P&L. See Section 3.6 of Chapter 3 for types of scenarios and a worked numerical example.

Scenario P&L features are forward-looking in a way that VaR is not. VaR uses backward-looking distributions; scenarios ask “what would happen *if*?” Changes in scenario P&L capture shifts in portfolio vulnerability that may not yet appear in VaR.

9.5.1 Feature Definitions

Let $\pi_{j,t}$ denote the PIT-lagged scenario P&L for scenario j at time t , with $j = 1, \dots, J$ scenarios.

Scenario P&L Features

(1) Worst-case rank:

$$f_t^{\text{worst-id}} = \arg \min_j \tilde{\pi}_{j,t} \quad (9.12)$$

Which scenario produces the largest loss today? This is a categorical feature—the *identity* of the worst scenario, not its magnitude.

(2) Scenario dispersion:

$$f_t^{\text{disp}} = \sqrt{\frac{1}{J} \sum_{j=1}^J (\tilde{\pi}_{j,t} - \bar{\pi}_t)^2} \quad (9.13)$$

where $\bar{\pi}_t = J^{-1} \sum_j \tilde{\pi}_{j,t}$ is the cross-scenario mean. High dispersion means the portfolio is directionally sensitive—performance varies sharply with the type of stress. Low dispersion means the portfolio is well-hedged or indifferent to the specific scenario.

(3) Worst-case identity change:

$$f_t^{\text{flip}} = \mathbf{1}[f_t^{\text{worst-id}} \neq f_{t-1}^{\text{worst-id}}] \quad (9.14)$$

A binary indicator: did the worst scenario change from yesterday? A flip signals a qualitative shift in portfolio vulnerability—the desk’s primary risk has rotated.

(4) Scenario P&L skewness:

$$f_t^{\text{skew}} = \frac{J^{-1} \sum_{j=1}^J (\tilde{\pi}_{j,t} - \bar{\pi}_t)^3}{(f_t^{\text{disp}})^3} \quad (9.15)$$

Negative skewness means a few scenarios produce very large losses relative to the rest—the portfolio has a fat left tail across scenarios.

- $f^{\text{worst-id}}$ — categorical; use one-hot encoding or target encoding for tree models.
- f^{disp} — continuous; high values signal directional exposure.
- f^{flip} — binary; captures qualitative regime shifts.
- f^{skew} — continuous; captures asymmetry in scenario outcomes.

9.6 Feature Family 5: Cross-Asset Flow

Component VaR Shares

Component VaR (Section 3.4 of Chapter 3) decomposes total VaR by position or asset class. The component VaR share of asset class c is $s_{c,t} = \text{CVaR}_{c,t} / \text{VaR}_t$. Shares sum to 1 (Euler decomposition).

Cross-asset flow features track how risk *migrates* between asset classes. These features are distinct from factor concentration (Section 9.3), which measures overall diversification. Cross-asset flow measures the *direction* of risk rotation.

9.6.1 Feature Definitions

Let $s_{c,t}$ denote the PIT-lagged component VaR share of asset class c at time t .

Cross-Asset Flow Features

(1) Component VaR share shift:

$$f_t^{\Delta s_c} = \tilde{s}_{c,t} - \tilde{s}_{c,t-h} \quad (9.16)$$

for each asset class c and horizon $h \in \{5, 21\}$ days. Positive values mean the desk is accumulating risk in asset class c relative to other classes.

(2) Rolling cross-asset VaR correlation:

$$f_t^{\rho_{c_1, c_2}} = \text{Corr}(\tilde{s}_{c_1, t-w+1:t}, \tilde{s}_{c_2, t-w+1:t}) \quad (9.17)$$

over a rolling window of $w = 63$ days, for key asset-class pairs (e.g., rates–credit, equity–commodities). High positive correlation means the desk’s risk in both classes moves together; high negative correlation suggests active rotation from one class to another.

- $f^{\Delta s_c}$ — direction of risk rotation by asset class.
- $f^{\rho_{c1,c2}}$ — persistence and co-movement of risk allocation decisions.

Risk Rotation Predicts Relative Performance

If the rates VaR share rises while the credit VaR share falls, dealers are rotating risk—accumulating rate exposure and shedding credit exposure. Under the intermediary pricing framework (Chapter 2), this shift in dealer balance sheets should predict *relative* performance between rates and credit. He, Kelly, and Manela (2017) show that a single intermediary-capital factor prices the cross-section of returns across asset classes. Cross-asset flow features test whether *changes* in risk allocation across classes carry incremental information beyond the aggregate constraint level.

9.7 Feature Family 6: Dealer Greeks (Project 2)

Dealer Greeks

A dealer's net gamma, vega, vanna, and charm determine whether hedging flows stabilize or destabilize prices. Short-gamma dealers amplify price moves; long-gamma dealers dampen them. See Sections 4.3 and 4.4 of Chapter 4 for definitions and the shock-absorber vs. shock-amplifier intuition.

These features belong to Phase 4B (Project 2: Book-Gamma Intraday Momentum). They are included here for completeness and because the shared infrastructure (Task 5, data pipeline) should accommodate them from the start. All Greek features are available in real time from SecDB's pricing engine, so PIT lagging is not required—but you must still ensure that features used for end-of-day predictions are computed at a consistent timestamp (e.g., 3:00pm snapshot).

9.7.1 Feature Definitions

Let Γ_t^{net} , $\mathcal{V}_t^{\text{net}}$, $\text{Vanna}_t^{\text{net}}$, and $\text{Charm}_t^{\text{net}}$ denote the aggregate net book-level Greeks from SecDB at time t .

Dealer Greek Features

(1) Net gamma, vega, vanna, charm:

$$f_t^\Gamma = \Gamma_t^{\text{net}}, \quad f_t^\mathcal{V} = \mathcal{V}_t^{\text{net}}, \quad f_t^{\text{vanna}} = \text{Vanna}_t^{\text{net}}, \quad f_t^{\text{charm}} = \text{Charm}_t^{\text{net}} \quad (9.18)$$

These are the raw aggregate sensitivities. The sign of gamma is the key feature: negative aggregate gamma predicts intraday momentum (Baltussen, Da, Lammers, and Martens, 2021; Section 4.6 of Chapter 4).

(2) Gamma-flip level:

$$S_t^{\text{flip}} = \text{strike at which aggregate } \Gamma^{\text{net}} \text{ changes sign} \quad (9.19)$$

The gamma-flip level is the underlying price where the desk transitions from long gamma (stabilizing) to short gamma (destabilizing) or vice versa.

(3) Distance to flip:

$$f_t^{\text{d-flip}} = \frac{S_t - S_t^{\text{flip}}}{S_t} \times 100\% \quad (9.20)$$

where S_t is the current underlying price. Small absolute values mean the market is near the flip level—a regime transition in hedging flow direction is imminent.

- $f^\Gamma, f^\mathcal{V}, f^{\text{vanna}}, f^{\text{charm}}$ — raw book-level sensitivities with correct dealer sign (not estimated from public data).
- S^{flip} — the gamma regime boundary.
- $f^{\text{d-flip}}$ — proximity to regime transition; nonlinear effects concentrate near zero.

9.8 Confound Checking

Omitted Variable Bias

If a predictor x and outcome y are both driven by an omitted variable z , the apparent predictive relationship between x and y is spurious. Including z as a control variable reveals whether x has *incremental* predictive power. This is the financial-ML analogue of the “confound” in causal inference.

A risk-system feature that merely repackages information already available in public data adds no value. VaR rises when markets are stressed—but so does the VIX. If your VaR-based signal vanishes after controlling for VIX, you have not discovered a new signal; you have built a noisy VIX proxy.

9.8.1 Control Variables

Add the following public factors as controls in every model:

1. **VIX level and change:** VIX_t and ΔVIX_t . The most common confound for any risk-based feature.
2. **Credit spread:** Investment-grade and high-yield OAS (option-adjusted spread). Confounds credit-related features.
3. **Term slope:** 10y–2y Treasury yield spread. Confounds rate-related features and captures macro regime.
4. **Equity market return:** S&P 500 trailing 5-day return. Confounds any feature correlated with market direction.
5. **Dollar index:** DXY level and change. Confounds FX-related features.

9.8.2 The Confound Test

Run two models (ridge baseline, then LightGBM—see Chapters 5 and 6):

1. **Model A:** Risk features only.
2. **Model B:** Risk features + public controls.

Compare the information coefficient (IC) and SHAP importance (Chapter 7) of each risk feature across Models A and B.

Confound Checking Is Not Optional

If your signal vanishes after adding public controls, it is redundant with publicly available information—you have not added value. The test is simple: compare IC with and without controls. A feature that survives controls has *incremental* information from proprietary risk data. A feature that does not is just noise around VIX or credit spreads. This is the feature-level analogue of the confound checks in asset pricing (Chapter 1): a “new” factor that is spanned by existing factors adds nothing.

9.9 Summary

Table 9.1 maps each feature family to its constituent features, source chapter, and project phase.

Feature Family	Features	Source Chapter	Project Phase
VaR Utilization	Level, Δh -day change, z-score	Ch. 3 §3.7	Phase 2 (priority)
Factor Concentration	HHI, top-3 share, Δ HHI	Ch. 3 §3.5	Phase 2 (priority)
VaR Dynamics	Δ VaR, CVaR by asset class, VaR momentum	Ch. 3 §3.3	Phase 2
Scenario P&L	Worst-case id, dispersion, flip indicator, skewness	Ch. 3 §3.6	Phase 2
Cross-Asset Flow	CVaR share shifts, rolling cross-asset correlation	Ch. 3 §3.4	Phase 2
Dealer Greeks	Net $\Gamma/\mathcal{V}$ /vanna/charm, flip level, distance to flip	Ch. 4 §4.3	Phase 4B

Table 9.1: Feature family reference. Priority families (VaR utilization and factor concentration) are tested first in Phase 2 per the design spec. All batch-computed features require PIT lagging (Section 9.1). Dealer Greeks are real-time and belong to Project 2.

Key takeaways:

1. Point-in-time discipline is non-negotiable. Enforce PIT lagging at the data ingestion layer. VaR for date T is available at $T + 1$ open.
2. Six feature families span the full set of SecDB risk outputs. VaR utilization and factor concentration are tested first (strongest theory: Adrian and Shin, 2014; He, Kelly, and Manela, 2017; Coval and Stafford, 2007).
3. Every feature must pass confound checks against public controls (VIX, credit spread, term slope). Features that do not survive controls are noisy proxies, not new signals.
4. The level, rate of change, and z-score of a quantity are three different features—they capture different aspects (constraint state, direction, unusualness) and should all be included.
5. Dealer Greek features (Project 2) are real-time and do not require PIT lagging, but they do require a consistent intraday timestamp for reproducibility.

Chapter 10

Labeling and Target Construction

Project Connection

In Phase 1 you build the label construction module (Task 6 in the implementation plan). The choice of label is as important as the choice of features—the label *is* your hypothesis. Triple-barrier labeling with volatility-scaled thresholds is the primary labeling method; meta-labeling adds a precision filter. Your prediction targets (VIX innovations, realized vol, drawdowns) come from the theoretical foundations in Chapters 1–2.

10.1 Why Labeling Matters

Supervised Learning Recap

Supervised learning maps inputs $\mathbf{X}$ to labels $\mathbf{y}$. The model minimizes $\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$ over training data. If $\mathbf{y}$ is poorly constructed—noisy, misaligned with the trading horizon, or insensitive to the path of prices—no model can recover a useful signal. Garbage in, garbage out applies with full force here.

Every supervised ML problem requires a label: the thing you predict. In most ML applications the label is obvious (cat vs. dog, spam vs. not-spam). In finance the label is a *design choice*, and that choice encodes your hypothesis.

The label defines the question. Predicting 5-day returns asks a different question than predicting whether a stop-loss will be hit before a profit target. The same features can yield a strong signal under one labeling scheme and pure noise under another.

10.1.1 Standard Return Labels

The simplest label is the forward return over a fixed horizon:

$$y_t = \frac{P_{t+h} - P_t}{P_t}, \quad (10.1)$$

where h is a fixed holding period (1 day, 5 days, 21 days). You then threshold: $y_t > \tau \Rightarrow +1$, $y_t < -\tau \Rightarrow -1$, else 0.

Three problems:

1. **Path dependence is ignored.** A position that hits -5% intraday and recovers to $+1\%$ at $t + h$ gets labeled $+1$, but no risk manager would have held through the drawdown.

2. **Fixed thresholds fail across asset classes.** A $\pm 1\%$ threshold is reasonable for equities but trivial for emerging-market FX and far too wide for G10 rates. Heteroskedasticity within a single asset makes a fixed τ equally inappropriate across time.
3. **Horizon selection is arbitrary.** Why 5 days? Why not 3 or 10? The choice of h implicitly assumes a holding period that may not match the signal's actual predictive horizon.

Labels as Strategy Blueprints

Think of the label as a compressed description of what a trading strategy would do. A fixed-horizon return label describes a “buy and hold for exactly h days” strategy—no stops, no targets, no early exit. That is not how anyone trades. The triple-barrier method (next section) labels data the way a strategy actually behaves: entry, stop-loss, profit-target, and maximum holding period.

10.2 Triple-Barrier Labeling

The triple-barrier method, introduced by Lopez de Prado (2018, Ch. 3), fixes every problem listed above. It defines three barriers around each entry point and labels based on which barrier the price path touches first.

Triple-Barrier Method

Given an entry at time t_0 with price P_{t_0} :

1. **Upper barrier** (profit-taking): $P_{t_0} (1 + m_u \cdot \sigma_t)$, where m_u is the upper multiplier and σ_t is a rolling volatility estimate.
2. **Lower barrier** (stop-loss): $P_{t_0} (1 - m_\ell \cdot \sigma_t)$, where m_ℓ is the lower multiplier.
3. **Vertical barrier** (expiration): a maximum holding period $T_{\max}$ bars after t_0 .

The label is:

$$y = \begin{cases} +1 & \text{if upper barrier is hit first,} \\ -1 & \text{if lower barrier is hit first,} \\ 0 & \text{if vertical barrier is reached before either horizontal barrier.} \end{cases}$$

10.2.1 Volatility-Scaled Thresholds

The barrier widths adapt to current market conditions through σ_t , typically estimated as an exponentially weighted moving standard deviation of log returns:

$$\sigma_t = \sqrt{\sum_{i=0}^{n-1} w_i (r_{t-i} - \bar{r}_t^w)^2}, \quad w_i = \frac{(1 - \lambda) \lambda^i}{1 - \lambda^n}, \quad (10.2)$$

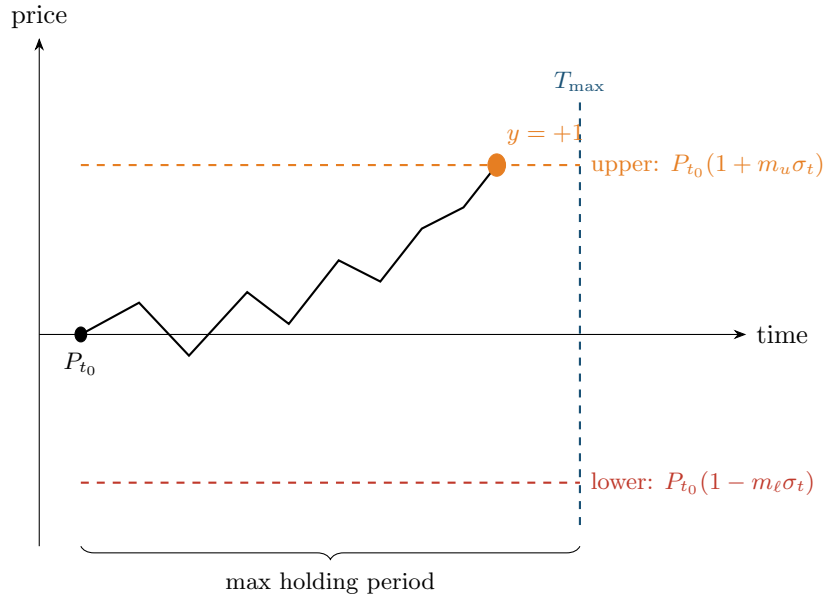
where $r_{t-i} = \ln P_{t-i} - \ln P_{t-i-1}$ are log returns, $\bar{r}_t^w = \sum_i w_i r_{t-i}$ is the weighted mean, and $\lambda = 2^{-1/\tau_h}$ for a half-life of τ_h days. Lopez de Prado recommends a span of approximately 100 bars (roughly $\tau_h \approx 50$ days for daily data).

Exponentially Weighted Moving Average

An EWMA gives more weight to recent observations. The half-life τ_h is the number of periods after which a past observation's weight drops to half. A 20-day half-life means an observation from 20 days ago contributes half as much as today's. Pandas implements this as `df.ewm(halflife=20).std()`.

Why volatility scaling matters: a 2% barrier is tight during a crisis (daily vol $\approx 3\%$) but loose during calm markets (daily vol $\approx 0.5\%$). Multiplying by σ_t normalizes the barrier width in “number of standard deviations,” making labels comparable across time and across asset classes.

10.2.2 Triple-Barrier Diagram



10.2.3 Worked Example: Step-by-Step Label Assignment

Worked Example:

Triple-Barrier Label Assignment Setup. Ten daily prices starting at $P_0 = 100$. Rolling volatility (EWMA, 20-day half-life) at entry: $\sigma_0 = 1.0\%$. Multipliers: $m_u = m_\ell = 2$. Maximum holding period: $T_{\max} = 5$ days.

Barriers. Upper: $100 \times (1 + 2 \times 0.01) = 102.00$. Lower: $100 \times (1 - 2 \times 0.01) = 98.00$. Vertical: day 5.

Price path.

Day	1	2	3	4	5
Price	100.40	100.80	101.50	101.20	102.10
Hit?	—	—	—	—	upper

On day 5 the price crosses 102.00 for the first time $\Rightarrow$ upper barrier is hit before the vertical barrier $\Rightarrow y = +1$.

Alternate path. If instead the prices were $\{99.80, 99.30, 98.50, 97.90, \dots\}$, the price crosses 98.00 on day 3 $\Rightarrow y = -1$.

Alternate path 2. Prices $\{100.20, 99.80, 100.10, 100.50, 100.30\}$. Neither barrier is breached within 5 days. At the vertical barrier: $P_5 = 100.30 > P_0 = 100$, so $y = 0$ (or optionally $y = \text{sign}(P_5 - P_0) = +1$, depending on configuration).

Why Triple-Barrier Labeling Works

Triple-barrier labeling aligns the label with how a trading strategy would actually behave: a position has a stop-loss, a profit target, and a maximum holding period. The label records *which constraint binds first*, not just the terminal return. This produces labels that are more informative and less noisy than fixed-horizon returns, because the label

reflects the path of prices, not just the endpoint.

10.3 Meta-Labeling

Triple-barrier labeling tells you the *outcome* of a trade. Meta-labeling (Lopez de Prado 2018, Ch. 3.6) adds a second layer: a model that predicts *whether to take the trade at all*.

Meta-Labeling

A two-stage labeling procedure:

1. **Primary model** predicts the *side*: long (+1) or short (−1). This can be any model—ML, fundamental, or rule-based (e.g., moving-average crossover, carry signal, mean-reversion filter).
2. **Secondary (meta) model** predicts whether the primary model is *correct*: 1 if the trade is profitable (per triple-barrier), 0 otherwise.

The meta-label is:

$$y_t^{\text{meta}} = \begin{cases} 1 & \text{if primary side} \times \text{triple-barrier label} > 0, \\ 0 & \text{otherwise.} \end{cases}$$

10.3.1 Precision–Recall Tradeoff

The primary model is designed for **high recall**: it flags most profitable opportunities, even at the cost of many false positives. The meta-model adds **precision**: it filters out trades where the primary model is likely wrong.

Precision and Recall

Precision = $TP / (TP + FP)$: of all trades the model takes, what fraction are profitable?

Recall = $TP / (TP + FN)$: of all profitable opportunities, what fraction does the model capture? A model that trades on every signal has recall = 1 but may have low precision. Meta-labeling sacrifices some recall for substantially higher precision.

In empirical tests using synthetic data and equity strategies, adding meta-labeling to a Bollinger Band mean-reversion strategy increased out-of-sample precision from 0.17 to 0.20 and accuracy from 17% to 63% (Singh and Joubert, Hudson & Thames research note). For a trend-following SMA crossover, out-of-sample precision rose from 0.48 to 0.54. These gains come from filtering false positives, not from discovering new signal.

10.3.2 Betting Size from Meta-Probabilities

The meta-model outputs a probability $p_t \in [0, 1]$. This directly maps to position size:

$$\text{size}_t = f(p_t) \times \text{side}_t, \quad (10.3)$$

where f is a monotone mapping (e.g., $f(p) = 2p - 1$ so that $p = 0.5 \Rightarrow$ no bet, $p = 1.0 \Rightarrow$ full position). The side comes from the primary model; the magnitude comes from the meta-model's confidence.

Meta-Labeling Requires a Reasonable Primary Model

Meta-labeling filters false positives from the primary model. If the primary model is random (accuracy $\approx 50\%$ with no structure), there is nothing systematic for the meta-model

to learn. Always verify that the primary model has above-chance directional accuracy before investing effort in meta-labeling. A ridge regression baseline (Chapter 5) is a reasonable first primary model.

10.4 Label Uniqueness and Concurrency

Why i.i.d. Assumptions Fail for Overlapping Labels

Standard ML assumes training samples are independent and identically distributed (i.i.d.). Triple-barrier labels with a 5-day maximum holding period create overlapping return windows: the label for day t depends on prices $\{P_{t+1}, \dots, P_{t+5}\}$, and the label for day $t + 1$ depends on $\{P_{t+2}, \dots, P_{t+6}\}$. Four of five days overlap. This makes consecutive labels highly correlated, violating the i.i.d. assumption. Ignoring this inflates apparent model performance, because the model “memorizes” shared return information rather than learning genuine patterns.

10.4.1 Concurrency and Uniqueness

Lopez de Prado (2018, Ch. 4) quantifies this overlap with two measures.

Concurrency Count

For a set of N labeled events, each spanning a time window $[t_i^{\text{start}}, t_i^{\text{end}}]$, the concurrency at bar t is:

$$c_t = \sum_{i=1}^N \mathbf{1} \left[t_i^{\text{start}} \leq t \leq t_i^{\text{end}} \right], \quad (10.4)$$

where $\mathbf{1}[\cdot]$ is the indicator function. c_t counts how many labels are “active” at time t —i.e., how many overlapping return windows include bar t .

Average Uniqueness

The average uniqueness of label i is:

$$\bar{u}_i = \frac{1}{|T_i|} \sum_{t \in T_i} \frac{1}{c_t}, \quad (10.5)$$

where $T_i = \{t : t_i^{\text{start}} \leq t \leq t_i^{\text{end}}\}$ is the set of bars spanned by label i . A label with $\bar{u}_i = 1$ has no overlap with any other label. A label with $\bar{u}_i = 0.25$ shares its information with roughly three other concurrent labels on average.

Worked Example:

Computing Average Uniqueness Three events with the following spans:

Event	Start	End	Bars
A	1	3	$\{1, 2, 3\}$
B	2	4	$\{2, 3, 4\}$
C	4	5	$\{4, 5\}$

Concurrency at each bar: $c_1 = 1$, $c_2 = 2$, $c_3 = 2$, $c_4 = 2$, $c_5 = 1$.

Average uniqueness:

$$\begin{aligned}\bar{u}_A &= \frac{1}{3} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{2} \right) = \frac{1}{3}(2.0) \approx 0.667, \\ \bar{u}_B &= \frac{1}{3} \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right) = 0.500, \\ \bar{u}_C &= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{1} \right) = 0.750.\end{aligned}$$

Event C is most unique; event B is least unique. In sample-weighted training, C receives the highest weight.

10.4.2 Using Uniqueness for Sample Weighting

There are two practical applications:

1. **Sample weights.** Set $w_i = \bar{u}_i$ and pass these weights to the model's loss function. Both LightGBM and scikit-learn accept a `sample_weight` argument. High-uniqueness samples contribute more to the gradient; low-uniqueness (highly overlapping) samples are down-weighted.
2. **Bagging adjustment.** When using bagged models (random forests, bagged trees), set `max_samples` = $\bar{u}$, the average uniqueness across all labels. This prevents bootstrap samples from containing many overlapping observations. Lopez de Prado further proposes *sequential bootstrapping*, which draws samples one at a time and reweights the sampling distribution after each draw to penalize overlap with already-selected observations.

10.5 Prediction Targets for the Project

The labeling method (triple-barrier, meta-labeling) defines *how* you label. The prediction target defines *what* you label. This section specifies the four targets from the design spec and explains why each is motivated by theory.

10.5.1 VIX Innovations

VIX Innovation

$$\Delta \text{VIX}_t = \text{VIX}_t - \text{VIX}_{t-1}. \quad (10.6)$$

This is the daily change in implied volatility. Positive ΔVIX_t means the market's expectation of future volatility increased.

Adrian and Shin (2010): Dealer Repos Forecast VIX

Adrian and Shin show that changes in U.S. primary dealer repo positions forecast innovations in the VIX index. A forecasting regression of VIX changes on lagged repo growth produces $R^2 = 11.6\%$, compared to $R^2 = 8.9\%$ using only the lagged VIX level—a relative improvement of 30%. The predictive power comes primarily through the *volatility risk premium* (the gap between implied and realized volatility), not through actual volatility changes. Interpretation: when dealers expand their balance sheets (repo growth > 0), risk appetite rises and implied volatility compresses; when dealers contract (repo growth < 0), risk appetite falls and implied volatility spikes.

Source: Adrian, T. and Shin, H.S. (2010), "Liquidity and Leverage," *Journal of Financial Intermediation*, 19(3), 418–437.

Why this is the cleanest target: VIX innovations are a single, tradeable, observable time series with a direct theoretical link to the risk data you have in SecDB. Adrian–Shin used stale quarterly Fed data; SecDB gives you *daily* dealer-level risk outputs. If the quarterly signal works, the daily signal should be strictly stronger.

10.5.2 Realized Volatility

Realized Volatility

$$RV_t = \sqrt{\sum_{i=1}^n r_{t,i}^2}, \quad (10.7)$$

where $r_{t,i}$ is the i -th intraday return on day t and n is the number of intraday intervals. For daily data without intraday granularity, the simpler proxy is $RV_t = |r_t|$ or $RV_t = r_t^2$.

Why Volatility Is Easier to Predict Than Returns

Returns have very low signal-to-noise ratios (SNR). A Sharpe ratio of 1.0 on daily returns corresponds to $SNR \approx 1/\sqrt{252} \approx 0.063$ —the signal is 6% of the noise. Volatility is more persistent (high autocorrelation), more predictable (GARCH models capture 30–50% of variance), and does not require predicting direction. Predicting RV_{t+1} is a strictly easier problem than predicting r_{t+1} .

Realized volatility is the second-priority target. It is useful both as a standalone prediction (vol-timing strategies) and as an input to volatility-adjusted position sizing. Predicting it with risk-cube features tests whether SecDB data captures information about future realized risk beyond what standard GARCH models provide.

10.5.3 Drawdowns by Most-Concentrated Asset Class

Maximum Drawdown over Horizon

$$DD_{t,h} = \max_{0 \leq s \leq h} \frac{\max_{0 \leq u \leq s} P_{t+u} - P_{t+s}}{\max_{0 \leq u \leq s} P_{t+u}}, \quad (10.8)$$

where h is the forward-looking horizon. This measures the worst peak-to-trough decline in the window $[t, t+h]$.

The design spec identifies factor concentration (HHI of VaR decomposition; see Chapter 9) as a priority feature. The natural target to pair it with is drawdowns in the most-concentrated asset class: when risk is concentrated and then unwinds (fire sales per Coval and Stafford 2007, as discussed in Chapter 2), the concentrated asset class should experience the largest drawdown.

This target is noisier than VIX innovations. Use it as a secondary target, not a starting point.

10.5.4 Cross-Asset Momentum Reversals

Momentum Reversal Indicator

Label $y_t = +1$ if the trailing 12-month winner underperforms the trailing 12-month loser over the next h days (reversal), and $y_t = -1$ otherwise (continuation). Compute across asset classes (equities, rates, credit, FX) rather than within.

Cross-asset momentum reversals are the most complex target. The hypothesis: when VaR utilization is near the limit, forced deleveraging causes recent winners to reverse as dealers unwind their most profitable (and most concentrated) positions. He, Kelly, and Manela (2017, discussed in Chapter 2) show that a single intermediary SDF prices assets across classes, so cross-asset reversals should be predictable from cross-asset risk data.

This target has the longest horizon and the fewest independent observations. Reserve it for Phase 4A if VIX innovations and realized vol produce positive results.

Target Priority Order

Start with VIX innovations (cleanest target per Adrian–Shin, single observable series, direct theory link) and realized volatility (higher SNR than returns, more predictable). Only move to drawdowns and momentum reversals after establishing that SecDB features contain signal on the easier targets. If VIX innovations and realized vol both show zero IC after purged cross-validation (Chapter 11), the harder targets will not rescue the project—that is the Week 13 checkpoint.

10.6 Practical Considerations

10.6.1 Choosing Barrier Parameters

The triple-barrier method has three parameters: m_u , m_ℓ , and $T_{\max}$. Guidelines from practice:

- **Symmetric barriers** ($m_u = m_\ell$) are the default. Start with $m_u = m_\ell = 2$ (i.e., barriers at $\pm 2\sigma$). Asymmetric barriers encode a view that profits and losses are treated differently—only use this if you have a specific reason.
- **Maximum holding period.** Match $T_{\max}$ to the signal’s expected decay. For daily risk-cube features, $T_{\max} = 5$ to 21 trading days is reasonable. Too short and most labels hit the vertical barrier ($y = 0$); too long and labels become noisy.
- **Sensitivity analysis.** Always test at least 3 parameter combinations (e.g., $m = \{1.5, 2.0, 2.5\}$, $T_{\max} = \{5, 10, 21\}$) and report results for all. A signal that only appears at one parameter setting is fragile. Log every combination in the experiment tracker for honest Deflated Sharpe Ratio accounting (Chapter 11).

10.6.2 Handling Label Imbalance

Triple-barrier labels often produce imbalanced classes. In low-volatility periods, most labels hit the vertical barrier ($y = 0$), creating a dominant class. Options:

1. **Merge** $y = 0$ into the sign of the terminal return, reducing to a binary $\{-1, +1\}$ problem.
2. **Upsample** the minority class with replacement before training.
3. **Class weights.** Set $w_c = N/(k \cdot N_c)$, where N is total samples, k is number of classes, and N_c is class count. Scikit-learn and LightGBM support this natively.

10.7 Summary

Concept	Key Takeaway
Fixed-horizon labels	Simple but ignore path, fail across asset classes, and assume an arbitrary holding period.
Triple-barrier labeling	Three barriers (profit, stop, time) with volatility-scaled thresholds. Label = which barrier is hit first. Aligns labels with actual strategy behavior.
Meta-labeling	Primary model predicts side; secondary model predicts whether primary is correct. Trades recall for precision; meta-probability $\rightarrow$ position size.
Label uniqueness	Overlapping return windows violate i.i.d. Average uniqueness $\bar{u}_i = \frac{1}{ T_i } \sum_{t \in T_i} 1/c_t$ gives sample weights for training.
VIX innovations	Cleanest target. Adrian–Shin (2010): dealer repo growth predicts ΔVIX with R^2 improvement from 8.9% to 11.6%.
Realized volatility	Higher SNR than returns. Tests whether SecDB data beats GARCH.
Drawdowns	Pair with factor concentration (HHI). Noisier; use as secondary target.
Momentum reversals	Cross-asset, longest horizon. Reserve for Phase 4A.

The labeling choices made here flow directly into the validation framework (Chapter 11): purged cross-validation must respect the overlapping windows created by triple-barrier labels, and the Deflated Sharpe Ratio must account for every target–parameter combination tested. Build the label module (Task 6) before any signal testing, and commit to the experiment log from the first run.

Chapter 11

Validation Frameworks

Project Connection

This chapter is the validation stack from Phase 1 (Task 7 in the implementation plan). Every number in your project runs through this gauntlet: purged CV, DSR, CPCV, ridge baseline. This is what separates credible work from “intern ran a GBM.” Master this chapter and you can defend your results against any methodological challenge. The modules you build—`src/validation/purged_cv.py`, `src/validation/cpcv.py`, `src/validation/deflated_sharpe.py`, `src/validation/hairecut_sharpe.py`, `src/validation/baseline.py`—are the most important code in the entire project.

11.1 Why Standard K-Fold CV Fails for Time Series

K-Fold Cross-Validation

Standard K -fold CV partitions n observations into K roughly equal groups uniformly at random. For each fold k , the model trains on the remaining $K - 1$ folds and is evaluated on fold k . The final performance estimate is the average across all K folds. This works when observations are independent and identically distributed (i.i.d.)—e.g., classifying images where one cat photo tells you nothing about the next.

Autocorrelation

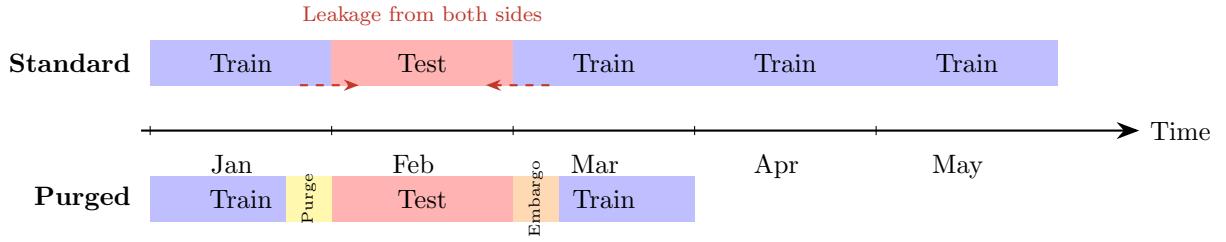
A time series $\{x_t\}$ is **autocorrelated** if $\text{Corr}(x_t, x_{t+h}) \neq 0$ for some lag $h > 0$. Financial returns exhibit weak autocorrelation (typically $|\rho_1| < 0.05$), but features built from returns—rolling means, exponential moving averages, volatility estimates—can have autocorrelation exceeding 0.95 at lag 1 if the lookback window is long.

Standard K -fold CV randomly partitions observations, ignoring their temporal order. This creates a specific, devastating problem: **train-test leakage through time**.

Consider a model trained on January and March data, tested on February. The training set contains observations from *both sides* of the test period. Any feature with temporal persistence—a 21-day rolling mean, a 63-day volatility estimate, a VaR utilization z-score—will carry information from the future into the training set. The model learns patterns that include forward-looking data, inflating test-set performance.

The problem is worse than it first appears. Financial labels themselves have temporal extent. A triple-barrier label (Chapter 10) starting at time t might not resolve until $t + 15$ trading days. If

observation t is in the training set and observation $t + 5$ is in the test set, the training label for t contains information about returns through $t + 15$ —which overlaps with the test observation’s feature window.



Why Random Splits Are Toxic for Finance

Imagine you are predicting tomorrow’s weather, and your training set includes the day after tomorrow’s temperature. Your model would look brilliant—not because it learned atmospheric physics, but because it peeked at the answer sheet. Standard CV on time series does exactly this. The model sees the “answer” (future data adjacent to the test period) in training, then is “tested” on the period sandwiched in between. The resulting performance estimate is not just wrong—it is systematically too optimistic, which is the worst kind of wrong.

11.2 Purged K-Fold CV with Embargo

López de Prado (2018, Ch. 7) introduced purged K -fold CV to solve the leakage problem. It has two components.

Purging

Let observation i have feature time t_i and a label that depends on returns from t_i to $t_i + h_i$ (the **label horizon**). Let the test fold span the time interval $[t_{\text{start}}, t_{\text{end}}]$. **Purge** from the training set any observation j whose label horizon overlaps with the test period:

$$\text{Remove } j \text{ from training if } [t_j, t_j + h_j] \cap [t_{\text{start}}, t_{\text{end}}] \neq \emptyset \quad (11.1)$$

- t_j — the timestamp of training observation j .
- h_j — the label horizon of observation j (e.g., the holding period of a triple-barrier label).
- $[t_{\text{start}}, t_{\text{end}}]$ — the time window of the test fold.

Purging removes training observations whose labels were influenced by returns that occur during the test period.

Embargo

After the end of each test fold at time t_{end} , remove an additional h_e observations from the training set:

$$\text{Remove } j \text{ from training if } t_j \in (t_{\text{end}}, t_{\text{end}} + h_e] \quad (11.2)$$

where h_e is the **embargo length**, typically set as a percentage of the total sample:

$$h_e = \lceil \text{pct}_{\text{embargo}} \times T \rceil \quad (11.3)$$

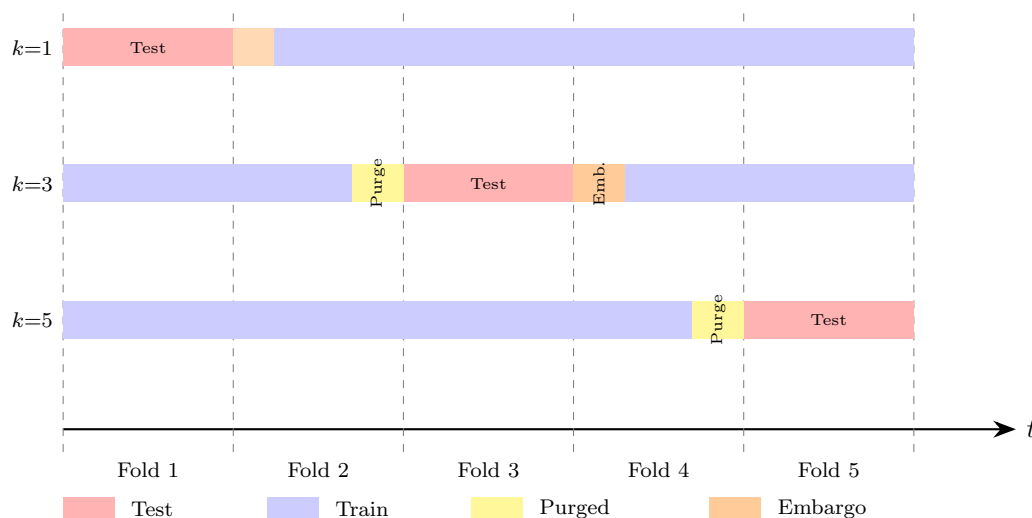
- h_e — the number of observations removed after each test fold boundary.

- $\text{pct}_{\text{embargo}}$ — the embargo fraction (commonly 1–5%).
- T — total number of observations.

The embargo accounts for serial correlation in features that persists beyond the label horizon.

Embargo Must Match Feature Lookback

Even with purging, if your features have long memory—a 63-day rolling volatility, a 21-day exponential moving average—the embargo must be at least as long as the longest feature lookback window. If you use a 63-day rolling standard deviation as a feature and set embargo to 5 days, the first training observation after the test fold still contains 58 days of overlap with test-period returns. Set $h_e \geq \max_k(\text{lookback}_k)$ where k indexes your features.



Worked Example:

Purged CV on 1,250 Daily Observations Suppose you have $T = 1,250$ daily observations (5 years), $K = 5$ folds, and embargo = 2% ($h_e = \lceil 0.02 \times 1,250 \rceil = 25$ days). Your labels have a 10-day horizon.

Fold boundaries: Each fold contains $1,250/5 = 250$ observations.

- Fold 1: obs 1–250 Fold 2: obs 251–500 Fold 3: obs 501–750
- Fold 4: obs 751–1000 Fold 5: obs 1001–1250

Testing fold 3 (obs 501–750). The training set starts as folds 1, 2, 4, 5 (1,000 obs).

Purge: Remove training obs whose 10-day label horizon overlaps [501, 750]:

- Before: obs 491–500 in fold 2 have labels extending to obs 501–510 $\Rightarrow$ remove 10 obs.
- After: obs 751–760 in fold 4 have features at times 751–760, but their labels do not reach back into the test fold, so no purging needed on the trailing side. However, *features* of obs 751–760 might use returns from the test fold (e.g., a 21-day rolling mean at $t = 755$ uses returns from $t = 735$ to $t = 755$). This is handled by the embargo.

Embargo: Remove obs 751–775 from training (25 obs after the test fold boundary).

Effective training size: $1,000 - 10 - 25 = 965$ observations (vs. 1,000 in standard CV).

The cost is modest: 3.5% fewer training observations. The benefit is enormous: elimination of look-ahead bias that can inflate Sharpe ratios by 50% or more.

11.3 Combinatorial Purged CV (CPCV)

Combinatorics: n Choose k

The binomial coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the number of ways to choose k items from n without regard to order. For example, $\binom{6}{2} = 15$.

Standard purged K -fold CV has a limitation: with $K = 5$ folds, you get only 5 Sharpe ratio estimates. That is too thin a distribution to draw reliable conclusions about your strategy's robustness.

Combinatorial Purged Cross-Validation (CPCV), introduced by López de Prado (2018, Ch. 12), solves this by systematically constructing all possible train-test splits from N sequential groups, choosing k as the test set.

CPCV

Partition T observations into N contiguous, non-overlapping groups of approximately equal size. For each combination of k groups selected as the test set (out of N total groups), train on the remaining $N - k$ groups with purging and embargo applied. This yields:

$$\text{Number of splits} = \binom{N}{k} \quad (11.4)$$

Each split produces one out-of-sample (OOS) Sharpe ratio estimate. The number of unique backtest paths (non-overlapping sequences through all time periods) is:

$$\varphi(N, k) = \frac{k}{N} \binom{N}{k} \quad (11.5)$$

- N — number of contiguous time groups.
- k — number of groups used as the test set in each split.
- $\binom{N}{k}$ — total combinations tested.
- $\varphi(N, k)$ — the number of distinct backtest paths, each covering the full time period.

Worked Example:

CPCV with $N = 6$, $k = 2$ Partition 1,250 observations into $N = 6$ groups of ~ 208 observations each. Choose $k = 2$ groups as test.

Number of splits: $\binom{6}{2} = 15$.

Number of paths: $\varphi(6, 2) = \frac{2}{6} \times 15 = 5$.

One specific combination: test groups $\{2, 5\}$, train on groups $\{1, 3, 4, 6\}$. After purging and embargo, the model trains on groups 1, 3, 4, 6 (minus purged/embargoed observations near group 2 and group 5 boundaries), and is tested on groups 2 and 5.

Each of the 6 groups appears as a test group in exactly $\binom{5}{1} = 5$ of the 15 splits, ensuring uniform coverage.

The 15 test-set combinations:

$\{1, 2\}$	$\{1, 3\}$	$\{1, 4\}$	$\{1, 5\}$	$\{1, 6\}$
$\{2, 3\}$	$\{2, 4\}$	$\{2, 5\}$	$\{2, 6\}$	
$\{3, 4\}$	$\{3, 5\}$	$\{3, 6\}$		
$\{4, 5\}$	$\{4, 6\}$			
$\{5, 6\}$				

From these 15 splits, you construct 5 non-overlapping backtest paths, each covering all

6 time groups. This gives you a distribution of 5 full-sample Sharpe ratios, compared to the single path from standard walk-forward or the 5 fold-level estimates from standard K -fold.

CPCV Gives You a Distribution, Not a Point Estimate

Standard CV gives you one average performance estimate. CPCV gives you a *distribution* of Sharpe ratios from a single history. If the distribution is tight and centered above zero, the signal is robust. If it is wide and straddles zero, you are likely fitting noise. This distribution is the input to the Probability of Backtest Overfitting (Section 11.6).

Scaling up: with $N = 10$ and $k = 2$, you get $\binom{10}{2} = 45$ splits and $\varphi(10, 2) = 9$ paths. With $N = 12$ and $k = 2$, you get 66 splits and 11 paths. More splits mean a richer distribution, but each split has a smaller test set and more purging overhead. In practice, $N = 6$ to $N = 10$ with $k = 2$ is a reasonable range for daily financial data.

11.4 The Deflated Sharpe Ratio

The Standard Normal CDF

The cumulative distribution function of the standard normal distribution is $\Phi(z) = \Pr(Z \leq z)$ where $Z \sim \mathcal{N}(0, 1)$. Its inverse $\Phi^{-1}(p)$ gives the z -value at which the CDF equals p . For example, $\Phi^{-1}(0.975) = 1.96$ and $\Phi^{-1}(0.95) = 1.645$.

Skewness and Kurtosis

For a random variable X with mean μ and standard deviation σ :

- **Skewness:** $\hat{\gamma}_3 = \frac{1}{T} \sum_{t=1}^T \left(\frac{x_t - \mu}{\sigma} \right)^3$. Negative skewness means the left tail is heavier (large losses more likely than large gains). Hedge fund returns typically have $\hat{\gamma}_3 \in [-1, -0.3]$.
- **Excess kurtosis:** $\hat{\gamma}_4 = \frac{1}{T} \sum_{t=1}^T \left(\frac{x_t - \mu}{\sigma} \right)^4 - 3$. Positive excess kurtosis means fatter tails than the normal distribution. Financial returns typically have $\hat{\gamma}_4 \in [1, 5]$.

You tested 30 feature configurations. The best has Sharpe ratio 1.2. Is it real, or did you just get lucky by picking the best out of many attempts?

This is the **multiple testing problem** applied to strategy selection. Bailey and López de Prado (2014) developed the **Deflated Sharpe Ratio** (DSR) to answer exactly this question.

11.4.1 Expected Maximum Sharpe Under the Null

If you run N independent trials on data with no true signal, the expected maximum Sharpe ratio among them is not zero—it grows with N . The **False Strategy Theorem** gives the benchmark:

Expected Maximum Sharpe Ratio (False Strategy Theorem)

If N independent strategies are tested on the same data, each with true Sharpe ratio zero, the expected maximum observed Sharpe ratio is approximately:

$$SR_0 \approx \sqrt{\widehat{V[SR_n]}} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right] \quad (11.6)$$

- SR_0 — the expected maximum Sharpe ratio under the null (no skill).
- $\mathbf{V}[\widehat{SR}_n]$ — the cross-sectional variance of Sharpe ratio estimates across the N trials. Under the null with i.i.d. returns, $\mathbf{V}[\widehat{SR}_n] \approx 1/(T-1)$.
- $\gamma \approx 0.5772$ — the Euler–Mascheroni constant.
- $e \approx 2.718$ — Euler’s number.
- N — the number of independent strategy trials.
- $\Phi^{-1}(\cdot)$ — inverse standard normal CDF.

For a rough approximation when N is moderate to large, $SR_0 \approx \sqrt{2 \ln N/T}$ (from extreme value theory for Gaussians), but equation (11.6) is more accurate and accounts for the cross-sectional variance of observed Sharpe ratios.

11.4.2 The DSR Formula

Deflated Sharpe Ratio

The DSR is the probability that the observed Sharpe ratio $\widehat{SR}^*$ exceeds the expected maximum under the null:

$$DSR = \Phi \left(\frac{(\widehat{SR}^* - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 SR_0 + \frac{\hat{\gamma}_4 - 1}{4} SR_0^2}} \right) \quad (11.7)$$

- $\widehat{SR}^*$ — the observed (non-annualized) Sharpe ratio of the best strategy.
- SR_0 — the expected maximum Sharpe under the null, from equation (11.6).
- T — the number of return observations (sample size).
- $\hat{\gamma}_3$ — skewness of the strategy’s returns.
- $\hat{\gamma}_4$ — kurtosis of the strategy’s returns (not excess kurtosis; use $\hat{\gamma}_4 = \text{excess kurtosis} + 3$).
- $\Phi(\cdot)$ — standard normal CDF.

DSR ranges from 0 to 1. A DSR of 0.95 means: even after accounting for the number of trials, non-normality, and sample size, there is a 95% probability that the strategy has genuine skill. $DSR < 0.5$ means the observed Sharpe is more likely a product of selection bias than real signal.

The denominator is the standard deviation of the Sharpe ratio estimator, adjusted for non-normal returns:

$$\hat{\sigma}(\widehat{SR}) = \sqrt{\frac{1 - \hat{\gamma}_3 SR_0 + \frac{\hat{\gamma}_4 - 1}{4} SR_0^2}{T-1}} \quad (11.8)$$

When returns are normal ($\hat{\gamma}_3 = 0$, $\hat{\gamma}_4 = 3$), this simplifies to $\hat{\sigma}(\widehat{SR}) = \sqrt{(1 + SR_0^2/2)/(T-1)}$, the standard Lo (2002) result.

Worked Example:

Computing the Deflated Sharpe Ratio You ran 30 experiments on 5 years of daily data ($T = 1,250$). The best strategy has annualized Sharpe 1.20, with return skewness $\hat{\gamma}_3 = -0.5$ and kurtosis $\hat{\gamma}_4 = 4.0$.

Step 1: De-annualize. $\widehat{SR}^* = 1.20/\sqrt{252} \approx 0.0756$ (daily).

Step 2: Compute SR_0 . Assume $\mathbf{V}[\widehat{SR}_n] \approx 1/(T-1) = 1/1249 \approx 0.000801$, so $\sqrt{\mathbf{V}} \approx$

0.0283. With $N = 30$:

$$\begin{aligned}\Phi^{-1}(1 - 1/30) &= \Phi^{-1}(0.9667) \approx 1.834 \\ \Phi^{-1}(1 - 1/(30 \cdot e)) &= \Phi^{-1}(0.9877) \approx 2.249 \\ \text{SR}_0 &\approx 0.0283 \times [(1 - 0.5772)(1.834) + 0.5772(2.249)] \\ &\approx 0.0283 \times [0.775 + 1.299] \\ &\approx 0.0283 \times 2.074 \approx 0.0587\end{aligned}$$

Step 3: Compute the denominator.

$$1 - (-0.5)(0.0587) + \frac{4.0 - 1}{4}(0.0587)^2 = 1 + 0.0294 + 0.000258 \approx 1.0296$$

Step 4: Compute DSR.

$$\text{DSR} = \Phi\left(\frac{(0.0756 - 0.0587)\sqrt{1249}}{\sqrt{1.0296}}\right) = \Phi\left(\frac{0.0169 \times 35.34}{1.0147}\right) = \Phi(0.589) \approx 0.72$$

Interpretation: DSR = 0.72. After accounting for 30 trials, there is only a 72% probability that the Sharpe of 1.20 reflects genuine skill rather than selection bias. This is below the 0.95 threshold most researchers would require for confidence. You cannot claim a credible signal at this level.

Bailey and López de Prado (2014)

With 45 independent trials on 5 years of daily data ($T \approx 1,250$), the expected maximum Sharpe ratio under the null is approximately 1.87 (annualized). This means a reported Sharpe of 1.0 after 45 trials has DSR ≈ 0.02 —virtually certain to be spurious. Even a Sharpe of 1.5 requires fewer than ~ 10 trials to maintain DSR above 0.95 on this sample size. The implication: track every experiment, because the number of trials is the single biggest driver of whether your results are credible.

Every Experiment Counts as a Trial

Every configuration you test—different feature subsets, different targets, different hyperparameters, different label horizons—counts as a trial for DSR purposes. If you try 5 feature sets $\times$ 4 targets $\times$ 3 model types, that is $N = 60$ trials, not 1. If you do not log failed experiments, your DSR is dishonestly high, because you are understating N . This is why the experiment tracker (Task 9, `src/tracking/tracker.py`) is not optional bureaucracy—it is the integrity mechanism that makes DSR honest.

11.5 Haircut Sharpe Ratio

p -Values and Hypothesis Testing

The p -value is the probability of observing a test statistic at least as extreme as the one obtained, assuming the null hypothesis is true. A p -value of 0.01 means: if there were truly no effect, you would see a result this extreme only 1% of the time. When you run multiple tests, the probability that *at least one* yields a small p -value by chance increases rapidly. Multiple testing corrections adjust p -values upward to account for this.

Harvey and Liu (2015) take a different approach to the same problem: instead of computing a probability (DSR), they compute a **haircut**—the percentage reduction applied to a reported Sharpe ratio to account for multiple testing.

11.5.1 Sharpe-to- t -Statistic Conversion

The foundation is the relationship between the Sharpe ratio and the t -statistic:

$$t = \text{SR} \times \sqrt{T} \quad (11.9)$$

where SR is the non-annualized Sharpe ratio and T is the number of observations. Testing whether a Sharpe ratio is significantly different from zero is equivalent to testing whether the mean excess return is significantly different from zero, which is a standard t -test.

11.5.2 Three Correction Methods

Harvey and Liu (2015) apply three progressively less conservative multiple testing corrections.

Bonferroni Correction

Given M tests with individual p -values $p_1, \dots, p_M$, the Bonferroni-adjusted p -values are:

$$p_i^{\text{Bonf}} = \min(M \cdot p_i, 1) \quad (11.10)$$

Every p -value is inflated by the total number of tests. This controls the **family-wise error rate** (FWER): the probability of even one false rejection. It is the most conservative correction.

Holm Step-Down Correction

Sort the M p -values in ascending order: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(M)}$. The adjusted p -values are:

$$p_{(i)}^{\text{Holm}} = \min\left(\max_{j \leq i} [(M - j + 1) \cdot p_{(j)}], 1\right) \quad (11.11)$$

Holm applies decreasing penalties: the smallest p -value is multiplied by M (same as Bonferroni), the second-smallest by $M - 1$, and so on. It also controls FWER but is uniformly more powerful (less conservative) than Bonferroni.

BHY (Benjamini–Hochberg–Yekutieli) FDR Correction

Sort p -values descending: $p_{(M)} \geq p_{(M-1)} \geq \dots \geq p_{(1)}$. The adjusted p -values are:

$$p_{(i)}^{\text{BHY}} = \min\left(\frac{M \cdot c(M)}{i} \cdot p_{(i)}, 1\right) \quad (11.12)$$

where $c(M) = \sum_{j=1}^M 1/j$ is the M -th harmonic number. For $M = 20$: $c(20) \approx 3.55$.

BHY controls the **false discovery rate** (FDR)—the expected fraction of rejections that are false—rather than the stricter FWER. It is the least conservative of the three.

11.5.3 Computing the Haircut

For each correction method, the haircut percentage on a given strategy's Sharpe ratio is:

$$\text{Haircut \%} = \frac{\text{SR}_{\text{reported}} - \text{SR}_{\text{adjusted}}}{\text{SR}_{\text{reported}}} \times 100 \quad (11.13)$$

where SR_{adjusted} is the Sharpe ratio corresponding to the corrected p -value. If the adjusted p -value is no longer significant (exceeds the significance level α), the adjusted Sharpe is effectively zero and the haircut is 100%.

Worked Example:

Haircut Sharpe with 20 Tests You report a strategy with annualized $SR = 1.50$ from $T = 1,250$ daily observations, and you have tested $M = 20$ strategies total.

Step 1: Convert to t -statistic. Daily $SR = 1.50/\sqrt{252} = 0.0945$. Then $t = 0.0945 \times \sqrt{1,250} = 3.34$.

Step 2: Single-test p -value. $p = 2(1 - \Phi(3.34)) \approx 0.00084$ (two-sided).

Step 3: Bonferroni. $p^{\text{Bonf}} = 20 \times 0.00084 = 0.0168$. Still significant at $\alpha = 0.05$. The adjusted t -statistic is $\Phi^{-1}(1 - 0.0168/2) = 2.39$, giving adjusted $SR_{\text{ann}} = (2.39/\sqrt{1,250}) \times \sqrt{252} = 1.07$. Haircut: $(1.50 - 1.07)/1.50 = 28.5\%$.

Step 4: Holm. If this strategy's p -value ranks, say, 1st of 20, the Holm adjustment for the top-ranked is identical to Bonferroni: $20 \times 0.00084 = 0.0168$. Haircut: $\approx 28.5\%$ (same for the best strategy).

Step 5: BHY. With $c(20) = 3.55$: $p^{\text{BHY}} = (20 \times 3.55/1) \times 0.00084 = 0.0597$. Marginally insignificant at $\alpha = 0.05$. Adjusted $t = \Phi^{-1}(1 - 0.0597/2) = 1.88$, giving adjusted $SR_{\text{ann}} = 0.84$. Haircut: $(1.50 - 0.84)/1.50 = 43.7\%$.

Summary:

Method	Adjusted p	Adjusted SR	Haircut %
None (single test)	0.00084	1.50	0%
Bonferroni	0.0168	1.07	28.5%
Holm (rank 1)	0.0168	1.07	28.5%
BHY	0.0597	0.84	43.7%

A Sharpe of 1.50 after 20 tests survives Bonferroni and Holm (still significant at 5%), but the BHY adjustment renders it borderline. The 50% rule-of-thumb haircut that practitioners sometimes apply is in the right ballpark for BHY but too harsh for Bonferroni in this case.

Harvey, Liu, and Zhu (2016)

Cataloging 316 factors tested in the academic literature, Harvey, Liu, and Zhu show that the appropriate t -statistic hurdle for a newly proposed factor is $t > 3.0$, not the conventional 1.96. Using this criterion, only 9 of 313 return-correlated variables survive. A Sharpe ratio of 1.0 annualized on 5 years of daily data gives $t = (1.0/\sqrt{252}) \times \sqrt{1,250} \approx 2.23$ —below the 3.0 threshold. The implication: most published “factors” are likely false discoveries.

Harvey–Liu–Zhu vs. DSR: Complementary Tools

DSR and Haircut Sharpe address the same problem from different angles. DSR computes a probability that your observed Sharpe exceeds the expected maximum under the null. Haircut Sharpe computes the adjusted Sharpe after penalizing for multiple tests. Use both. If $DSR > 0.95$ and the haircutted Sharpe is still economically meaningful (say, above 0.5 annualized after costs), you have a defensible result.

11.6 Probability of Backtest Overfitting (PBO)

Bailey, Borwein, López de Prado, and Zhu (2014) propose a direct, model-free measure of whether a strategy selection process is overfit: the **Probability of Backtest Overfitting** (PBO).

11.6.1 Combinatorially Symmetric Cross-Validation (CSCV)

The procedure works on the $T \times N$ matrix of trial P&L, where T is the number of time periods and N is the number of strategy configurations tested.

CSCV and PBO

Step 1. Partition the T time periods into S non-overlapping, contiguous blocks of approximately equal size (S must be even).

Step 2. Form all $\binom{S}{S/2}$ combinations of $S/2$ blocks. For each combination c :

1. The selected $S/2$ blocks form the in-sample (IS) set J_c .
2. The remaining $S/2$ blocks form the out-of-sample (OOS) set $\bar{J}_c$.
3. Compute the performance metric (Sharpe ratio) for each of the N strategies on both IS and OOS.
4. Let $n_c^* = \arg \max_n \text{SR}_n^{\text{IS}}(c)$ be the strategy that performed best in-sample.
5. Compute its OOS relative rank: $\omega_c = (\text{rank of } n_c^* \text{ in OOS})/N$.

Step 3. Compute the logit of the relative rank:

$$\lambda_c = \ln\left(\frac{\omega_c}{1 - \omega_c}\right) \quad (11.14)$$

Step 4. The **Probability of Backtest Overfitting** is:

$$\text{PBO} = \Pr(\lambda < 0) = \frac{\#\{c : \lambda_c < 0\}}{\binom{S}{S/2}} \quad (11.15)$$

- S — number of temporal blocks (typically 8–16; must be even).
- $\binom{S}{S/2}$ — number of IS/OOS partitions. For $S = 10$: $\binom{10}{5} = 252$.
- n_c^* — the strategy that performed best in-sample for partition c .
- ω_c — the relative rank of n_c^* in the OOS sample (0 = worst, 1 = best).
- $\lambda_c < 0$ means the IS-optimal strategy ranked below median OOS.
- $\text{PBO} > 0.50$ means the IS-best strategy is more likely to underperform than outperform OOS—the selection process is overfit.

What PBO Really Measures

PBO answers a simple question: “If I pick the strategy that looks best in-sample, what fraction of the time does it rank below average out-of-sample?” If $\text{PBO} = 0.60$, then 60% of the time, the strategy you would choose based on backtesting actually underperforms the median strategy in fresh data. Your selection process is not identifying skill—it is identifying noise, and the noise looks different in every partition.

For $S = 10$, you get 252 IS/OOS partitions—a rich enough sample to estimate PBO reliably. The rule of thumb: $\text{PBO} < 0.30$ is encouraging; $\text{PBO} > 0.50$ is a red flag; $\text{PBO} > 0.70$ means your strategy selection is almost certainly overfit and you should not proceed.

11.7 Experiment Tracking for Honest DSR

The DSR, Haircut Sharpe, and PBO all depend on knowing N —the number of trials. If you do not track trials systematically, you will understate N and overstate significance. This is not a theoretical concern; it is the most common form of unintentional p-hacking in quantitative research.

The Experiment Log Is as Important as the Model Code

Every configuration you test must be logged, including failures. The log should record:

Field	Example
Experiment ID	exp-042
Timestamp	2026-06-15 14:30
Features	VaR utilization z-score, factor HHI
Target	VIX 5-day innovation
Model	LightGBM, 100 trees, max_depth=4
CV method	Purged 5-fold, embargo=2%
Raw Sharpe (annualized)	0.87
DSR (cumulative N)	0.68 ($N = 42$)
Notes	Weak on credit targets

This log is the input to `src/tracking/tracker.py` (Task 9). When you compute DSR at experiment 42, $N = 42$. If you had deleted experiments 10–25 because they “didn’t work,” you would compute DSR with $N = 27$ and get a higher, dishonest value.

What counts as a new trial?

- Different feature subset — yes, new trial.
- Different target variable — yes, new trial.
- Different model type (ridge vs. GBM) — yes, new trial.
- Different hyperparameter configuration — yes, if you select based on OOS performance. If you use an inner CV loop for hyperparameter tuning and only the outer fold counts for selection, then only the outer configurations count.
- Re-running the same configuration after a bug fix — not a new trial (the previous run was invalid, not a genuine test of a different hypothesis).

11.8 The Ridge Baseline: Your Credibility Floor

Every result you report must be accompanied by a ridge regression baseline on the identical feature set. This is not optional.

Why Ridge, Not OLS

OLS overfits when the number of features approaches the number of observations (Chapter 5). Ridge adds an ℓ_2 penalty that shrinks coefficients toward zero, giving a stable baseline that is hard to overfit. If your GBM does not beat ridge on the same features and the same purged CV, the GBM is fitting noise—the nonlinear interactions it is detecting are not real, they are artifacts.

The baseline protocol:

1. Standardize all features to zero mean and unit variance.
2. Fit ridge regression with λ chosen by inner purged CV (3-fold within each outer training fold).

3. Evaluate on the same outer test fold as your GBM.
4. Report ridge Sharpe alongside GBM Sharpe in every table and chart.

If GBM Sharpe exceeds ridge Sharpe by a meaningful margin (e.g., > 0.2 annualized) and this improvement is consistent across CPCV paths, there is evidence of genuine nonlinear signal. If $\text{GBM} \approx \text{ridge}$, the signal is linear and a simpler model suffices. If $\text{GBM} < \text{ridge}$, the GBM is overfitting.

The Ridge Test in Your Checkpoint

At the Week 13 checkpoint (Task 16), the decision criteria explicitly include “GBM beats ridge.” If the GBM does not outperform the ridge baseline, the checkpoint assessment pivots toward Project 2 or the hybrid option. This is not punitive—it is information. A ridge-competitive result means the signal is linear, which is still valuable but may not justify the complexity of a tree-based model. Present it honestly: “The signal exists and is well-captured by a linear model.”

11.9 The Sample-Size Problem

Five years of daily data gives you approximately 1,250 observations. This is not a lot.

Bailey, Borwein, López de Prado, and Zhu (2014)

On 5 years of daily data ($T \approx 1,250$), approximately 45 independent trials are sufficient for the expected maximum Sharpe ratio under the null to reach ~ 1.87 (annualized). After 45 trials, a Sharpe of 1.0 is almost certainly noise. After 100 trials, even a Sharpe of 1.5 becomes dubious. The fundamental constraint is not computational—it is informational: there are only 1,250 independent observations, and each trial extracts a bit more of the finite signal from the noise floor.

This creates a tension: you want to explore many hypotheses (more trials $\Rightarrow$ better chance of finding something real), but each trial inflates N (more trials $\Rightarrow$ higher bar for significance). Three mitigations:

11.9.1 Theory-Motivated Features

Features grounded in economic theory (intermediary asset pricing, He–Krishnamurthy 2013, Chapter 2) are not the same as features generated by a data-mining search. The DSR framework assumes trials are independent draws from the null. If your feature is motivated by a specific causal hypothesis—“VaR utilization proxies for dealer risk capacity, which theory says should price assets”—it is more defensible than “I tried 200 technical indicators and this one worked.” This does not let you skip DSR, but it strengthens the narrative when presenting to a skeptical audience.

11.9.2 Panel Structure

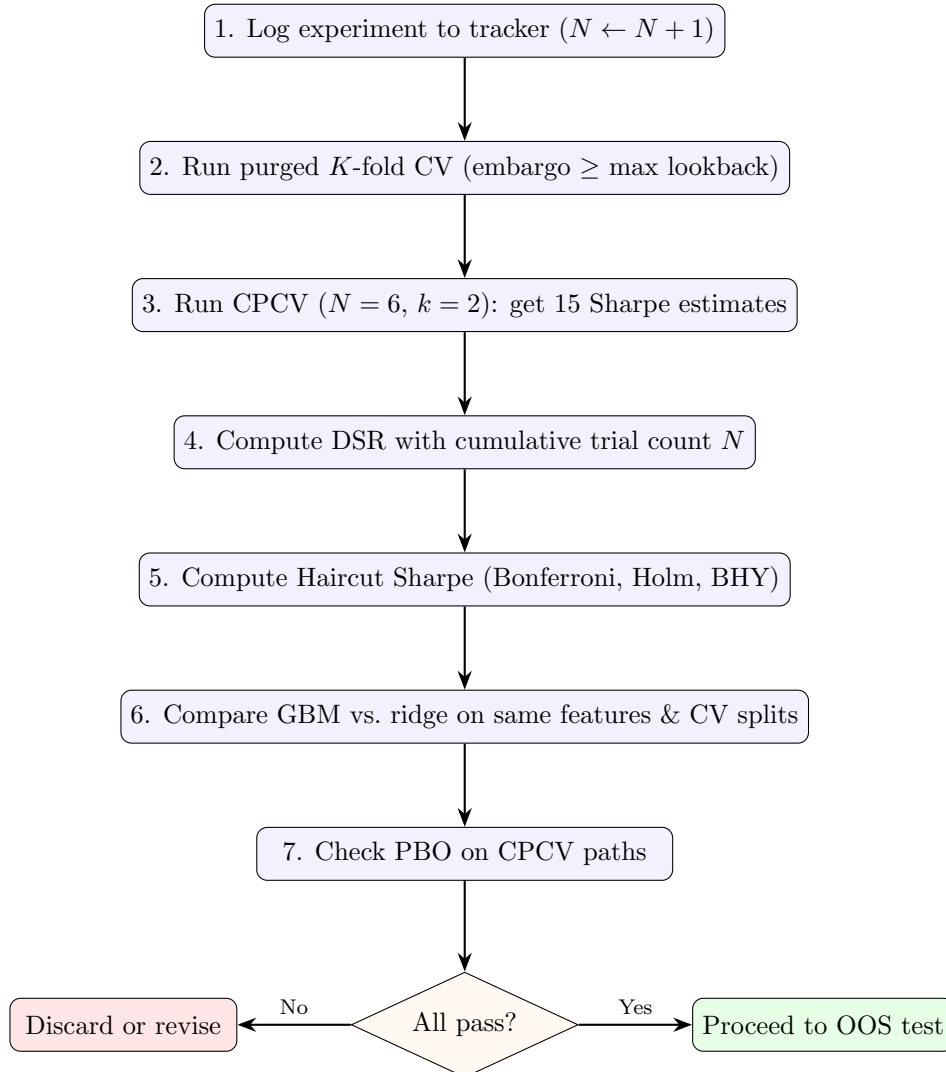
Instead of treating each asset class as a separate experiment, use a panel regression (Chapter 8) with asset-class fixed effects. A single panel model that tests VaR utilization across equities, rates, FX, and credit is *one trial*, not four. This preserves N while increasing the effective sample size to $1,250 \times K_{\text{assets}}$, improving statistical power without inflating the multiple-testing penalty.

11.9.3 Pre-Registration

Before looking at results, write down your hypotheses and the features you will test. Specify the decision criteria. This is the scientific analog of “pre-registering” a clinical trial. It does not change the statistics, but it gives you intellectual credibility when you claim that your 5 experiments were the only 5 you ran, rather than the 5 survivors out of 50.

11.10 Putting It All Together: The Validation Pipeline

The full validation pipeline for every experiment in your project is:



A result passes the gauntlet if:

1. Purged CV Sharpe > 0 consistently across folds.
2. DSR > 0.95 (or at minimum > 0.50 , with honest N).
3. Haircutted Sharpe is economically meaningful (> 0.5 annualized after costs).
4. GBM beats ridge (or ridge alone is acceptable as a simpler model).
5. PBO < 0.50 (ideally < 0.30).

Only results that survive this pipeline should touch the reserved holdout data in Phase 5 (Chapter 12). The holdout is a one-shot test—you do not iterate on it. If you send a noise-fitted model to holdout, you waste your only chance at a clean OOS result.

11.11 Summary

1. Standard K -fold CV fails for time series because random partitioning creates train-test leakage through temporally autocorrelated features and overlapping label horizons.
2. **Purged CV** removes training observations whose labels overlap with the test period. **Embargo** removes additional observations after the test boundary to handle serial correlation in features.
3. **CPCV** generates $\binom{N}{k}$ train-test splits from N contiguous groups, producing a distribution of Sharpe ratios rather than a single estimate. With $N = 6$, $k = 2$: 15 splits, 5 paths.
4. The **Deflated Sharpe Ratio** accounts for multiple testing, non-normality, and sample size: $\text{DSR} = \Phi\left(\frac{(\widehat{\text{SR}}^* - \text{SR}_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3 \text{SR}_0 + (\hat{\gamma}_4-1) \text{SR}_0^2/4}}\right)$. Require $\text{DSR} > 0.95$.
5. The **expected maximum Sharpe under the null** grows with the number of trials N . With 45 trials on 5 years of daily data, $\mathbb{E}[\max \text{SR}] \approx 1.87$ annualized.
6. **Haircut Sharpe** applies Bonferroni, Holm, or BHY corrections. Harvey–Liu–Zhu (2016): a new factor needs $t > 3.0$, not 1.96. Only 9 of 313 published factors survive.
7. **PBO** = $\Pr(\lambda < 0)$ where λ is the logit of the IS-optimal strategy's OOS rank. $\text{PBO} > 0.50 = \text{overfit}$.
8. The **ridge baseline** is your credibility floor. GBM must beat ridge on identical features and CV splits.
9. **Every experiment must be logged**. The experiment tracker ensures honest trial counting for DSR.
10. With $T \approx 1,250$ daily observations, approximately 45 trials exhaust a Sharpe of 1.0. Mitigate with theory-motivated features, panel structure, and pre-registration.

Key Results Reference

Paper	Key Claim	Headline Result
López de Prado (2018), Ch. 7, 12	Standard CV leaks in time series; purging and embargo fix it	Purged CV + CPCV: the correct CV framework for financial ML
Bailey & López de Prado (2014)	Selection bias inflates Sharpe ratios	DSR: 45 trials on 5yr daily data $\Rightarrow \mathbb{E}[\max \text{SR}] \approx 1.87$; Sharpe 1.0 has $\text{DSR} \approx 0.02$
Harvey & Liu (2015)	Backtested Sharpe must be haircutted	Three corrections: Bonferroni, Holm, BHY; 50% rule of thumb is imprecise
Harvey, Liu, & Zhu (2016)	Most published factors are false discoveries	$t > 3.0$ hurdle; only 9 of 313 factors survive; 316+ factors tested
Bailey, Borwein, LdP, Zhu (2014)	Backtest overfitting is quantifiable	PBO via CSCV; $\text{PBO} > 0.50 = \text{more likely overfit than not}$

Chapter 12

Backtesting Methodology

Project Connection

This chapter covers the backtesting engine from Phase 1 (Task 8) and the walk-forward OOS test in Phase 5 (Task 22). Transaction-cost-aware P&L, turnover, and capacity estimation are “what the trading floor asks about first”—the practical questions that determine whether a signal is tradeable. The modules you build here—`src/backtest/engine.py`, `src/backtest/metrics.py`, `src/backtest/reporting.py`—translate statistical significance into economic significance.

Chapter 11 gave you the statistical toolkit: purged CV, DSR, CPCV, PBO, ridge baseline. Those tools answer *is this signal real?* This chapter answers the follow-up question: *if it is real, can anyone trade it?* A signal with IC = 0.05 and a Sharpe of 1.2 before costs is worthless if turnover eats the spread. Every number you report in your project must survive both the validation gauntlet and the transaction-cost gauntlet.

12.1 Transaction-Cost-Aware P&L

Basis Points

One basis point (bp) = 0.01% = 0.0001. A 5 bp cost means you pay 0.05% of notional each time you trade. Financial practitioners quote spreads, fees, and slippage in basis points because percentages are too coarse for large-notional positions.

Signal as Position Weight

Throughout this chapter, $\hat{y}_t$ denotes the signal (model prediction) at time t , interpreted as a *signed position weight*: $\hat{y}_t > 0$ means long, $\hat{y}_t < 0$ means short, and the magnitude determines position size. In your project, $\hat{y}_t$ comes from the LightGBM or ridge model trained on risk-cube features (Chapter 9). We assume $\hat{y}_t$ is scaled so that $|\hat{y}_t| \leq 1$.

12.1.1 Gross P&L

The simplest P&L measure ignores trading costs entirely:

$$\text{PnL}_t^{\text{gross}} = \hat{y}_t \cdot r_{t+1}, \quad (12.1)$$

where:

- $\hat{y}_t$ — signal (position weight) formed at end of day t , using information available up to t .
- r_{t+1} — realized return from close of day t to close of day $t + 1$.

The product is the return earned by holding $\hat{y}_t$ units through the next period. Note the strict time alignment: $\hat{y}_t$ must use only data from t and earlier. Using r_t instead of r_{t+1} is the most common backtest bug and produces spectacularly inflated results.

The Off-by-One That Ends Careers

If your backtest uses $\hat{y}_t \cdot r_t$, the signal is multiplied by a return that has already been realized. This is look-ahead bias, and it will produce Sharpe ratios above 3.0 routinely. *Every* backtest implementation must be audited for this. The fix: always pair the signal timestamp with the *next-period* return. In your pipeline, the point-in-time module (`src/data/point_in_time.py`) enforces this alignment.

12.1.2 Net P&L

Trading costs are proportional to the *change* in position:

$$\text{PnL}_t^{\text{net}} = \hat{y}_t \cdot r_{t+1} - |\hat{y}_t - \hat{y}_{t-1}| \cdot c, \quad (12.2)$$

where:

- $|\hat{y}_t - \hat{y}_{t-1}|$ — absolute change in position (the amount traded).
- c — half-spread cost per unit of notional, in the same units as r_{t+1} .

The cost term $|\hat{y}_t - \hat{y}_{t-1}| \cdot c$ captures the spread you pay each time the position changes. If $\hat{y}_t = \hat{y}_{t-1}$, you pay nothing. If the signal flips from $+1$ to -1 , you pay $2c$ (a full round-trip).

12.1.3 Parameterized Spreads by Asset Class

Costs vary enormously across asset classes. Typical half-spread estimates for liquid instruments:

Asset Class	Half-Spread (bps)	Example Instrument
G10 FX (spot)	1–2	EUR/USD
US Treasuries (on-the-run)	0.5–1	10y UST
Equity index futures	0.5–1	E-mini S&P 500
IG credit (CDX)	2–5	CDX.IG
EM FX	5–15	USD/BRL
Single-name CDS	5–10	IG single names

Report Sharpe at Multiple Cost Levels

Never report a single Sharpe ratio. Compute net Sharpe at $c = 0, 1, 2, 5, 10$, and 20 bps. Plot the **Sharpe-vs-cost curve**: a monotonically decreasing function that shows exactly where the signal dies. The **breakeven cost** c^* is the spread at which net Sharpe = 0. This is the single number the desk cares about most. If $c^* > 5$ bps for liquid rates or FX, the signal is plausibly tradeable. If $c^* < 1$ bp, it is not.

Worked Example:

Gross vs. Net P&L Over 5 Days
Suppose $c = 5$ bps = 0.0005, and $\hat{y}_0 = 0$ (flat at start).

t	$\hat{y}_t$	r_{t+1} (%)	$ \Delta\hat{y}_t $	Gross (%)	Net (%)
1	+0.80	+0.30	0.80	+0.240	+0.200
2	+0.60	-0.10	0.20	-0.060	-0.070
3	-0.50	-0.40	1.10	+0.200	+0.145
4	-0.50	+0.20	0.00	-0.100	-0.100
5	+0.30	+0.50	0.80	+0.150	+0.110
			$\Sigma = 2.90$	$\Sigma = +0.430$	$\Sigma = +0.285$

Day-by-day:

- **Day 1:** Enter long 0.80. Gross = $0.80 \times 0.30\% = 0.240\%$. Cost = $0.80 \times 0.05\% = 0.040\%$. Net = 0.200% .
- **Day 3:** Flip from +0.60 to -0.50 costs $1.10 \times 0.05\% = 0.055\%$. The flip is expensive—but the short earns 0.200% gross because the market falls.
- **Day 4:** No trade ($\Delta\hat{y} = 0$), so gross = net. Holding a short into a rising market loses 0.100% .

Cumulative gross P&L = $+0.430\%$; cumulative net = $+0.285\%$. Costs consumed $(0.430 - 0.285)/0.430 = 33.7\%$ of gross returns. This cost drag is typical for a moderately active signal on liquid instruments. On a 5 bp spread, one-third of gross evaporated. At 10 bps it would be two-thirds.

Breakeven cost. The breakeven cost c^* is the cost level where cumulative net P&L = 0. Here, cumulative $|\Delta\hat{y}| = 2.90$, cumulative gross = 0.430% . Solving $0.430\% - 2.90 \times c^* = 0$ gives $c^* = 0.430/2.90 = 0.148\% \approx 15$ bps. This signal survives costs up to 15 bps—comfortable for G10 FX or rates futures, marginal for EM instruments.

12.2 Turnover

Turnover

Daily turnover is the absolute change in position weight:

$$\tau_t = |\hat{y}_t - \hat{y}_{t-1}|. \quad (12.3)$$

Average daily turnover over T days:

$$\bar{\tau} = \frac{1}{T} \sum_{t=1}^T |\hat{y}_t - \hat{y}_{t-1}|. \quad (12.4)$$

Annualized turnover = $252 \times \bar{\tau}$. A signal with $\bar{\tau} = 0.10$ trades the equivalent of 10% of its position daily, or roughly $25 \times$ annual turnover.

Why Turnover Matters

Transaction costs are proportional to turnover. A signal with high IC but high turnover may have lower net Sharpe than a weaker signal with low turnover. The net Sharpe of any signal can be decomposed as:

$$SR^{\text{net}} \approx SR^{\text{gross}} - \frac{252 \cdot \bar{\tau} \cdot c}{\sigma_{\text{ret}}},$$

where σ_{ret} is the annualized volatility of gross returns. The second term is the Sharpe-ratio drag from costs.

Turnover is controlled by signal smoothness. High-frequency signals (e.g., daily z-score of VaR utilization) tend to have higher turnover than slow signals (e.g., 63-day rolling Herfindahl index).

Quantifying the cost drag. Suppose your signal produces an annualized gross Sharpe of 1.2 with return volatility $\sigma_{\text{ret}} = 10\%$, so annualized gross return $= 1.2 \times 10\% = 12\%$. With $\bar{\tau} = 0.10$ and $c = 5$ bps:

$$\text{Annual cost} = 252 \times 0.10 \times 0.0005 = 1.26\%.$$

Net return $= 12\% - 1.26\% = 10.74\%$. Net Sharpe $= 10.74/10 = 1.07$. Costs shaved 0.13 off the Sharpe ratio. Double the turnover to $\bar{\tau} = 0.20$ and the cost doubles: net Sharpe drops to 0.95. Triple it and you lose 0.38 Sharpe points. The relationship is linear and merciless.

Two practical levers to control turnover:

1. **Signal filtering.** Apply exponential smoothing: $\hat{y}_t^{\text{smooth}} = \alpha \hat{y}_t + (1 - \alpha) \hat{y}_{t-1}^{\text{smooth}}$. This reduces turnover at the cost of slower reaction.
2. **Trade buffer.** Only trade when $|\hat{y}_t - \hat{y}_{t-1}| > \delta$ for some threshold δ . This eliminates small, cost-inefficient position adjustments.

Turnover as a Tax on Instability

Think of turnover as a tax you pay for changing your mind. A signal that flips daily from long to short pays the spread twice per day— $2c$ per round-trip. At 5 bps half-spread, that is 10 bps daily, or roughly 25% annually. No signal survives that. The practical sweet spot for daily signals is $\bar{\tau} \in [0.02, 0.15]$: enough activity to capture opportunities, low enough to keep costs manageable.

12.3 Capacity Estimation

Market Impact

When you trade a large order, you move the price against yourself. Buying pushes the price up; selling pushes it down. This *market impact* is a cost above and beyond the bid-ask spread. It depends on how large your order is relative to the market's daily volume.

A signal that works on \$1M of capital may not work on \$1B. **Capacity** is the dollar amount at which market impact erodes the signal to zero net Sharpe.

12.3.1 The Square-Root Market Impact Model

The empirically best-supported market impact model is the square-root law (Almgren and Chriss, 2001; Bouchaud, 2010):

$$\Delta p = \eta \sigma \sqrt{\frac{v}{V}}, \quad (12.5)$$

where:

- Δp — price impact as a fraction of price.
- σ — daily volatility of the instrument.
- v — number of shares (or notional) you trade.
- V — average daily volume (ADV) of the instrument.
- η — dimensionless constant, empirically $\eta \approx 0.1$ – 1.0 depending on asset class and urgency.

The square root makes impact concave: trading 4% of ADV costs twice as much (per unit) as trading 1% of ADV, not four times. This concavity has a practical implication: splitting a

large order into smaller pieces and spreading them over time reduces total impact. This is the theoretical foundation of execution algorithms (VWAP, TWAP) used on trading desks.

Numerical example. Consider a US Treasury future with $\sigma = 0.50\%$ daily and $V = \$30\text{B}$ ADV. Trading $v = \$300\text{M}$ ($= 1\%$ of ADV) with $\eta = 0.3$:

$$\Delta p = 0.3 \times 0.005 \times \sqrt{0.01} = 0.3 \times 0.005 \times 0.1 = 0.00015 = 1.5 \text{ bps.}$$

At $v = \$3\text{B}$ ($= 10\%$ of ADV): $\Delta p = 0.3 \times 0.005 \times \sqrt{0.1} \approx 4.7 \text{ bps}$. Ten times the volume, only three times the impact—that is the square root at work.

12.3.2 Order-of-Magnitude Capacity Reasoning

For your project, exact capacity is unknowable. Use order-of-magnitude reasoning:

1. Estimate the strategy's daily trade notional: $\text{Notional} = \text{AUM} \times \bar{\tau}$.
2. Compare to ADV of the target instruments. Rule of thumb: you can trade up to 1–5% of ADV without excessive impact.
3. Compute implied AUM capacity: $\text{AUM}^* = (0.01 \text{ to } 0.05) \times \text{ADV} / \bar{\tau}$.

Example. A cross-asset signal on G10 rates futures (combined ADV $\approx \$50\text{B}$ notional) with $\bar{\tau} = 0.05$. At 1% of ADV: capacity $= 0.01 \times 50\text{B} / 0.05 = \10B . At 5% of ADV: $\$50\text{B}$. This is a high-capacity strategy—liquid instruments with moderate turnover.

Compare a signal on single-name CDS (ADV $\approx \$50\text{M}$) with $\bar{\tau} = 0.20$: capacity $= 0.01 \times 50\text{M} / 0.20 = \2.5M . Institutional-scale capital cannot trade this.

Capacity Estimates Are Rough

The square-root model is a useful approximation, not a precise tool. Impact depends on execution algorithm, time horizon, market conditions, and counterparty relationships—none of which you observe in a backtest. Use order-of-magnitude reasoning: is this a \$10M strategy or a \$10B strategy? Do not report capacity to spurious precision. Saying “capacity is \$847M” signals that you do not understand the uncertainty. Saying “capacity is order \$1B for liquid rates, order \$10M for single-name CDS” signals that you do.

12.4 Walk-Forward Out-of-Sample Testing

In-Sample vs. Out-of-Sample

In-sample (IS) data is used for model training and selection. **Out-of-sample (OOS)** data is reserved and never used until final evaluation. The distinction matters because any data used for decisions—even indirectly through hyperparameter tuning or feature selection—is no longer truly out-of-sample.

The validation tools in Chapter 11 (purged CV, DSR, CPCV, PBO) all operate on the in-sample data. They provide strong evidence that your signal is not overfit. But the final, definitive test is the **holdout**—data reserved since Phase 1 (Task 4), untouched through months of experimentation.

12.4.1 The Sacred Holdout

In your project, you reserve the most recent 3–6 months of data as the holdout at the very beginning (Phase 1, Week 3). This data is:

1. **Locked.** The holdout dates are written to a config file. The pipeline module refuses to return holdout data until explicitly unlocked.
2. **Invisible.** You never look at holdout returns, not even summary statistics. Any information leakage—even knowing that “the holdout period was volatile”—biases your decisions.
3. **One-shot.** You run the holdout test exactly once, in Phase 5 (Task 22). There is no iteration. If the result is disappointing, you report it honestly.

The Holdout Is Not a Second Chance

If your model fails the holdout test, you do *not* go back, re-tune, and test again. That would make the holdout into just another validation fold. The entire point of the holdout is that it is the one piece of data your decisions never touched. Iterating on it destroys this property permanently. Chapter 11 emphasized this: only results that survive the full validation gauntlet (purged CV, DSR, PBO) should touch the holdout.

12.4.2 One-Shot Holdout Protocol

The Phase 5 holdout test follows a strict protocol:

1. **Freeze the model.** The model specification, hyperparameters, and feature set are fixed. No changes after this point.
2. **Train on all IS data.** Retrain the final model on the full in-sample period (not a single CV fold).
3. **Generate predictions on holdout.** Run the frozen model on holdout data. Record predictions.
4. **Compute all metrics.** IC, ICIR, Sharpe (gross and net at multiple cost levels), Sortino, hit rate, max drawdown, turnover.
5. **Apply DSR and Haircut Sharpe.** Use the experiment count from your log (`experiments/experiment` as N). Compute DSR on the holdout Sharpe. Apply the Haircut (Chapter 11).
6. **Report.** Include holdout results alongside IS results. If holdout Sharpe $<$ IS Sharpe (it almost always is), that is expected. If holdout Sharpe ≤ 0 , report it and discuss why.

12.4.3 Rolling-Window Walk-Forward

The one-shot holdout is the gold standard, but it gives a single number from a single period. The rolling-window walk-forward provides a richer picture:

1. Fix a training window of length W (e.g., 756 days $\approx$ 3 years).
2. Train the model on days $[1, W]$. Predict day $W + 1$.
3. Roll forward: train on $[2, W + 1]$, predict $W + 2$. Repeat.
4. Stitch all one-step-ahead predictions into a single OOS return series.

This produces a fully out-of-sample prediction for every day in the dataset (after the initial training window), at the cost of retraining the model $T - W$ times. The stitched return series can be analyzed for IC, Sharpe, and all other metrics.

In practice, you do not retrain every single day. A common compromise: retrain monthly (every 21 trading days) and use the same model for the next 21 predictions. This reduces computation by $\sim 21\times$ while maintaining the walk-forward discipline. The key requirement is that *no training set ever includes data from the prediction period or later*.

Walk-forward vs. holdout. These are complements, not substitutes. The walk-forward gives a rich time series of OOS performance across multiple market regimes. The holdout gives a single, pristine test on data your decisions never touched. Report both. If they agree, your result is robust. If walk-forward looks good but the holdout fails, suspect that walk-forward

parameters (window length W , retraining frequency) were tuned to look good—a subtler form of overfitting.

Rolling-Window vs. Expanding-Window

A **rolling** window (fixed length W) discards old data as new data arrives. An **expanding** window (train on all available data up to t) keeps everything. Rolling windows are better when you suspect regime change or structural breaks—old data from a different regime may hurt more than help. Expanding windows are better when sample size is the binding constraint (your case, with $T \approx 1,250$). Report both if feasible.

12.5 IC, ICIR, and Performance Metrics

Spearman Rank Correlation

The Spearman rank correlation between two variables is the Pearson correlation computed on their ranks rather than their raw values. It measures monotonic association without assuming linearity. Range: $[-1, +1]$. In finance, rank-based measures are preferred because they are robust to outliers in returns.

12.5.1 Information Coefficient (IC)

Information Coefficient

The Information Coefficient at time t is the Spearman rank correlation between the model's cross-sectional predictions and the realized outcomes:

$$\text{IC}_t = \rho_{\text{Spearman}}(\hat{y}_{i,t}, r_{i,t+1}), \quad i = 1, \dots, N_t, \quad (12.6)$$

where $\hat{y}_{i,t}$ is the prediction for asset i at time t , $r_{i,t+1}$ is the realized return, and N_t is the number of assets in the cross-section. For a single-asset time-series signal (no cross-section), IC reduces to the rank correlation between the signal and the next-period return computed over rolling windows.

IC benchmarks for cross-sectional equity signals (Gu, Kelly, and Xiu, 2020):

- $\text{IC} < 0.02$: noise.
- $\text{IC} \in [0.02, 0.05]$: weak but potentially tradeable with low cost.
- $\text{IC} \in [0.05, 0.10]$: strong. Most published factors live here.
- $\text{IC} > 0.10$: exceptional. Verify you do not have a bug.

12.5.2 ICIR

Information Coefficient Information Ratio

$$\text{ICIR} = \frac{\overline{\text{IC}}}{\sigma_{\text{IC}}} = \frac{\frac{1}{T} \sum_{t=1}^T \text{IC}_t}{\sqrt{\frac{1}{T-1} \sum_{t=1}^T (\text{IC}_t - \overline{\text{IC}})^2}}, \quad (12.7)$$

where $\overline{\text{IC}}$ is the time-series mean and σ_{IC} is the time-series standard deviation of the IC sequence.

ICIR measures the *consistency* of predictive power. A signal with $\overline{\text{IC}} = 0.04$ and $\sigma_{\text{IC}} = 0.02$ ($\text{ICIR} = 2.0$) is far more reliable than one with $\overline{\text{IC}} = 0.04$ and $\sigma_{\text{IC}} = 0.08$ ($\text{ICIR} = 0.5$). An

ICIR above 0.5 is considered good; above 1.0 is very strong.

12.5.3 Performance Metric Reference

Metric	Formula / Description	When to Use
IC	Spearman $\rho(\hat{y}_{i,t}, r_{i,t+1})$	Cross-sectional signal quality
ICIR	$\overline{\text{IC}}/\sigma_{\text{IC}}$	Signal consistency over time
Sharpe Ratio	$\text{SR} = \frac{\mu_{\text{ret}}}{\sigma_{\text{ret}}} \times \sqrt{252}$	Risk-adjusted return (annualized)
Sortino Ratio	$\frac{\mu_{\text{ret}}}{\sigma_{\text{downside}}} \times \sqrt{252}$	Penalizes downside vol only; use if return distribution is skewed
Hit Rate	$\frac{\#\{\text{PnL}_t > 0\}}{T}$	Fraction of profitable days; aim for > 52%
Max Drawdown	$\max_{s \leq t} (\text{cumPnL}_s - \text{cumPnL}_t)$	Worst peak-to-trough loss; survival metric
Calmar Ratio	$\frac{\text{Annualized Return}}{\text{Max Drawdown}}$	Return per unit of worst-case pain
Turnover	$\bar{\tau} = \frac{1}{T} \sum \hat{y}_t - \hat{y}_{t-1} $	Cost exposure; lower is cheaper
Breakeven Cost	c^* where $\text{SR}^{\text{net}}(c^*) = 0$	Tradeability threshold

Lead with IC and Sharpe, Support with the Rest

In your presentation (Chapter 14), lead with IC, ICIR, and net Sharpe at realistic costs. Support with Sortino, hit rate, and max drawdown. Always show the ridge baseline alongside GBM. If the desk asks one question, it will be about breakeven cost. If they ask two, the second is about max drawdown.

12.6 P&L Decomposition by Regime

Project Connection: Regime Decomposition

In Phase 4A (Task 17, if you pass the Week 13 checkpoint), you fit a GMM on macro features to classify regimes. The regime decomposition in this section uses that GMM output. Even before Phase 4A, you can do a simpler version: split P&L by VIX tercile (low/medium/high) or by calendar period (e.g., pre/post a known stress event). The point is to check whether your signal's Sharpe is broad-based or concentrated.

Aggregate metrics hide regime dependence. A signal with a Sharpe of 1.5 that earned 90% of its P&L during the 2020 COVID crash is a crisis-alpha signal, not a steady earner. If your desk does not want concentrated crisis exposure, that Sharpe is misleading.

The regime decomposition is straightforward:

1. Classify each day into a regime using the GMM from Chapter 13 (e.g., “calm,” “transition,” “crisis”).

2. Split the cumulative P&L series by regime. Sum $P\&L_t^{\text{net}}$ separately for each regime.
3. Compute Sharpe, IC, and hit rate within each regime.
4. Plot the regime-conditional cumulative P&L curves.

Interpretation rules:

- If $> 70\%$ of cumulative P&L comes from a single regime that occupies $< 30\%$ of the sample: the signal is **fragile**. It is a regime bet, not a robust predictor.
- If P&L is roughly proportional to time spent in each regime: the signal is **broad-based**. This is what you want.
- If the signal is profitable in calm regimes and loses money in crises: it is likely a short-volatility strategy—harvesting a risk premium that blows up exactly when the desk needs it most.

The Regime P&L Plot

Imagine a cumulative P&L chart where you color the background by regime: blue for calm, yellow for transition, red for crisis. If the line only goes up during red periods, your signal is a crisis detector—useful, but fragile. If it goes up steadily regardless of background color, the signal is robust. This single chart communicates more than a page of summary statistics. Always include it in your research report (Chapter 14).

12.7 Signal Decay Post-Publication

McLean and Pontiff (2016): Post-Publication Signal Decay

Studying 97 anomalies published in top finance journals, McLean and Pontiff (2016, *Journal of Finance*) find:

- **26% decay out-of-sample:** anomaly returns are 26% lower in the period after the original sample ends but before the paper is published. This captures data-mining bias—the original in-sample results were partly lucky.
- **58% decay post-publication:** anomaly returns are 58% lower after the paper is published. The additional 32 percentage points ($58\% - 26\%$) are attributed to informed trading by investors who read the paper and arbitrage the signal away.
- **Decay is worse for liquid, low-idiosyncratic-risk stocks:** precisely the stocks that are cheapest to trade, and therefore most susceptible to arbitrage.

The implication is severe: a published signal with an in-sample Sharpe of 1.0 should be expected to deliver roughly $1.0 \times (1 - 0.58) = 0.42$ post-publication—before your own transaction costs. After the Haircut Sharpe correction from Chapter 11 and realistic transaction costs, most published anomalies are economically zero.

The 26% OOS decay and the 58% post-publication decay address different problems. The 26% is the cost of data mining: the original researchers tried many specifications and reported the best one. This is exactly what DSR corrects for. The additional 32 percentage points are the cost of *crowding*: once the paper is public, capital flows into the strategy, compresses the spread, and arbitrages the signal away. Decay is worse for liquid, low-idiosyncratic-risk stocks—precisely the instruments where arbitrage is cheapest.

12.7.1 Why Proprietary Data May Be More Durable

The McLean-Pontiff decay is driven by *publication*: once the signal is public, capital floods in and arbitrages it away. Signals built on proprietary data (like SecDB risk cubes) have a

structural advantage:

1. **Data barrier.** External researchers use stale quarterly Fed Z.1 data to proxy dealer balance-sheet constraints (He, Kelly, and Manela, 2017; Chapter 2). You have daily, cross-asset risk outputs. Even if the conceptual insight is known, no one outside the firm can replicate the exact signal.
2. **Execution barrier.** The signal is designed for a specific desk’s trading infrastructure. Replication by outsiders requires both the data and the execution capability.
3. **Continuous updating.** You observe the risk system daily. Public anomalies are static once published.

12.7.2 The Shared-Factor Caveat

Durability is not guaranteed. If your risk-cube signal is correlated with a published factor—say, it loads on momentum or value—then the published-factor component shares that factor’s decay. In your project, confound checks (Phase 2, Task 15) regress your signal’s returns on known public factors (Fama-French, Carhart, intermediary capital ratio). If the alpha survives these regressions, the proprietary component is genuinely incremental. If it does not, you are repackaging a known signal with expensive data.

The Durability Test

A signal’s expected durability depends on three questions: (1) Is the data proprietary or public? (2) Is the signal correlated with published factors? (3) Are the instruments liquid enough for the signal to be arbitrated away? Signals built on proprietary data, orthogonal to published factors, and trading in moderately liquid instruments have the best chance of survival. Your risk-cube signals score well on (1) and potentially on (2), which is why the confound checks matter.

12.8 Common Backtest Pitfalls

Beyond look-ahead bias (Section 12.1), several other pitfalls routinely produce inflated backtests.

1. **Survivorship bias.** Testing only on instruments that exist today, excluding those that delisted, defaulted, or were acquired. A CDS strategy tested only on names that survived ignores the names that blew up—exactly the tail events that matter most.
2. **Stale pricing.** Using end-of-day marks for instruments that trade infrequently (e.g., corporate bonds, EM sovereign CDS). The “returns” are partly mark-to-model artifacts, not executable prices. If you cannot trade at the price in your backtest, the backtest is fiction.
3. **Free-option bias.** Assuming you can trade at the close price after observing close-of-day data. In reality, you observe the close, compute your signal, and trade at the *next available* price. The gap between close and next-open can be significant in volatile markets.
4. **Ignoring short-selling constraints.** Assuming you can short any instrument at zero cost. In practice, shorting is expensive (borrow fees), sometimes impossible (recall risk), and subject to regulatory constraints. If your signal requires short positions, document the borrow cost assumption.
5. **Selection bias in signal construction.** Testing 50 feature variants and reporting the best one without adjusting for multiple testing. This is the problem DSR solves (Chapter 11), but only if you honestly log every trial.

Your backtesting engine should include checks for the first three. The point-in-time module handles look-ahead; the pipeline should flag stale prices (zero returns for consecutive days); and the engine should use next-period returns by construction.

12.9 Putting It All Together: The Backtest Report

Your backtesting engine (`src/backtest/reporting.py`) should produce a standardized report for every experiment. Standardization matters: when you have tested 30 signal variants over Phase 2, a consistent format lets you compare results at a glance rather than re-reading ad hoc notebooks. The desk will not read 30 notebooks. They will scan a table.

The report includes:

1. **Signal summary.** Feature set, model type (ridge/GBM), label type, training window.
2. **Gross and net P&L.** Cumulative P&L chart with cost levels overlaid.
3. **Sharpe-vs-cost curve.** Net Sharpe at $c = 0, 1, 2, 5, 10, 20$ bps. Breakeven cost c^* .
4. **IC time series.** Rolling IC with mean, standard deviation, ICIR.
5. **Performance table.** Sharpe, Sortino, hit rate, max drawdown, Calmar, turnover.
6. **Validation results.** Purged CV Sharpe, DSR, CPCV distribution, PBO (from Chapter 11).
7. **SHAP summary.** Top features by mean $|\text{SHAP}|$ value (Chapter 7).
8. **Ridge baseline comparison.** GBM vs. ridge on identical features and CV splits.
9. **Regime decomposition.** P&L by regime (once Chapter 13 infrastructure is built).
10. **Experiment ID.** Logged to `experiments/experiment_log.csv` with all parameters for honest DSR trial counting.

This report is generated for every signal variant tested during Phase 2. At Phase 5, the holdout results are appended as a final section, completing the story from hypothesis to out-of-sample evidence.

Project Connection: The Report as Deliverable

The backtest report format above maps directly to the research report you write in Phase 5 (Task 23). Each section of the report corresponds to a slide or a page in your final presentation (Chapter 14). Build the reporting module early in Phase 1, run it automatically for every experiment, and your Phase 5 consolidation becomes assembly rather than reconstruction.

12.10 Summary

1. **Gross P&L** is $\hat{y}_t \cdot r_{t+1}$. **Net P&L** subtracts $|\hat{y}_t - \hat{y}_{t-1}| \cdot c$ for half-spread cost c . Always report both.
2. **Turnover** $\bar{\tau}$ measures trading intensity. Sharpe degrades linearly in $\bar{\tau} \times c$. Target $\bar{\tau} \in [0.02, 0.15]$ for daily signals.
3. The **breakeven cost** c^* is the spread at which net Sharpe = 0. This is the single number the desk asks for.
4. **Capacity** follows the square-root impact model: $\Delta p = \eta \sigma \sqrt{v/V}$. Use order-of-magnitude reasoning, not false precision.
5. The **holdout** is sacred: reserved since Phase 1, tested once in Phase 5, never iterated upon.
6. **Rolling-window walk-forward** stitches one-step-ahead OOS predictions across time. Use expanding windows when sample size is binding ($T \approx 1,250$).

7. **IC** measures cross-sectional prediction quality. **ICIR** measures its consistency. Benchmarks: $IC \in [0.02, 0.05]$ is typical; $ICIR > 0.5$ is good.
8. **Regime decomposition** splits P&L by market regime. If $> 70\%$ of P&L comes from $< 30\%$ of time: the signal is fragile.
9. **McLean and Pontiff (2016)**: anomaly returns decay 26% out-of-sample and 58% post-publication across 97 published signals. Proprietary data signals may be more durable, but only if orthogonal to published factors.
10. Every experiment generates a **standardized backtest report**: P&L, costs, IC, validation, SHAP, ridge baseline, regime decomposition, experiment ID.

Key Results Reference

Paper	Key Claim	Headline Result
McLean & Pontiff (2016, JF)	Published anomalies decay post-publication	26% OOS decay (data mining); 58% post-publication decay (arbitrage) across 97 signals
López de Prado (2018), Ch. 14–15	Backtesting must account for costs and overfitting	Transaction-cost-aware P&L; backtest report framework; holdout discipline
Bailey, Borwein, LdP, Zhu (2014)	Backtest overfitting is quantifiable and endemic	~45 trials on 5yr daily data exhaust Sharpe 1.0; most published backtests are overfit
Almgren & Chriss (2001)	Market impact follows a square-root law	$\Delta p = \eta \sigma \sqrt{v/V}$; capacity is order-of-magnitude, not precise

Chapter 13

Regime Modeling

Project Connection

In Phase 4A (Task 17), you fit a Gaussian Mixture Model on macro features to classify market regimes and decompose your signal's performance by regime. This chapter teaches GMM regime classification, the Two Sigma template for interpreting regimes, ADWIN for online drift detection, Markov-switching models for regime persistence, and—critically—the Asness et al. (2017) caveat on why regime-conditional strategies often disappoint in practice.

13.1 Why Regime-Awareness Matters

A signal that earns $IC = 0.04$ unconditionally might earn $IC = 0.10$ in calm markets and $IC = -0.03$ during crises. If the positive performance comes entirely from one market regime, the signal is *fragile*: the next crisis could wipe out years of steady gains. Regime decomposition answers a simple but important question: *is your signal's Sharpe driven by one regime, or is it broad-based?*

Three practical uses of regime modeling in your project:

1. **Diagnostic.** Decompose backtest Sharpe by regime. A signal that works only in steady-state periods is less valuable than one that works across regimes.
2. **Risk management.** Scale position sizes down when the model assigns high probability to a crisis regime.
3. **Feature enrichment.** Feed regime probabilities as additional features into your LightGBM model (Chapter 6).

Regimes as Weather

Think of regimes like weather seasons. A farmer who only knows the *average* annual rainfall will plant the same crop year-round. One who distinguishes wet and dry seasons adapts. Your signal is the crop; regimes are the seasons. The goal is not to predict regimes (that is hard)—it is to *check whether your crop survives all seasons*.

13.2 Gaussian Mixture Models for Regime Classification

Multivariate Gaussian and Mixture Models

A d -dimensional Gaussian density is

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right),$$

where $\boldsymbol{\mu} \in \mathbb{R}^d$ is the mean and $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is the covariance matrix (symmetric positive-definite). A *mixture model* says the data come from several such Gaussians, each chosen with some probability. If this is unfamiliar, review Bishop (2006), Ch. 2 and 9.

Gaussian Mixture Model

A K -component Gaussian Mixture Model (GMM) models the density of observations $\mathbf{x} \in \mathbb{R}^d$ as

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (13.1)$$

where:

- K is the number of mixture components (regimes),
- $\pi_k \geq 0$ is the **mixing weight** for component k , with $\sum_{k=1}^K \pi_k = 1$,
- $\boldsymbol{\mu}_k \in \mathbb{R}^d$ is the **mean** of component k ,
- $\boldsymbol{\Sigma}_k \in \mathbb{R}^{d \times d}$ is the **covariance** of component k .

The parameters $\boldsymbol{\theta} = \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K$ are estimated by maximum likelihood. Because the log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^T \log \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

has no closed-form solution (the log wraps a sum), we use the Expectation–Maximization (EM) algorithm (Dempster, Laird, and Rubin, 1977).

13.2.1 The EM Algorithm

EM alternates between two steps until convergence:

EM for Gaussian Mixtures

E-step (soft labels). For each observation $\mathbf{x}_t$ and component k , compute the *responsibility*:

$$\gamma_{tk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}. \quad (13.2)$$

γ_{tk} is the posterior probability that observation t belongs to component k .

M-step (update parameters). Using the responsibilities as weights:

$$N_k = \sum_{t=1}^T \gamma_{tk}, \quad (13.3)$$

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{t=1}^T \gamma_{tk} \mathbf{x}_t, \quad (13.4)$$

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{t=1}^T \gamma_{tk} (\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})(\mathbf{x}_t - \boldsymbol{\mu}_k^{\text{new}})^\top, \quad (13.5)$$

$$\pi_k^{\text{new}} = \frac{N_k}{T}. \quad (13.6)$$

N_k is the effective number of points assigned to component k .

Repeat E-step and M-step until $|\ell^{(i+1)} - \ell^{(i)}| < \epsilon$. EM guarantees $\ell(\boldsymbol{\theta})$ is non-decreasing at every iteration (Dempster, Laird, and Rubin, 1977).

13.2.2 Choosing K : BIC

More components always improve in-sample fit, so you cannot pick K by maximizing the log-likelihood. The Bayesian Information Criterion penalizes complexity:

$$\text{BIC} = -2 \ell(\hat{\boldsymbol{\theta}}) + p \log T, \quad (13.7)$$

where p is the number of free parameters and T is the sample size. For a K -component GMM in d dimensions with full covariance matrices, $p = K(1 + d + d(d+1)/2) - 1$. Choose the K that *minimizes* BIC. Two Sigma used cross-validated log-likelihood with similar results. For your project, $K \in \{2, 3, 4\}$; $K = 3$ or 4 is typical.

13.2.3 Feature Selection for Regime GMM

The regime GMM takes macro observables, not your alpha features. Typical choices:

Feature	Source	Captures
VIX level	CBOE	Equity risk appetite
IG credit spread	ICE BofA	Credit stress
10y–2y term slope	Treasury	Growth/policy expectations
DXY (USD index)	Fed	Dollar regime
Realized cross-asset corr.	Computed	Contagion / diversification

Standardize all features to zero mean and unit variance before fitting. This prevents the GMM from being dominated by whichever feature has the largest scale.

Do Not Use Your Alpha Features as Regime Inputs

If you feed your risk-cube signals into the regime GMM *and then* condition on regimes, you introduce circularity. The regime model should use independent macro variables. Your alpha features go into the LightGBM prediction model; the regime model provides an orthogonal lens for decomposing performance.

Worked Example:

One EM Iteration on a 2D, 3-Component GMM Suppose $d = 2$ features (VIX, credit spread), $K = 3$ components, and $T = 5$ observations. After initialization, the current parameters are:

$$\pi_1 = 0.5, \quad \pi_2 = 0.3, \quad \pi_3 = 0.2.$$

$$\boldsymbol{\mu}_1 = \begin{pmatrix} 14 \\ 1.0 \end{pmatrix}, \quad \boldsymbol{\mu}_2 = \begin{pmatrix} 22 \\ 2.5 \end{pmatrix}, \quad \boldsymbol{\mu}_3 = \begin{pmatrix} 35 \\ 5.0 \end{pmatrix}.$$

All covariances initialized to $\boldsymbol{\Sigma}_k = \mathbf{I}_2$ for simplicity.

E-step for $\mathbf{x}_1 = (15, 1.2)^\top$:

Compute unnormalized responsibilities:

$$\tilde{\gamma}_{1,1} = \pi_1 \cdot \mathcal{N}(\mathbf{x}_1 \mid \boldsymbol{\mu}_1, \mathbf{I}) = 0.5 \times \mathcal{N}((15, 1.2)^\top \mid (14, 1.0)^\top, \mathbf{I}).$$

$$\|\mathbf{x}_1 - \boldsymbol{\mu}_1\|^2 = (15 - 14)^2 + (1.2 - 1.0)^2 = 1.04.$$

$$\mathcal{N}(\cdot) = \frac{1}{2\pi} \exp(-1.04/2) = 0.0942.$$

$$\tilde{\gamma}_{1,1} = 0.5 \times 0.0942 = 0.0471.$$

Similarly, $\|\mathbf{x}_1 - \boldsymbol{\mu}_2\|^2 = (15 - 22)^2 + (1.2 - 2.5)^2 = 50.69$, giving $\tilde{\gamma}_{1,2} = 0.3 \times 1.32 \times 10^{-12} \approx 0$. And $\|\mathbf{x}_1 - \boldsymbol{\mu}_3\|^2 = 414.44$, giving $\tilde{\gamma}_{1,3} \approx 0$.

Normalize:

$$\gamma_{1,1} = \frac{0.0471}{0.0471 + 0 + 0} \approx 1.00, \quad \gamma_{1,2} \approx 0, \quad \gamma_{1,3} \approx 0.$$

Observation $\mathbf{x}_1$ is assigned almost entirely to component 1 (low-VIX, low-spread = steady state). Repeat for all T observations to complete the E-step, then update parameters via Eqs. (13.3)–(13.6).

Takeaway. Points far from a component's mean get near-zero responsibility. This is how the GMM “self-organizes” into clusters without hard labels.

13.3 The Two Sigma Regime Template

Two Sigma applied a GMM to 17 factor returns and identified four market conditions that recur historically. Their template provides a useful vocabulary for interpreting your own GMM output.

Four Macro Regimes (Two Sigma, 2021)

Regime	Characteristics	Asset behavior
Steady State	Low VIX, tight spreads, moderate growth. Dominates the 2010s decade.	Equity, credit, and most style factors earn positive returns.
Crisis	VIX spikes, spreads widen, cross-asset correlations surge. 1987, 2008 GFC, 2020 COVID.	Equity and credit suffer large drawdowns; defensive factors outperform.
Inflation	Rising CPI, steepening yield curve, commodity strength. Concentrated in 1970s–early 1980s.	Local inflation factor posts double-digit mean returns; equities and bonds underperform.
Walking on Ice	Equity returns above average but volatility ~ 1.6 pp above long-run mean. Often precedes or follows crises.	Positive equity returns mask elevated fragility; bubble risk.

13.3.1 Regime-Conditional Sharpe Decomposition

Once you assign regime probabilities to each date in your backtest, compute regime-conditional Sharpe ratios:

$$SR_k = \frac{\bar{r}_k}{\sigma_k}, \quad \text{where } \bar{r}_k = \frac{\sum_t \gamma_{tk} r_t}{\sum_t \gamma_{tk}}, \quad \sigma_k^2 = \frac{\sum_t \gamma_{tk} (r_t - \bar{r}_k)^2}{\sum_t \gamma_{tk}}. \quad (13.8)$$

Report these alongside your unconditional Sharpe. The presentation (Chapter 14) should include a table like:

	Steady State	Crisis	Inflation	Walking on Ice
SR	1.20	−0.30	0.45	0.85
Fraction of days	55%	10%	15%	20%

If SR is positive only in Steady State (55% of days), your signal is fragile. If positive across three or four regimes, it is robust.

Sharpe Ratio and Annualization

The Sharpe ratio $SR = \bar{r}/\sigma$ measures risk-adjusted return. For daily data, annualize via $SR_{\text{annual}} = SR_{\text{daily}} \times \sqrt{252}$. This chapter computes regime-conditional Sharpe ratios on daily strategy returns, then annualizes. See Chapter 12 for the full backtesting framework.

13.4 ADWIN for Concept Drift Detection

The GMM approach is *offline*: you fit the model to all available data and then label each date. In production, you need *online* detection of distribution shifts. ADWIN (Adaptive Windowing; Bifet and Gavaldà, 2007) provides exactly this.

ADWIN Algorithm

ADWIN maintains a variable-length window W of recent observations. At each time step, it tests all possible partitions of W into two contiguous sub-windows W_0 (older) and W_1

(newer). A drift is declared when the absolute difference between the sub-window means exceeds a threshold derived from the Hoeffding bound:

$$|\hat{\mu}_{W_0} - \hat{\mu}_{W_1}| \geq \epsilon_{\text{cut}}, \quad \text{where } \epsilon_{\text{cut}} = \sqrt{\frac{1}{2m} \cdot \ln \frac{4|W|}{\delta}}, \quad (13.9)$$

$m = (1/|W_0| + 1/|W_1|)^{-1}$ is the harmonic mean of the sub-window sizes, $|W|$ is the total window length, and $\delta \in (0, 1)$ is a confidence parameter controlling sensitivity. When drift is detected, all observations in W_0 are discarded.

How to use ADWIN in your project:

1. Feed your model's *prediction errors* ($\hat{y}_t - y_t$) into ADWIN.
2. When ADWIN fires, it means the error distribution has shifted—your model's relationship with the data has changed.
3. Trigger a model retrain on the post-drift window only.

The parameter δ controls the trade-off: $\delta = 0.01$ (more sensitive, more false positives, faster detection) vs. $\delta = 0.0001$ (fewer false positives, slower detection). For daily financial data with ~ 252 observations per year, $\delta = 0.002$ is a reasonable starting point.

ADWIN as a Smoke Detector

ADWIN is like a smoke detector for your model. It does not tell you *what* regime you are in (the GMM does that). It tells you that something has *changed*—your model's errors suddenly look different from what they were. When the alarm sounds, you investigate: has the market entered a new regime, or did your feature pipeline break?

13.5 Markov-Switching Models

Hidden Markov Models

A Hidden Markov Model (HMM) has two layers: (1) a latent state sequence $s_t \in \{1, \dots, K\}$ evolving as a Markov chain, and (2) observed emissions $\mathbf{x}_t$ whose distribution depends on s_t . The key assumption: s_t depends on s_{t-1} only (Markov property). If unfamiliar, review Murphy (2012), Ch. 17.

Hamilton (1989) introduced Markov-switching regression to economics, modeling U.S. GNP growth as an autoregression whose mean shifts between two states: expansion and contraction. The model is:

$$y_t = \mu_{s_t} + \sum_{i=1}^p \phi_i(y_{t-i} - \mu_{s_{t-i}}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (13.10)$$

where $s_t \in \{0, 1\}$ is the latent state with transition matrix:

$$\mathbf{P} = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}, \quad p_{ij} = \Pr(s_t = j \mid s_{t-1} = i), \quad p_{i0} + p_{i1} = 1. \quad (13.11)$$

Hamilton (1989): Markov-Switching GNP

Applied to U.S. quarterly GNP growth (1951Q1–1984Q4) with an AR(4) specification, Hamilton estimated:

- Mean growth in expansion ($s_t = 1$): $\hat{\mu}_1 = 1.16\%$ per quarter.
- Mean growth in recession ($s_t = 0$): $\hat{\mu}_0 = -0.36\%$ per quarter.
- Persistence: $\hat{p}_{11} = 0.90$ (expansions last on average $1/(1 - 0.90) \approx 10$ quarters), $\hat{p}_{00} = 0.75$ (recessions last ≈ 4 quarters).

The model's filtered probabilities closely matched NBER recession dates without using them as inputs.

13.5.1 GMM vs. Markov-Switching: When to Use Which

	GMM	Markov-Switching
Temporal structure	None. Each $\mathbf{x}_t$ classified independently.	Markov chain: s_t depends on s_{t-1} .
Regime persistence	Not modeled. Can oscillate between regimes at high frequency.	Explicitly modeled via transition probabilities.
Estimation	EM on mixture (fast, scales well).	Hamilton filter (slower, requires forward-backward recursion).
Output	Posterior probabilities γ_{tk} per observation.	Filtered $\Pr(s_t = k \mid \mathbf{x}_{1:t})$ and smoothed $\Pr(s_t = k \mid \mathbf{x}_{1:T})$.
Python	<code>sklearn.mixture.GaussianMixture</code>	<code>statsmodels.tsa.regime_switching</code>
Best for	Quick regime labeling; diagnostic decomposition.	When regime persistence matters; macro modeling.

Use GMM for Your Project

GMM treats each observation independently; Markov-switching models regime persistence. For your project, you are performing a *post-hoc diagnostic*: “did my signal’s Sharpe come from one regime or many?” You are not forecasting regimes in real time. For this purpose, GMM is simpler, faster, and sufficient. If you later want to build a regime forecasting overlay for position sizing, switch to Markov-switching.

13.6 The Factor Timing Caveat

Before you get excited about regime-conditional strategies, read the cautionary evidence.

Asness, Chandra, Iltanen, and Israel (2017)

In “Contrarian Factor Timing is Deceptively Difficult,” the authors test whether buying factors after drawdowns (contrarian timing via value spreads) improves multi-factor portfolios. Key findings:

- For a 3-factor portfolio (HML + UMD + BAB), value-spread timing *decreases* the Sharpe ratio from 0.55 to 0.54.
- For individual factors, timing helps momentum (Sharpe 0.26 $\rightarrow$ 0.33) but hurts defensive/low-beta (Sharpe 0.53 $\rightarrow$ 0.40).
- The timing signal is correlated with the value factor itself, so in a portfolio that already holds value, timing adds redundant and intermittent exposure rather than genuine new information.
- Results are reported gross of transaction costs. Net of costs, the case for timing weakens further.

The authors conclude that contrarian factor timing is “a weak addition for long-term investors holding well-diversified factors including value.”

13.6.1 Implications for Your Project

The Asness et al. result does *not* say regime modeling is useless. It says:

1. **Conditional > unconditional is the wrong comparison.** You must compare the regime-conditional strategy to the unconditional strategy *after accounting for any additional factor exposures* the conditioning introduces. If your regime overlay tilts toward value or momentum, you are adding a factor bet, not alpha.
2. **Document honestly.** Report both conditional and unconditional Sharpe. If the conditional strategy wins, show that it is not explained by loading on known factors.
3. **The bar is high.** Even sophisticated practitioners at AQR found that timing adds little. If your regime decomposition shows that the signal works only in one regime, the honest conclusion may be to report that fragility rather than to build a timing strategy around it.

Regimes Are Only Known in Hindsight

Regime-conditional strategies look better in backtests than in practice because the regime is only known with certainty in hindsight. Your GMM assigns *probabilities*, not hard labels. Today’s observation might be 60% Steady State and 40% Walking on Ice. If you build a strategy that goes fully risk-on in Steady State and risk-off otherwise, you are implicitly assuming perfect regime classification. In practice: (a) use soft probabilities, not hard assignments; (b) test with lagged regime labels (yesterday’s classification applied today); (c) measure how often the GMM changes its mind on the most recent 20 days of data when one new observation arrives.

13.7 Practical Implementation Checklist

For Task 17 in your project plan, follow this sequence:

1. **Assemble macro features.** Pull VIX, IG credit spread, 10y–2y slope, DXY, and realized 60-day cross-asset correlation. Standardize to z -scores.
2. **Fit GMMs for $K \in \{2, 3, 4, 5\}$.** Use `sklearn.mixture.GaussianMixture` with `covariance_type='full'` and `n_init=10` (multiple random restarts to avoid local optima). Record BIC for each K .
3. **Select K .** Pick the K that minimizes BIC. Inspect the regimes manually: do they correspond to recognizable market conditions? If $K = 3$ and $K = 4$ have similar BIC, prefer the one with more interpretable clusters.
4. **Label each date.** Assign soft probabilities γ_{tk} , not hard labels. Store as columns in your feature DataFrame.
5. **Decompose signal performance.** Compute regime-conditional Sharpe (Eq. 13.8), IC, and hit rate. Report in a table.
6. **Test regime overlay.** If desired, scale positions by $\gamma_{t,\text{steady}}$ (probability of being in the favorable regime). Compare Sharpe to the unconditional strategy. Apply the Asness et al. sanity check: does the improvement survive controlling for additional factor exposures?
7. **Set up ADWIN.** Feed prediction residuals into ADWIN with $\delta = 0.002$. Log all detected drift points. Check whether they align with GMM regime transitions.

13.8 Summary

Concept	Key Takeaway
GMM regime classification	$p(\mathbf{x}) = \sum_k \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. EM alternates E-step (soft labels) and M-step (update parameters). Select K by BIC.
Two Sigma template	Four regimes—Steady State, Crisis, Inflation, Walking on Ice—fit to factor returns via GMM.
ADWIN	Online drift detection via adaptive windowing and Hoeffding bounds. Apply to prediction errors to trigger re-training.
Markov-switching	Hamilton (1989): HMM with regime-dependent parameters. Models persistence. Use GMM for post-hoc decomposition; Markov-switching for real-time regime forecasting.
Factor timing caveat	Asness et al. (2017): contrarian timing of factors via value spreads barely helps (3-factor Sharpe $0.55 \rightarrow 0.54$) and can hurt diversification. Regime conditioning is diagnostic first, strategy second.
Your project	Phase 4A, Task 17: fit GMM on macro features, decompose signal by regime, report conditional Sharpe honestly. Use soft probabilities, test with lagged labels, control for factor exposure overlap.

Chapter 14

Presenting Quantitative Research

Project Connection

This is preparation for Week 20—the final presentation. Everything in this chapter maps directly to Phase 5 deliverables: the research report (Task 23) and the presentation itself (Task 24). The framing advice here draws on how the best academic papers and desk presentations communicate quantitative results. You have built infrastructure (Chapter 11), engineered features (Chapter 9), labeled targets (Chapter 10), validated honestly (Chapter 11), and backtested with costs (Chapter 12). This chapter teaches you how to *communicate* all of that so a desk head trusts your work in 20 minutes.

14.1 Framing: Causal Hypothesis + ML Testing

Why Framing Determines Reception

A quantitative result is not evaluated in a vacuum. The audience—whether a trading desk, a research seminar, or a hiring committee—decides within the first 60 seconds whether to engage or to look for holes. The frame you set in that first minute determines which mode they enter. This section teaches you to set the right frame.

There are two ways to open a quant presentation.

- **Good:** “He–Krishnamurthy (2013) predicts that intermediary balance-sheet constraints price risk across asset classes. We tested this with daily SecDB risk data and found that factor-concentration changes predict next-week drawdowns with $IC = 0.05$.”
- **Bad:** “We fed 47 risk features into a gradient-boosted tree and it predicted returns.”

The first framing invites engagement. It says: *there is a well-grounded economic hypothesis, and we used modern statistical tools to test it*. The audience evaluates the hypothesis, then checks whether the test is sound.

The second invites skepticism. It says: *we did a fishing expedition and found a pattern*. The audience immediately asks “how many features did you try?” and reaches for the Deflated Sharpe Ratio (Chapter 11).

The ML Is the Test, Not the Story

Frame your work as testing a well-grounded economic hypothesis with modern statistical tools. The theory (Chapter 2) provides the *why*; the ML (Chapters 5 and 6) provides the *how precisely*. Lead with the theory. Save the model details for the methodology slide.

Kelly and Xiu (2023) survey the financial ML literature and note that the most impactful papers—Gu, Kelly, and Xiu (2020), Freyberger, Neuhierl, and Weber (2020), Feng, Giglio, and Xiu (2020)—all begin with an asset-pricing question, not a modeling trick. The ML is evaluated by whether it answers the economic question better than existing methods, not by its raw R^2 .

Practical framing template. For each signal family you present, use this three-sentence structure:

1. **Theory:** “[Paper] predicts that [mechanism] should cause [observable effect].”
2. **Data advantage:** “SecDB gives us [specific data] that external researchers cannot observe.”
3. **Result:** “We find [IC / Sharpe / direction], consistent [or inconsistent] with the hypothesis.”

Never Lead with Model Complexity

If the first thing your audience hears is “we used a 500-tree LightGBM ensemble with Bayesian hyperparameter optimization,” you have lost them. They will spend the next 19 minutes looking for overfitting rather than listening to your results. Complexity is a *liability* until you have established that simpler methods cannot do the job—which is exactly why you always show the ridge baseline (Section 14.3).

14.2 The Primary Metrics Triad

What These Metrics Measure

If you have not seen IC, Sharpe, or turnover before, review Chapter 12 for formal definitions and computation. This section focuses on how to *report* them, not how to compute them.

Every signal should be judged on three dimensions simultaneously. Cherry-picking one is misleading.

The Metrics Triad

1. **Information Coefficient (IC):** Does the signal predict direction? Report the mean rank correlation between signal and subsequent return, plus its standard deviation across time. Example: $IC = 0.04 \pm 0.12$ (monthly, cross-sectional).
2. **Sharpe Ratio (SR):** Does the signal make money? Report both the raw annualized Sharpe and the DSR-adjusted value (Chapter 11). Example: $SR = 1.2$, $DSR = 0.8$ after 35 trials.
3. **Turnover:** Is it tradeable? Report average daily one-way turnover and the breakeven transaction cost. Example: turnover = 18%/day, breakeven cost = 12 bps.

Why all three matter. A signal with $IC = 0.08$ but daily turnover of 60% is untradeable—transaction costs will eat the alpha. A signal with $SR = 2.0$ but $DSR = 0.3$ is probably overfit. A signal with low turnover and a decent Sharpe but $IC = 0.00$ is just a static factor exposure, not a timing signal.

Worked Example:

Reporting the Triad for VaR Utilization Suppose your VaR-utilization signal (Chapter 9) produces the following backtest results after purged cross-validation (Chapter 11):

- Mean IC: 0.045 (rank correlation of signal with next-5-day returns, averaged across 48 monthly cross-sections)
- Std of IC: 0.11 (so $IC-t = 0.045/(0.11/\sqrt{48}) \approx 2.83$)
- Raw SR: 1.35 (annualized, long-short quintile portfolio)
- DSR: 0.91 (after accounting for 28 experiments logged in your experiment tracker, Chapter 11)
- Daily one-way turnover: 14%
- Breakeven cost: $\text{Annual alpha}/(\text{turnover} \times 252 \times 2) = 8.1\%/(0.14 \times 252 \times 2) \approx 11.5 \text{ bps}$

You would report this as:

Metric	VaR Utilization	Ridge Baseline
IC (mean $\pm$ std)	0.045 ± 0.11	0.031 ± 0.13
IC- t	2.83	1.69
SR (annualized)	1.35	0.92
DSR (28 trials)	0.91	0.58
Turnover (%/day)	14%	9%
Breakeven cost (bps)	11.5	14.8

Key observations: (1) the GBM beats ridge on IC and Sharpe, so the nonlinearity adds value; (2) the DSR remains positive even after 28 trials; (3) turnover is moderate and the breakeven cost of 11.5 bps is realistic for institutional execution. The ridge baseline's higher breakeven cost (14.8 bps) reflects lower turnover despite lower alpha—it trades less but earns less.

The Triad as a Sanity Triangle

Think of IC, Sharpe, and turnover as the vertices of a triangle. A signal that scores well on all three is tradeable, predictive, and profitable. If any vertex collapses, the triangle breaks: high IC + high Sharpe + high turnover = costs kill you; high IC + low turnover + low Sharpe = the signal is real but the effect size is too small to monetize; high Sharpe + low IC = you are harvesting a risk premium, not predicting. Always check all three.

14.3 Ridge Baseline on Every Chart

Why Ridge Is the Right Baseline

Ridge regression (Chapter 5) is the simplest model that can handle correlated features without blowing up. It is linear, fully interpretable, and hard to overfit. If a gradient-boosted tree cannot beat ridge on identical features, the nonlinearity is not helping—the problem is either linear or your tree is overfitting. This was discussed in detail in Chapter 5, Section 5.8.

The rule. Every cumulative P&L chart, every IC time series, every Sharpe bar chart has *two* lines:

- **Solid line:** your model (LightGBM or whatever you used)
- **Dashed line:** ridge baseline on the identical feature set

Three possible outcomes:

1. **Clear separation (GBM $\gg$ ridge).** The nonlinearity adds value. Show where the lines diverge and use SHAP (Section 14.4) to explain *why* the tree found something ridge missed—typically an interaction or a threshold effect.
2. **Lines overlap (GBM $\approx$ ridge).** The problem is approximately linear. This is *not* a failure—it means your features are good and the relationship is simple. Say so honestly: “Ridge captures the signal; the GBM adds marginal improvement that does not survive transaction costs.” Then use ridge as your production model. It is simpler, faster, and more robust.
3. **Ridge wins (GBM $<$ ridge).** The tree overfit. Do not present the tree results. Present ridge, explain why you tried trees, and report that they did not help. This is an honest negative result (Section 14.6).

The Ridge Baseline Is Your Credibility Insurance

Showing the ridge baseline on every chart is the single most effective way to build credibility with a quantitative audience. It signals three things: (1) you understand the bias-variance tradeoff; (2) you are not hiding behind model complexity; (3) you would have reported honestly if the simple model won. Kelly and Xiu (2023) note that the most convincing ML-in-finance papers are those that clearly demonstrate when and why ML adds value over linear benchmarks.

14.4 SHAP Waterfall Plots

SHAP Background

Chapter 7 covers SHAP values in detail: Shapley’s axioms, TreeSHAP computation, and the difference between global importance (mean $|\text{SHAP}|$) and local explanation (waterfall). This section focuses specifically on how to *present* waterfall plots to a non-technical audience.

For the top 3 most interesting predictions your model makes, show a SHAP waterfall plot. These convert “trust me, the model works” into “here is exactly what drove this prediction.”

How to read a waterfall plot. A waterfall plot for a single observation shows:

1. The **base value** $\mathbb{E}[f(\mathbf{X})]$: the model’s average prediction across the dataset.
2. Each feature’s **contribution**: a bar pushing the prediction up (red) or down (blue) from the base value.
3. The **final prediction** $f(\mathbf{x}_i)$: the sum of the base value and all feature contributions.

Presenting waterfalls to a desk audience. Pick three predictions that illustrate different mechanisms:

1. **A strong correct prediction.** “On 2024-03-15, the model predicted a 2.1% drawdown in IG credit. The main driver was a spike in factor-concentration (HHI up 40% week-over-week, Chapter 9). The drawdown materialized at 1.8%.”
2. **A strong wrong prediction.** “On 2024-07-02, the model predicted a drawdown that did not happen. SHAP shows the driver was VaR utilization at 93%—but a Fed announcement reversed the move. The model cannot see policy announcements.”

3. **A prediction where GBM and ridge disagree.** “Ridge predicted no move; the GBM predicted a reversal. SHAP shows the driver was an interaction between factor concentration and cross-asset flow (Chapter 9)—a nonlinearity that ridge cannot capture.”

Do Not Show 20 Waterfall Plots

One well-chosen waterfall is worth more than twenty. If you show too many, the audience loses the thread. Pick three. Have backup slides with additional examples in case someone asks “what about [date]?” The global summary plot (mean $|\text{SHAP}|$ bar chart) belongs in the methodology section, not the results section.

Explaining to non-technical audiences. Avoid the word “Shapley.” Say: “This chart breaks down the model’s prediction into the contribution of each input. Red bars push the prediction up; blue bars push it down. The biggest bar tells you what mattered most for this particular day.” That is sufficient. If someone asks how SHAP is computed, you can explain the game-theory foundation (Chapter 7), but do not volunteer it.

14.5 One Slide Per Claim

Slide Design Principles

The “one slide per claim” principle comes from scientific presentation best practices (Tufte, 2001; Schwabish, 2014). Each slide should communicate exactly one takeaway. If the audience remembers only the title of each slide, they should understand your full argument.

Slide structure. Every slide follows the same template:

- **Title = claim.** Not “Results” but “Factor concentration predicts IG drawdowns with IC = 0.05.” The title is the takeaway.
- **Body = one chart.** The chart proves the claim. No multi-chart slides—they split attention and invite the audience to look at the wrong chart while you are talking about the right one.
- **Footer = metric.** A single line with the key number: “IC = 0.045, SR = 1.35, DSR = 0.91, $n = 1,247$.”

Suggested 10-slide deck. For the Phase 5 presentation (Task 24), structure your deck as follows:

1. **Title + thesis.** “Risk as Alpha: Testing Intermediary Asset Pricing with Daily SecDB Data.” One sentence summarizing the core finding.
2. **Theory.** He–Krishnamurthy (2013), Adrian–Etula–Muir (2014): dealer constraints price risk. SecDB gives daily data where academics use quarterly. (Chapter 2)
3. **Data and methodology.** Feature families (Chapter 9), validation framework (Chapter 11), backtesting with costs (Chapter 12). One flow diagram.
4. **Result: Signal Family A.** Cumulative P&L chart with ridge baseline. Footer: IC, Sharpe, DSR, turnover.
5. **Result: Signal Family B.** Same format.
6. **Result: Signal Family C.** Same format. (If only two families survived, use this slide for the combined model.)

7. **Ridge vs. GBM comparison.** Side-by-side bar chart of IC and Sharpe for each signal family. Footer: “GBM adds [X]% IC over ridge.”
8. **Regime decomposition.** Signal Sharpe by regime (Chapter 13). Footer: “Signal works in [N] of [K] regimes.”
9. **Capacity and costs.** Sharpe vs. transaction cost curve (Chapter 12). Footer: breakeven cost.
10. **Next steps.** What you would do with 6 more months. Extensions, additional data, live paper trading.

If They Only Read the Titles

Write your slide titles so that reading just the titles in sequence tells the complete story. Test this: list your 10 titles. If a colleague can understand your argument from the titles alone, the deck is well structured. If they cannot, rewrite the titles until they can.

14.6 What Didn’t Work as Credibility

Documented negative results are your strongest credibility signal. A presentation that reports only successes triggers the audience’s overfitting alarm. One that says “we tested five signal families; three worked, two didn’t” signals intellectual honesty.

What counts as a valuable negative result.

- “Cross-asset flow features (Chapter 9) showed no predictive power for VIX innovations after purged CV (Chapter 11). $IC = 0.003 \pm 0.14$, not significantly different from zero.”
- “Scenario P&L dispersion predicted drawdowns in-sample ($IC = 0.06$) but failed walk-forward validation (Chapter 12). We attribute this to a structural break in the scenario set after 2022.”
- “The GBM did not beat ridge on VaR-dynamics features. The relationship appears to be linear.”

Each of these is genuinely informative. They tell the audience what the *data* look like, not just what the *model* found.

Where to put negatives in your presentation. Include one slide between the results slides and the regime decomposition slide (after slide 6, before slide 8 in the suggested deck). Title it honestly: “Signal families that did not survive validation.” Show a table:

Feature Family	IC (purged CV)	DSR	Why It Failed
Cross-asset flow	0.003	−0.12	No signal after detrending
Scenario P&L	0.061 (IS) / 0.008 (OOS)	0.04	Structural break in scenarios

Negatives Build Trust Because They Are Costly

Reporting negatives is costly: it uses scarce presentation time and shows the limits of your work. The audience knows this. The fact that you are *willing* to pay that cost signals that the positives are real. A presenter who hides failures is optimizing for impression, not information. A desk head can tell the difference.

14.7 Handling Desk Questions

Know Your Backup Slides

For every question below, have a backup slide with the detailed chart or table. Your verbal answer should be two sentences. If they want more, flip to the backup slide. Prepare 5–8 backup slides covering the scenarios below.

The following questions come up in almost every desk presentation of a quantitative signal. Prepare two-sentence answers for each.

“What’s the capacity?” *Answer:* “At current turnover of $[X]\%$ /day, the signal breaks even at $[Y]$ bps per side. Institutional execution costs for $[\text{asset class}]$ are approximately $[Z]$ bps, so estimated capacity is $\$[N]$ million before market impact becomes material.”

Backup slide: Sharpe ratio vs. transaction cost curve (Chapter 12). Show the point where Sharpe hits zero.

“What happens in a crisis?” *Answer:* “We decomposed performance by regime (Chapter 13). The signal earned $SR = [A]$ in steady-state periods and $SR = [B]$ during stress. It $[\text{does not}]$ work in both regimes.”

Backup slide: Regime-conditional cumulative P&L chart with GMM regime shading.

“Why not just a linear model?” *Answer:* “We tested ridge on identical features. The GBM improves IC by $[X]\%$ and Sharpe by $[Y]\%$. SHAP identifies two interaction effects that ridge cannot capture: $[\text{feature A} \times \text{feature B}]$ and $[\text{threshold in feature C}]$.”

Backup slide: Ridge vs. GBM bar chart (Section 14.3) plus SHAP dependence plot for the key interaction (Chapter 7).

“How is this different from $[\text{public factor}]$?” *Answer:* “We ran confound checks by regressing our signal on $[\text{Fama–French} / \text{Barra} / \text{known macro factors}]$. The residual IC is $[X]$, which means $[Y]\%$ of the signal is orthogonal to published factors.”

Backup slide: Table of alpha after controlling for known factors (Chapter 8).

“Is this overfit?” *Answer:* “We tracked $[N]$ experiments in total. The Deflated Sharpe Ratio is $[X]$, which means the result survives correction for multiple testing. Walk-forward out-of-sample Sharpe is $[Y]$, consistent with purged CV estimates.”

Backup slide: CPCV distribution plot (Chapter 11) and walk-forward equity curve.

Never Say “I Don’t Know” Without a Follow-Up

If you genuinely do not know the answer, say: “I haven’t tested that specific scenario. My best estimate based on $[\text{related analysis}]$ is $[X]$. I can run that test and follow up by $[\text{date}]$.” This is honest, shows you understand what test would answer the question, and commits to a deliverable. Never just say “I don’t know” and move on.

14.8 Signal Decay and Durability

Why Signals Decay

Signal decay is the phenomenon where a predictive pattern weakens over time. This happens for two reasons: (1) *statistical decay*—the pattern was partially spurious (data mining); (2) *economic decay*—other market participants discover and trade the pattern, arbitraging it away. McLean and Pontiff (2016) disentangle these two channels.

McLean–Pontiff: Published Signals Lose 26% OOS, 58% Post-Publication

McLean and Pontiff (2016, *Journal of Finance*) study 97 cross-sectional return predictors documented in academic publications. They find:

- **Out-of-sample decay:** Portfolio returns are **26% lower** in the post-sample period (after the original paper’s sample ends but before publication). This is an upper bound on the data-mining component.
- **Post-publication decay:** Returns decline by a further **32%** (58% total) after the paper is published. This additional decay (58%–26% = 32%) is attributed to publication-informed trading: arbitrageurs read the paper and trade the signal.
- **Implication:** Roughly half of the apparent decay is real (arbitrage), and half is data mining.

What this means for your project. Your project uses proprietary SecDB data. This creates a partial shield against the McLean–Pontiff decay mechanism:

1. **Competitors cannot replicate internal risk signals.** External researchers approximate dealer positions from public data (13F filings, CFTC reports, Fed Z.1). These proxies are quarterly, lagged, and noisy (Chapter 2). SecDB data is daily and precise. If your signal’s alpha comes from the *timeliness* of the data, competitors with quarterly data cannot replicate it.
2. **But correlation with published factors reduces the shield.** If your VaR-utilization signal is 80% correlated with Adrian–Etula–Muir’s (2014) intermediary leverage factor, then 80% of the signal shares that factor’s post-publication decay. The confound checks (Chapter 8) directly measure this overlap.
3. **Structural signals decay slower.** A signal based on regulatory constraints (VaR limits, capital requirements) decays only if the regulation changes. Basel III has been stable since 2013. A signal based on behavioral patterns (crowding, herding) decays faster because behavior adapts.

Be Honest About Durability

A signal that decays is still valuable if you are the first to trade it. The question is not “will this signal last forever?” but “how long is the edge, and can we monetize it before it decays?” Present a realistic assessment: “Based on the proprietary data advantage and low correlation with published factors ($r = [X]$), we estimate a useful life of $[Y]$ years before competitors can approximate the signal with public data.”

Presenting decay honestly. In your report (Task 23) and presentation (Task 24), include a brief discussion of durability. Frame it around three questions:

1. **Data advantage:** Can an external researcher replicate this signal from public data? If yes, the McLean–Pontiff 58% decay applies in full.

2. **Factor overlap:** What fraction of the signal’s variance is explained by published factors? Report the R^2 from the confound regression (Chapter 8).
3. **Structural vs. behavioral:** Is the signal driven by regulatory constraints (slow to change) or trader behavior (fast to adapt)?

14.9 The Research Report

Task 23 calls for a full desk-ready research report. This is a written document, not slides. Structure it as follows:

1. **Executive summary** (1 page). Thesis, data, key result, capacity estimate. A busy desk head reads only this page.
2. **Hypothesis and motivation** (1–2 pages). Economic theory (Chapter 2), data advantage, prior literature.
3. **Data and features** (2–3 pages). Feature families (Chapter 9), point-in-time construction (Chapter 12), sample period, holdout reservation.
4. **Methodology** (2–3 pages). Labeling (Chapter 10), validation framework (Chapter 11), backtesting engine (Chapter 12), model specifications.
5. **Results** (3–5 pages). One subsection per signal family. Each subsection: IC table, cumulative P&L chart (with ridge baseline), SHAP summary, regime decomposition.
6. **Negative results** (1 page). What did not work and why.
7. **Robustness** (1–2 pages). CPCV distribution, DSR, walk-forward OOS, rolling IC stability, confound checks.
8. **Capacity and costs** (1 page). Sharpe vs. cost curve, breakeven cost, estimated dollar capacity.
9. **Limitations and next steps** (1 page). Sample size caveats, data coverage gaps, suggested extensions.

One chart per claim. Every chart in the report should support exactly one claim, stated in the chart’s caption. Bad caption: “Figure 3: Results.” Good caption: “Figure 3: Factor-concentration HHI predicts IG credit drawdowns (IC = 0.05, DSR = 0.91). Solid line: Light-GBM; dashed line: ridge baseline.”

Do Not Iterate After Unblinding the Holdout

Chapter 12 made this point, but it bears repeating in the presentation context. The walk-forward OOS test on the reserved holdout (Task 22) is a *one-shot* exercise. You run it once, report the result, and do not go back to tune the model. If the OOS result is disappointing, report it honestly. “In-sample SR = 1.35, out-of-sample SR = 0.72” is a legitimate result. “We re-optimized on the holdout until OOS Sharpe improved” is fraud.

14.10 Common Presentation Mistakes

1. **Leading with R^2 .** In finance, R^2 values are tiny by design. Gu, Kelly, and Xiu (2020) report monthly out-of-sample R^2 of 0.40% for their best model. If you lead with “ $R^2 = 0.003$,” your audience hears “this model explains nothing.” Lead with IC and Sharpe instead.

2. **Reporting in-sample metrics only.** Every metric you present should be out-of-sample (purged CV or walk-forward). In-sample metrics belong in the appendix, if anywhere.
3. **Omitting the number of trials.** If you report $SR = 1.2$ without saying how many experiments you ran, the audience cannot assess whether the result survives multiple-testing correction. Bailey, Borwein, and Lopez de Prado (2014) show that even with a true $SR = 0$, running ~ 45 independent trials on 5 years of daily data will produce a maximum observed SR near 1.0 by pure chance. Always report the trial count and the DSR.
4. **Showing only the best signal.** If you tested five signal families and show only the one that worked, you are p-hacking by omission. Show all five. The ones that failed are credibility (Section 14.6).
5. **Ignoring transaction costs.** A signal with $SR = 2.0$ gross and $SR = 0.3$ net is not a good signal. Always report net Sharpe. Chapter 12 covers cost modeling in detail.
6. **Confusing statistical significance with economic significance.** An IC- t of 3.5 on a signal that generates 20 bps/year of alpha is statistically significant but economically irrelevant for a desk that needs 200 bps to justify the infrastructure.

14.11 Tying It All Together

This learning guide has covered a complete research pipeline:

1. **Theory** (Chapters 1 and 2): Why dealer balance-sheet constraints should predict returns.
2. **Market context** (Chapters 3 and 4): How risk systems and dealer positioning create the data you observe.
3. **Statistical methods** (Chapters 5, 6, and 8): Ridge, LightGBM, and panel regression—the tools you use to test the hypothesis.
4. **Interpretation** (Chapter 7): SHAP, MDI, MDA—how to explain *why* the model predicts what it predicts.
5. **Feature engineering** (Chapter 9): Turning raw SecDB outputs into testable signals.
6. **Labeling** (Chapter 10): Triple-barrier, meta-labeling, and standard return targets.
7. **Validation** (Chapter 11): Purged CV, CPCV, DSR, Haircut Sharpe—the honest scorecard.
8. **Backtesting** (Chapter 12): Transaction-cost-aware P&L and walk-forward OOS.
9. **Regimes** (Chapter 13): Is your signal robust across market conditions?
10. **Presentation** (this chapter): How to communicate all of the above so that the work speaks for itself.

The connecting thread is the same thesis that appeared in Chapter 2: intermediary constraints price risk, SecDB data captures those constraints daily, and the question is whether the data advantage translates into tradeable alpha. Everything else—the infrastructure, the modeling, the validation, the presentation—is in service of answering that question rigorously.

14.12 Summary

Concept	Key Takeaway
Framing	Lead with the economic hypothesis, not the ML model. The ML is the test, not the story.
Metrics triad	Report IC (predictiveness), Sharpe (profitability), and turnover (tradeability) together. Cherry-picking one is misleading.
Ridge baseline	Every chart has two lines: model and ridge. If ridge wins, say so. This is your credibility insurance.
SHAP waterfalls	Pick three predictions. Show what drove each one. Convert “trust me” into “here’s the mechanism.”
One slide per claim	Title = claim, body = one chart, footer = metric. 10 slides total. If you read only the titles, the story should be clear.
Negative results	Report what didn’t work. It shows intellectual honesty and prevents the audience from wondering what you are hiding.
Desk Q&A	Prepare two-sentence answers for: capacity, crisis behavior, linear vs. nonlinear, public factor overlap, overfitting. Have backup slides with detailed charts.
Signal decay	McLean–Pontiff: 26% OOS decay, 58% post-publication. Proprietary data partially shields against publication-informed trading, but factor overlap reduces the shield.
Research report	Structured document: executive summary, hypothesis, data, methodology, results (with ridge), negatives, robustness, capacity, next steps. One chart per claim.
Common mistakes	Do not lead with R^2 . Report OOS metrics. State the trial count. Show all signals tested. Report net-of-cost Sharpe.